

Spring 2026 CSE 538 - Natural Language Processing Assignment 1

Due date: 11:59 pm Mar 2, 2026

Credits: This code is part of the starter package of the assignment/s used in NLP course at Stony Brook University. This assignment has been designed, implemented and revamped by many NLP TAs to varying degrees. In chronological order of TAship they include Heeyoung Kwon, Jun Kang, Mohaddeseh Bastan, Harsh Trivedi, Matthew Matero, Nikita Soni, Sharvil Katariya, Yash Kumar Lal, Adithya V. Ganesan, Sounak Mondal, Md. Saqib Hasan, Dhruv Verma, Nishit Jain, Jasdeep Grover. Thanks to all of them!

Disclaimer/License: This code is only for school assignment purpose, and **any version of this should NOT be shared publicly on github or otherwise even after semester ends**. Public availability of answers devalues usability of the assignment and work of several TAs who have contributed to this. We hope you'll respect this restriction.

Overview

In this assignment, you will:

Generate the co-occurrence matrix for GloVe: Implement a method to create a co-occurrence matrix from the provided text corpus using a sliding window approach. This matrix will capture how often words appear together in a specified window.

Implement the GloVe model with least-squares loss: Develop the GloVe model to train word embeddings using the least-squares loss function, as described in the original GloVe paper. The loss will be computed based on the co-occurrence matrix.

Tune the parameters for word embeddings: Experiment with different hyperparameters (such as vector size, learning rate, and window size) to optimize the quality of the learned word embeddings.

Apply the best-learned word embeddings to word analogy tasks: Use the trained GloVe embeddings to perform word analogy tasks, assessing their ability to capture semantic relationships between words.

Use word vectors with a logistic regression model: Integrate GloVe word vectors into a logistic regression model to evaluate their performance in a downstream supervised learning task.

Note: Please make your code edits only in the designated TODO(students) sections of the notebook. Ensure that you run all preceding cells before executing subsequent ones. Provide clear comments to explain your code, and use meaningful variable names.

How to use this notebook:

- This notebook is best viewed and executed in Google Colab.
- Please upload the .ipynb version of this notebook in Google Drive on your SBU CS account.
- Double click and select Open with Colab

- Upload the files provided in the current working directory of the Colab notebook

Please use the following Google Colab Tutorial in case you are not familiar with the tool: [Link](#)

Please start the assignment early since it takes a significant amount of time to train each model.

Setting up the data and needed libraries

```
# Install required libraries
```

```
!pip install --upgrade matplotlib numpy datasets transformers
!apt-get install git-lfs

Requirement already satisfied: matplotlib in
/usr/local/lib/python3.12/dist-packages (3.10.0)
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (52 kB)
----- 52.8/52.8 kB 1.9 MB/s eta
0:00:00
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-
packages (2.0.2)
Collecting numpy
  Downloading numpy-2.4.2-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Requirement already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting datasets
  Downloading datasets-4.6.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Collecting transformers
  Downloading transformers-5.2.0-py3-none-any.whl.metadata (32 kB)
Traceback (most recent call last):
  File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/base_comman
d.py", line 179, in exc_logging_wrapper
    status = run_func(*args)
             ^^^^^^^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/req_command
.py", line 67, in wrapper
    return func(self, options, args)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/commands/instal
l.py", line 377, in run
    requirement_set = resolver.resolve(
                     ^^^^^^^^^^^^^^^
  File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
```

```

resolvelib/resolver.py", line 95, in resolve
    result = self._result = resolver.resolve(
                                ^^^^^^^^^^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib/resolv
ers.py", line 546, in resolve
    state = resolution.resolve(requirements, max_rounds=max_rounds)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib/resolv
ers.py", line 397, in resolve
    self._add_to_criteria(self.state.criteria, r, parent=None)
File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib/resolv
ers.py", line 173, in _add_to_criteria
    if not criterion.candidates:
        ^^^^^^^^^^^^^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/resolvelib/struct
s.py", line 156, in __bool__
    return bool(self._sequence)
           ^^^^^^^^^^^^^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/found_candidates.py", line 174, in __bool__
    return any(self)
           ^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/found_candidates.py", line 162, in <genexpr>
    return (c for c in iterator if id(c) not in
self._incompatible_ids)
                                   ^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/found_candidates.py", line 116, in
_iter_built_with_inserted
    candidate = func()
               ^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/factory.py", line 185, in _make_candidate_from_link
    base: Optional[BaseCandidate] =
self._make_base_candidate_from_link(
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/factory.py", line 231, in make_base_candidate_from_link
```

```

self._link_candidate_cache[link] = LinkCandidate(
    ^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/candidates.py", line 303, in __init__
    super().__init__(
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/candidates.py", line 158, in __init__
    self.dist = self._prepare()
    ^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/candidates.py", line 247, in _prepare
    self._check_metadata_consistency(dist)
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/resolution/
resolvelib/candidates.py", line 229, in _check_metadata_consistency
    list(dist.iter_dependencies(list(dist.iter_provided_extras())))
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/metadata/import
lib/_dists.py", line 227, in iter_dependencies
    elif any(req.marker.evaluate(context) for context in contexts):
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/metadata/import
lib/_dists.py", line 227, in <genexpr>
    elif any(req.marker.evaluate(context) for context in contexts):
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/packaging/markers
.py", line 310, in evaluate
    current_environment = cast("dict[str, str]",
default_environment())
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/packaging/markers
.py", line 249, in default_environment
    "platform_machine": platform.machine(),
    ^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/platform.py", line 1017, in machine
    def machine():

```

KeyboardInterrupt

During handling of the above exception, another exception occurred:

Traceback (most recent call last):

```
File "/usr/local/bin/pip3", line 10, in <module>
```

```

    sys.exit(main())
          ^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/main.py",
line 80, in main
    return command.main(cmd_args)
           ^^^^^^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/base_command.py", line 100, in main
    return self._main(args)
           ^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/base_command.py", line 232, in _main
    return run(options, args)
           ^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/cli/base_command.py", line 215, in exc_logging_wrapper
    logger.critical("Operation cancelled by user")
File "/usr/lib/python3.12/logging/__init__.py", line 1586, in critical
    self._log(CRITICAL, msg, args, **kwargs)
File "/usr/lib/python3.12/logging/__init__.py", line 1684, in _log
    self.handle(record)
File "/usr/lib/python3.12/logging/__init__.py", line 1700, in handle
    self.callHandlers(record)
File "/usr/lib/python3.12/logging/__init__.py", line 1762, in callHandlers
    hdlr.handle(record)
File "/usr/lib/python3.12/logging/__init__.py", line 1028, in handle
    self.emit(record)
File
"/usr/local/lib/python3.12/dist-packages/pip/_internal/utils/logging.py", line 177, in emit
    self.console.print(renderable, overflow="ignore", crop=False,
style=style)
File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/rich/console.py",
line 1685, in print
    render_options = self.options.update(
                      ^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/rich/console.py",
line 983, in options
    max_height=self.size.height,
           ^^^^^^^^^^
File

```

```

"/usr/local/lib/python3.12/dist-packages/pip/_vendor/rich/console.py",
line 1003, in size
    if self.is_dumb_terminal:
        ^^^^^^^^^^^^^^^^^^^^^
File
"/usr/local/lib/python3.12/dist-packages/pip/_vendor/rich/console.py",
line 975, in is_dumb_terminal
    _term = self._environ.get("TERM", "")
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "<frozen _collections_abc>", line 807, in get
File "<frozen os>", line 715, in __getitem__
File "<frozen os>", line 795, in decode
KeyboardInterrupt
^C
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git-lfs is already the newest version (3.0.2-1ubuntu0.3).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.

# Download datafile
!wget http://mattmahoney.net/dc/text8.zip
!unzip text8.zip
!rm text8.zip

--2026-03-01 17:53:19-- http://mattmahoney.net/dc/text8.zip
Resolving mattmahoney.net (mattmahoney.net)... 20.119.76.151
Connecting to mattmahoney.net (mattmahoney.net)|20.119.76.151|:80...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 31344016 (30M) [application/zip]
Saving to: 'text8.zip'

text8.zip          100%[=====>] 29.89M  8.46MB/s   in
3.5s

2026-03-01 17:53:24 (8.46 MB/s) - 'text8.zip' saved
[31344016/31344016]

Archive: text8.zip
  inflating: text8

```

Importing needed libraries and setting up random seeds

```

# All import statements

import collections
from collections import Counter

import json

```

```

import numpy as np
from scipy.spatial import distance
from scipy import sparse

import torch
import torch.nn as nn
import torch.nn.init as init
import torch.optim as optim
from torch.utils.data import DataLoader, Dataset
from itertools import chain

import math
import time

from tqdm import tqdm

import os
import pickle

from matplotlib import pyplot as plt
from datasets import load_dataset

# Setting up all the seeds for repeatable experiments
# DO NOT CHANGE
np.random.seed(1234)
torch.manual_seed(1234)

<torch._C.Generator at 0x7b6ede708350>

```

Generating the Data

To train GloVe word vectors, we need to generate a co-occurrence matrix from the text corpus. The co-occurrence matrix counts how often two words appear together within a specified window size. GloVe records co-occurrences of all words within a sliding window around the current word.

Example: Consider the following sentence:

"The quick brown fox jumps over the lazy dog."

Assume a **window size** of 4, that means for every word, we will look at up to 4 words to the left and 4 words to the right.

For the word **"fox"**:

Context Window: ["quick", "brown", "jumps", "over", "the", "lazy"]

Update Co-occurrence Matrix:

```
Co-occurrence of ("fox", "quick") → +1
Co-occurrence of ("fox", "brown") → +1
Co-occurrence of ("fox", "jumps") → +1
Co-occurrence of ("fox", "over") → +1
Co-occurrence of ("fox", "the") → +1
Co-occurrence of ("fox", "lazy") → +1
```

Similarly, continue sliding the window for each word, recording how often each word co-occurs with the words in its context.

Note: The matrix will be symmetric, since the co-occurrence of ("word1", "word2") is the same as ("word2", "word1").

The two functions given below are helper functions and will assist in fetching the data from the file streams.

```
# Read the data into a list of strings
def read_data(filename):
    with open(filename) as file:
        text = file.read()
        # [word1, word2, word3, ... ]
        data = [token.lower() for token in text.strip().split()]
    return data

# Build the dataset and return word indices and vocab mappings
def build_dataset(words, vocab_size):
    """
    Builds the vocabulary and word index mapping.

    Args:
    - words: A list of tokenized words from the corpus.
    - vocab_size: Maximum number of unique words to include in the
    vocabulary.

    Returns:
    - data: A list of word indices (corpus converted to word IDs).
    - vocab_token_to_id: A dictionary mapping each word to a unique
    ID.
    - vocab_id_to_token: A dictionary mapping each unique ID back to
    the word.
    """
    # Count the most common words in the corpus
    count = [['UNK', -1]] # UNK for unknown words
    count.extend(Counter(words).most_common(vocab_size - 1))

    # Create a dictionary mapping each word to a unique ID
    vocab_token_to_id = {word: i for i, (word, _) in enumerate(count)}

    data = [] # Corpus converted to word indices
```



```

unk_count = 0

# Convert words to their respective IDs
for word in words:
    if word in vocab_token_to_id:
        index = vocab_token_to_id[word]
    else:
        index = 0 # UNK token for unknown words
        unk_count += 1
    data.append(index)

# Update 'UNK' token count
count[0][1] = unk_count

# Create a reverse mapping: IDs to words
vocab_id_to_token = {i: word for word, i in
vocab_token_to_id.items()}

return data, count, vocab_token_to_id, vocab_id_to_token

```

Variable Description

data_index: The index of a word in the corpus.

window_size: Defines how many words to consider to the left and right of a target word.

cooccurrence_matrix: A sparse matrix to store co-occurrence counts, where each entry represents how often two words occur together within the window.

Please fill the TODO section in the code below to generate data batches.

```

class Dataset:
    def __init__(self, data, vocab_size, window_size=4):
        """
        Initialize the GloVe dataset.
        """
        self.data = data
        self.vocab_size = vocab_size
        self.window_size = window_size
        self.cooccurrence_matrix = sparse.lil_matrix((vocab_size,
vocab_size), dtype=np.float64)
        self.num_non_zero = 0 # To track the number of non-zero co-
occurrence entries

    def generate_cooccurrence_matrix(self):
        """
        Generate the word co-occurrence matrix.
        """
        data_size = len(self.data)

        # Iterate through each word in the dataset

```

```

    for index, word_id in enumerate(self.data):
        if index % 100000 == 0:
            print(f"Building cooccurrence matrix: on line
{index}/{data_size}")

            # Define the window to capture context words
            start = max(0, index - self.window_size)
            end = min(data_size, index + self.window_size + 1)

            ### TODO(students): start
            # Loop through context words within the window
            for context_index in range(start, end):
                # skip the center word itself
                if context_index == index:
                    continue
                # Distance between the words
                distance = abs(context_index-index)

                # Weight the co-occurrence by 1/distance (closer
words get more weight)
                increment = 1/distance

                # Since co-occurrence is bidirectional, update
both sides of the matrix
                context_word_id = self.data[context_index]
                self.cooccurrence_matrix[word_id, context_word_id]
+= increment
                self.cooccurrence_matrix[context_word_id, word_id]
+= increment

            ### TODO(students): end
            # Count the number of non-zero entries (co-occurrences)
            self.num_non_zero = len(self.cooccurrence_matrix.nonzero()[0])

            return self.cooccurrence_matrix

    def get_num_batches(self, batch_size):
        """
        Returns the total number of batches.
        """
        return (self.num_non_zero + batch_size - 1) // batch_size #
Ceiling division to account for any remainder

    def generate_batches(self, batch_size):
        """
        Generator that yields batches from the co-occurrence matrix.
        """
        # Get the indices of non-zero co-occurrence entries
        non_zero_indices = self.cooccurrence_matrix.nonzero()

```

```

num_non_zero = len(non_zero_indices[0])

# Loop over batches of size `batch_size`
for i in range(0, num_non_zero, batch_size):

    batch_indices = range(i, min(i + batch_size,
num_non_zero))
    center_words = []
    context_words = []
    cooccurrence_counts = []

    ### TODO(students): start
    # loop through the selected batch indices
    # retrieve the center & context word indices and their co-
occurrence count, and append them to the lists for batching
    for batch_index in batch_indices:
        # batch_index is the center word index.
        center_index = non_zero_indices[0][batch_index]
        context_index = non_zero_indices[1][batch_index]
        cooccurrence_count =
self.cooccurrence_matrix[center_index, context_index]

        center_words.append(center_index)
        context_words.append(context_index)
        cooccurrence_counts.append(cooccurrence_count)
    ### TODO(students): end
    yield (torch.tensor(center_words),
torch.tensor(context_words), torch.tensor(cooccurrence_counts,
dtype=torch.float64))

# running a sample experiment with a limited vocabulary size and
visualizing the co-occurrence matrix

# define parameters

# Set vocabulary size and context window size
vocab_size = 5 # Limit vocabulary size
window_size = 3 # Context window size

filename = "./text8" # Input text file

# Read the first 10 words from the text file
words = read_data(filename)[:10]

# Build dataset with word indices and vocab mappings
data, count, vocab_token_to_id, vocab_id_to_token =
build_dataset(words, vocab_size)

# Display sample data and vocabulary
print(f"Sample data (words): {words[:5]}")

```

```

print(f"Sample data (indices): {data[:5]}")
print(f"Vocabulary: {list(vocab_token_to_id.keys())[:5]}")
print(f"Count: {count[:5]}")

# Create the GloVeDataset
glove_dataset = Dataset(data, vocab_size, window_size)

# Generate the co-occurrence matrix
cooccurrence_matrix = glove_dataset.generate_cooccurrence_matrix()

# Print the co-occurrence matrix as a dense array
print(cooccurrence_matrix.toarray())

Sample data (words): ['anarchism', 'originated', 'as', 'a', 'term']
Sample data (indices): [1, 2, 3, 4, 0]
Vocabulary: ['UNK', 'anarchism', 'originated', 'as', 'a']
Count: [['UNK', 6], ('anarchism', 1), ('originated', 1), ('as', 1), ('a', 1)]
Building cooccurrence matrix: on line 0/10
[[32.          0.          0.66666667  1.66666667  3.66666667]
 [ 0.          0.          2.          1.          0.66666667]
 [ 0.66666667  2.          0.          2.          1.          ]
 [ 1.66666667  1.          2.          0.          2.          ]
 [ 3.66666667  0.66666667  1.          2.          0.          ]]

```

[Reference] co-occurrence matrix:

```

[[32.          0.          0.66666667  1.66666667  3.66666667]
 [ 0.          0.          2.          1.          0.66666667]
 [ 0.66666667  2.          0.          2.          1.          ]
 [ 1.66666667  1.          2.          0.          2.          ]
 [ 3.66666667  0.66666667  1.          2.          0.          ]]

```

Building the Model

GloVe: Least-Squares Loss

In GloVe, the core idea is to factorize a matrix of word-word co-occurrence counts. GloVe minimizes a weighted least-squares loss to learn word embeddings. The GloVe model uses the word co-occurrence matrix, where each entry represents how frequently a pair of words occurs together in the corpus. The loss function aims to ensure that the dot product of word vectors (plus their respective biases) approximates the logarithm of the co-occurrence counts.

The **loss function** is given by:

$$J = \sum_{i,j} f(X_{ij}) * (w_i^T * w_j + b_i + b_j - \log(X_{ij}))^2$$

Where:

w_i and w_j are the word vectors for word i and word j , respectively (center and context embeddings).

b_i and b_j are bias terms for word i and word j .

X_{ij} is the co-occurrence count between words i and j .

$f(X_{ij})$ is a weighting function to down-weight frequent co-occurrences, ensuring that rare pairs are not neglected.

The GloVe model seeks to minimize this loss function, which is based on the logarithmic difference between the actual co-occurrence count and the predicted one, weighted by the function $f(X_{ij})$ to ensure that frequent word pairs don't dominate the training process.

This least-squares objective allows GloVe to capture both semantic similarities and relationships between words, resulting in meaningful word vectors.

Training a GloVe Model:

Training a GloVe model with the default settings involves building the co-occurrence matrix from a corpus and using it to learn the word embeddings through gradient descent. Depending on the corpus size, training time can still be significant.

For a detailed explanation of the GloVe model and its theoretical background, you can refer to the [GloVe paper](#) and the related sections in [Stanford CS224n lecture notes](#).

Please implement the models and loss functions in the code below in the TODO sections

```
class GloVe(nn.Module):
    def __init__(self, V, embedding_dim, x_max=100, alpha=3 / 4):
        """
        V: Size of the vocabulary (number of unique words)
        embedding_dim: Dimensionality of the word vectors
        x_max: Weighting parameter
        alpha: Exponent for weighting function.
        """
        super(GloVe, self).__init__()
        self.V = V
        self.embedding_dim = embedding_dim
        self.x_max = x_max
        self.alpha = alpha

        # center and context words embeddings
        self.center_embeddings = nn.Embedding(num_embeddings=V,
        embedding_dim=embedding_dim)
        self.context_embeddings = nn.Embedding(num_embeddings=V,
        embedding_dim=embedding_dim)

        # center and context words biases
        self.center_biases = nn.Embedding(num_embeddings=V,
        embedding_dim=1)
```

```

        self.context_biases = nn.Embedding(num_embeddings=V,
embedding_dim=1)

        # Initialize weights - 2 ways of initializing

        # method 1
        # self.center_embeddings.weight.data.uniform_(-0.5 /
embedding_dim, 0.5 / embedding_dim)
        # self.context_embeddings.weight.data.uniform_(-0.5 /
embedding_dim, 0.5 / embedding_dim)
        # self.center_biases.weight.data.uniform_(-0.5 /
embedding_dim, 0.5 / embedding_dim)
        # self.context_biases.weight.data.uniform_(-0.5 /
embedding_dim, 0.5 / embedding_dim)

        # method 2, we'll use this for now
        for params in self.parameters():
            init.uniform_(params, a=-1, b=1)

    def forward(self, center_word, context_word, cooccurrence_count):
        """
        Forward method to compute GloVe loss.
        center_word: Indices of center words (batch of center words).
        context_word: Indices of context words (batch of context
words).
        cooccurrence_count: The actual co-occurrence count between
center and context words.
        """
        # Embedding lookups for center and context words
        center_word_embed = self.center_embeddings(center_word) #
(batch_size, embedding_dim)
        context_word_embed = self.context_embeddings(context_word) #
(batch_size, embedding_dim)

        # Bias lookups for center and context words
        center_bias = self.center_biases(center_word).squeeze() #
(batch_size)
        context_bias = self.context_biases(context_word).squeeze() #
(batch_size)

        # Call the least-squares loss function
        return self.compute_loss(center_word_embed,
context_word_embed, center_bias, context_bias, cooccurrence_count)

    def compute_loss(self, center_word_embed, context_word_embed,
center_bias, context_bias, cooccurrence_count):
        """
        Compute the least-squares GloVe loss.
        center_word_embed: Embeddings for center words (batch_size,
embedding_dim).

```

```

        context_word_embed: Embeddings for context words (batch_size,
embedding_dim).
        center_bias: Biases for center words (batch_size).
        context_bias: Biases for context words (batch_size).
        cooccurrence_count: The actual co-occurrence count between
center and context words.
    """

    ### TODO(students): start

    # Dot product of center and context word embeddings ( $w_i^T * w_j$  in the GloVe loss equation)
    dot_product = center_word_embed * context_word_embed
    dot_product = dot_product.sum(dim=1)

    # Weighting function for co-occurrence counts  $f(X_{ij})$  in the GloVe equation
    # This gives higher weights to lower co-occurrence counts and prevents too much emphasis on frequent pairs.
    weight_factor = (cooccurrence_count/self.x_max) ** self.alpha

    weight_factor[weight_factor > 1] = 1 # Clip the weight factor to 1 for high co-occurrence counts

    # Log of co-occurrence count  $\log(X_{ij})$  in the GloVe equation
    log_cooccurrence = torch.log(cooccurrence_count)

    # Compute the GloVe least-squares loss:
    #  $(w_i^T * w_j + b_i + b_j - \log(X_{ij}))^2 * f(X_{ij})$ 
    # Where  $w_i$  and  $w_j$  are word embeddings,  $b_i$  and  $b_j$  are biases,  $X_{ij}$  is co-occurrence count.
    loss = ((dot_product + center_bias + context_bias - log_cooccurrence) ** 2) * weight_factor

    ### TODO(students): end

    # Return the mean loss over the batch
    return torch.mean(loss)

def print_closest(self, validation_words, reverse_dictionary, top_k=8):
    """
    Print the closest words to the validation set words.
    validation_words: A list of word IDs to validate.
    reverse_dictionary: Mapping from word IDs to words (for printing).
    top_k: How many closest words to print for each validation word.
    """
    print('Printing closest words')

```

```

        # Get the embeddings for center words
        embeddings = self.center_embeddings.weight.data.cpu().numpy()
        # Normalize the embeddings for comparison
        norm = np.sqrt(np.sum(np.square(embeddings), axis=1,
keepdims=True))
        normalized_embeddings = embeddings / norm

        # Get the embeddings for the validation words
        validation_embeddings = normalized_embeddings[validation_words]

        # Compute cosine similarity between validation words and the
entire vocabulary
        similarity = np.matmul(validation_embeddings,
normalized_embeddings.T)

        for i in range(len(validation_words)):
            word_id = validation_words[i]

            # Lookup word from reverse_dictionary
            word = reverse_dictionary[word_id]

            # Get the nearest words by sorting the similarity matrix
            nearest = (-similarity[i, :]).argsort()[1:top_k+1] # Sort
and ignore the word itself

            # Lookup nearest words
            nearest_words = [reverse_dictionary.get(nearest[k], 'UNK')
for k in range(top_k)]

            print(f'{word}:', nearest_words)

```

Training and Data Loading Loops

The code below uses the models and losses built above and runs the actual training process.

```

class Trainer:
    def __init__(self, model, dataset, ckpt_save_path,
reverse_dictionary):
        """
        Trainer for the GloVe model.

        Args:
        - model: The GloVe model instance.
        - dataset: The GloVe dataset instance.
        - ckpt_save_path: Path to save model checkpoints.
        - reverse_dictionary: Dictionary to map word IDs to words (for
validation purposes).
        """

```



```

self.model = model
self.dataset = dataset
self.ckpt_save_path = ckpt_save_path
self.reverse_dictionary = reverse_dictionary

def training_step(self, center_word, context_word,
cooccurrence_count):
    """
    Perform a training step that computes the GloVe loss for a
    batch.

    Args:
    - center_word: Batch of center word indices.
    - context_word: Batch of context word indices.
    - cooccurrence_count: Co-occurrence counts for word pairs.

    Returns:
    - loss: The computed loss for the current batch.
    """
    loss = self.model(center_word, context_word,
cooccurrence_count)
    return loss

def train(self, epochs, batch_size=512, learning_rate=0.05,
ckpt_steps=1000, validation_words=None, device="cpu"):
    """
    Train the GloVe model over a dataset.

    Args:
    - epochs: Number of epochs.
    - batch_size: Size of each training batch.
    - learning_rate: Learning rate for the optimizer.
    - ckpt_steps: Number of steps between each checkpoint save.
    - validation_words: List of word IDs to validate and print
closest words.
    - device: Device to run training on (CPU or GPU).
    """
    # Optimizer and DataLoader setup
    optimizer = torch.optim.Adam(self.model.parameters(),
lr=learning_rate)
    self.model.to(device)
    self.model.train()
    self.losses = []

    # Track losses for checkpoints and printing
    record_losses, record_step_nos = [], []

    # Initialize step counter
    step = 0
    num_batches = self.dataset.get_num_batches(batch_size)

```

```

    print(f"Total Batches per Epoch: {num_batches}")

    # Iterate over the number of epochs
    for epoch in range(epochs):
        print(f"Starting Epoch {epoch + 1}/{epochs}")
        tqdm_iterator =
tqdm(self.dataset.generate_batches(batch_size), desc=f"Epoch {epoch +
1}/{epochs}", total=num_batches)
        epoch_loss = []
        for center_word, context_word, cooccurrence_count in
tqdm_iterator:
            center_word = center_word.to(device)
            context_word = context_word.to(device)
            cooccurrence_count = cooccurrence_count.to(device)

            optimizer.zero_grad()

            # Perform a single training step and compute the loss
            loss = self.training_step(center_word, context_word,
cooccurrence_count)
            loss.backward()
            optimizer.step()

            # Track the loss
            self.losses.append(loss.item())
            epoch_loss.append(loss.item())

            # Update progress bar
            tqdm_iterator.set_postfix({"Batch Loss": loss.item(),
"Avg Loss": sum(self.losses[-2000:]) / len(self.losses[-2000:])})
            step += 1

            # End of epoch: log average loss for the epoch
            avg_loss = sum(epoch_loss) / len(epoch_loss)
            print(f"Epoch {epoch + 1} completed - Average Loss:
{avg_loss:.4f}")

            # Save losses for plotting
            record_losses.append(avg_loss)
            record_step_nos.append(step)

            # Save checkpoint after each epoch
            self.save_ckpt(step)

            # Print closest words after each epoch
            self.model.print_closest(validation_words,
self.reverse_dictionary)

            # Final loss summary after all epochs
            print(f"Training completed after {epochs} epochs - Final

```

```

Average Loss: {avg_loss:.4f}")

    # Return tracked losses and steps for further analysis
    return list(zip(record_losses, record_step_nos))

def save_ckpt(self, curr_step):
    """
    Save a model checkpoint.

    Args:
    - curr_step: The current step number, used in naming the
    checkpoint file.
    """
    model_save_path =
f"{self.ckpt_save_path}/glove_model_step_{curr_step}.pt"
    torch.save(self.model.state_dict(), model_save_path)
    print(f"Checkpoint saved at step {curr_step} to
{model_save_path}")

```

Training Framework

The following run_training function will train a model for you as shown in the results below. Please use it for the various experiments you will perform. The parameters of the run_training() function include many hyperparameters which should be experimented with. Some examples include vector size, batch size, vocabulary size, epochs etc. Please use function call similar to the following cell for guidance on how to train the model.

```

# Helper function to create directory if it doesn't exist
def create_path(path):
    if not os.path.exists(path):
        os.mkdir(path)
        print(f"Created path: {path}")

# Main training function
def run_training(
    epochs=10,
    lr=0.05, # Learning rate for training
    checkpoint_model_path='./checkpoints', # Path to save checkpoints
    final_model_path='./final_model', # Path to save the final model
    vocab_size=int(1e5), # Size of the vocabulary
    batch_size=512, # Batch size for training
    embedding_size=128, # Dimensionality of word vectors
    window_size=4, # Window size for co-occurrence
    checkpoint_step=1000, # Number of steps after which a checkpoint
    is saved
):
    # Create directory for checkpoints
    checkpoint_model_path = f"{checkpoint_model_path}_glove/"
    create_path(checkpoint_model_path)

```

```

# Read the data - limiting it to 1M words
words = read_data("./text8")[:1000000]

# Build the dataset with a vocabulary size
data, count, vocab_token_to_id, vocab_id_to_token =
build_dataset(words, vocab_size)
print('Most common words (+UNK)', count[:5])
print('Sample data:', data[:10], [vocab_id_to_token[i] for i in
data[:10]])

glove_dataset = Dataset(data, vocab_size, window_size=window_size)

# Generate the co-occurrence matrix
print("Generating co-occurrence matrix...")
glove_dataset.generate_cooccurrence_matrix()
print("Co-occurrence matrix generated.")

# Get the generator for batches
batch_generator = glove_dataset.generate_batches(batch_size)

# Get the first batch using next()
center_word, context_word, cooccurrence_count =
next(batch_generator)

# Print the first 8 examples from the batch
# for i in range(8):
#     print(center_word[i].item(),
vocab_id_to_token[center_word[i].item()],
#         '->', context_word[i].item(),
vocab_id_to_token[context_word[i].item()])

# Validation set: select random words to evaluate similarity
during training
valid_size = 16 # Random set of words to evaluate similarity
valid_window = 100 # Only pick dev samples in the head of the
distribution
valid_examples = np.random.choice(np.arange(1, 100), 16,
replace=False)
print("Printing Validation Set Words")
for word_id in valid_examples:
    word = vocab_id_to_token.get(word_id, 'UNK')
    print(f"Word: {word}, ID: {word_id}")

# Initialize the GloVe model
model = GloVe(V=vocab_size, embedding_dim=embedding_size)

# Initialize the Trainer
trainer = Trainer(model, glove_dataset, checkpoint_model_path,
vocab_id_to_token)

```

```

# Choose the device (CPU or GPU)
device = torch.device("cuda:0" if torch.cuda.is_available() else
"cpu")
print(f'Device: {device}')

# Train the model using epochs instead of steps
plot_losses = trainer.train(
    epochs=epochs, # Train for a specified number of epochs
    batch_size=batch_size, # Batch size for training
    learning_rate=lr, # Learning rate
    ckpt_steps=checkpoint_step, # Save a checkpoint every
specified steps
    validation_words=valid_examples, # Words to validate against
    device=device # Training device (CPU or GPU)
)

# Save the final model and embeddings
create_path(final_model_path)
model_filepath = os.path.join(final_model_path,
'glove_final_model.pt')
torch.save(model.state_dict(), model_filepath)
print(f"Final model saved at {model_filepath}")

# Save embeddings (both center and context embeddings)
embeddings_filepath = os.path.join(final_model_path,
'glove_embeddings.pkl')
with open(embeddings_filepath, 'wb') as f:
    pickle.dump({
        'vocab_token_to_id': vocab_token_to_id,
        'center_embeddings':
model.center_embeddings.weight.detach().cpu().numpy(),
        'context_embeddings':
model.context_embeddings.weight.detach().cpu().numpy()
    }, f)
print(f"Embeddings saved at {embeddings_filepath}")

# Plot the loss over time
loss_values, training_steps_counter = zip(*plot_losses)
plt.plot(training_steps_counter, loss_values)
plt.xlabel("Training Step")
plt.ylabel("Average Training Loss")
plt.title("GloVe Training Loss")
plt.show()

```

The following cell shows a demo with much lesser training epochs, embedding size and vocabulary size to test your code. Please use values closer to function defaults in the above cell for experimentation in the following sections.

Please make sure to keep an eye on the Avg. Loss value being printed as the model trains. This value should gradually decrease if you have implemented your code well.

```

start_time = time.time()

run_training(
    epochs = 10,
    lr=0.01, # Learning rate
    checkpoint_model_path='./baseline_model_checkpoints', # Directory
to save model checkpoints
    final_model_path='./final_model', # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size
    batch_size=512, # Batch size for training
    embedding_size=128, # Embedding size (vector dimension)
    window_size=3, # Context window size for co-occurrence
    checkpoint_step=2000, # Save checkpoints after every 1000 steps
)

```

```

end_time = time.time()
total_time = (end_time - start_time) / 60
print(f"Training completed in {total_time:.2f} minutes")

```

```

Created path: ./baseline_model_checkpoints_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
Co-occurrence matrix generated.
Printing Validation Set Words
Word: but, ID: 40
Word: this, ID: 36
Word: these, ID: 65
Word: war, ID: 75
Word: many, ID: 56
Word: over, ID: 88
Word: d, ID: 58
Word: from, ID: 28
Word: states, ID: 76
Word: history, ID: 83
Word: which, ID: 34

```

Word: united, ID: 85
Word: used, ID: 63
Word: during, ID: 94
Word: they, ID: 49
Word: him, ID: 87
Device: cuda:0
Total Batches per Epoch: 4507
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 4507/4507 [00:24<00:00, 187.47it/s, Batch Loss=0.284, Avg Loss=0.273]

Epoch 1 completed - Average Loss: 0.6073

Checkpoint saved at step 4507 to

./baseline_model_checkpoints_glove//glove_model_step_4507.pt

Printing closest words

but: ['being', 'UNK', 'an', 'UNK', 'UNK', 'it', 'UNK', 'odour']

this: ['it', 'is', 'before', 'that', 'have', 'being', 'world', 'external']

these: ['gulette', 'beat', 'UNK', 'assa', 'facelift', 'taft', 'UNK', 'UNK']

war: ['controversial', 'gambling', 'UNK', 'proceeding', 'UNK', 'naked', 'zouave', 'espagne']

many: ['UNK', 'berith', 'UNK', 'lightvessels', 'xu', 'successful', 'pseud', 'UNK']

over: ['UNK', 'UNK', 'him', 'UNK', 'mss', 'has', 'UNK', 'econometrics']

d: ['their', 'roberval', 'UNK', 'into', 'chichester', 'amoebosoa', 'state', 'UNK']

from: ['UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'syndicates', 'now', 'intrinsic']

states: ['UNK', 'UNK', 'UNK', 'have', 'when', 'chaney', 'been', 'UNK']

history: ['minimum', 'floren', 'occultist', 'UNK', 'unwithering', 'cleland', 'reef', 'teleprinter']

which: ['at', 'with', 'venezuelan', 'UNK', 'x', 'is', 'UNK', 'frozen']

united: ['UNK', 'UNK', 'within', 'UNK', 'UNK', 'UNK', 'subgiant', 'UNK']

used: ['country', 'not', 'certify', 'sustenance', 'benford', 'UNK', 'UNK', 'UNK']

during: ['UNK', 'UNK', 'UNK', 'intuitively', 'UNK', 'UNK', 'sidechains', 'cal']

they: ['into', 'UNK', 'UNK', 'UNK', 'UNK', 'then', 'UNK', 'UNK']

him: ['UNK', 'UNK', 'to', 'over', 'rale', 'three', 'mylist', 'UNK']

Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 4507/4507 [00:23<00:00, 191.38it/s, Batch Loss=5.89, Avg Loss=1.14]

Epoch 2 completed - Average Loss: 0.6771

Checkpoint saved at step 9014 to

```
./baseline_model_checkpoints_glove//glove_model_step_9014.pt
Printing closest words
but: ['has', 'by', 'their', 'that', 'be', 'its', 'most', 'on']
this: ['see', 'however', 'would', 'following', 'states', 'only', 'it',
'which']
these: ['history', 'people', 'there', 'form', 'about', 'through',
'most', 'so']
war: ['while', 'many', 'like', 'i', 'both', 'british', 'still', 'use']
many: ['while', 'i', 'british', 'like', 'been', 'war', 'both',
'being']
over: ['into', 'then', 'directly', 'UNK', 'area', 'him', 'UNK',
'common']
d: ['charging', 'undermine', 'individual', 'five', 'always',
'geometric', 'UNK', 'age']
from: ['mantle', 'UNK', 'occupies', 'aynrand', 'combatant', 'UNK',
'sacd', 'recreations']
states: ['while', 'national', 'being', 'like', 'many', 'british', 'i',
'which']
history: ['these', 'when', 'about', 'form', 'during', 'any', 'even',
'made']
which: ['while', 'many', 'like', 'not', 'both', 'states', 'only', 'i']
united: ['both', 'after', 'many', 'great', 'because', 'often',
'several', 'while']
used: ['while', 'not', 'state', 'both', 'i', 'many', 'time', 'only']
during: ['her', 'these', 'people', 'between', 'history', 'new',
'through', 'city']
they: ['he', 'after', 'an', 'for', 'their', 'who', 'were', 'time']
him: ['title', 'then', 'her', 'long', 'new', 'given', 'since', 'set']
Starting Epoch 3/10
```

```
Epoch 3/10: 100%|██████████| 4507/4507 [00:24<00:00, 187.69it/s, Batch
Loss=0.829, Avg Loss=0.501]
```

Epoch 3 completed - Average Loss: 0.3558

Checkpoint saved at step 13521 to

```
./baseline_model_checkpoints_glove//glove_model_step_13521.pt
```

Printing closest words

```
but: ['by', 'where', 'that', 'as', 'these', 'local', 'austria',
'politics']
this: ['american', 'also', 'was', 'several', 'both', 'against',
'world', 'which']
these: ['there', 'most', 'were', 'first', 'including', 'history',
'during', 'though']
war: ['only', 'still', 'day', 'up', 'use', 'both', 'win', 'english']
many: ['only', 'although', 'called', 'war', 'would', 'other', 'use',
'both']
over: ['however', 'then', 'people', 'during', 'u', 'there', 'most',
'these']
d: ['american', 'see', 'world', 'english', 'with', 'television',
'six', 'c']
```


from: ['policy', 'UNK', 'is', 'disfavor', 'roman', 'his',
'schistosomiasis', 'nehru']
states: ['time', 'general', 'under', 'UNK', 'an', 'medical', 'UNK',
'religious']
history: ['most', 'including', 'found', 'about', 'during', 'known',
'there', 'these']
which: ['was', 'six', 'to', 'three', 'it', 'american', 'this', 'see']
united: ['UNK', 'UNK', 'australasia', 'hating', 'UNK', 'balances',
'azerbaijan', 'gifted']
used: ['called', 'only', 'however', 'has', 'then', 'death', 'many',
'into']
during: ['there', 'people', 'most', 'list', 'though', 'history',
'including', 'known']
they: ['their', 'were', 'economic', 'public', 'my', 'produced',
'brought', 'greek']
him: ['may', 'film', 'more', 'new', 'common', 'than', 'much',
'australia']
Starting Epoch 4/10

Epoch 4/10: 100%[██████████] 4507/4507 [00:24<00:00, 187.47it/s, Batch
Loss=2.8, Avg Loss=0.634]

Epoch 4 completed - Average Loss: 0.3776

Checkpoint saved at step 18028 to

./baseline_model_checkpoints_glove//glove_model_step_18028.pt

Printing closest words

but: ['were', 'they', 'their', 'i', 'being', 'all', 'like', 'first']

this: ['deontological', 'dongell', 'leach', 'UNK', 'wires', 'doyon',
'pons', 'depriving']

these: ['movement', 'found', 'various', 'major', 'addition',
'traditional', 'public', 'people']

war: ['only', 'use', 'some', 'would', 'world', 'many', 'both',
'state']

many: ['used', 'been', 'other', 'into', 'all', 'only', 'were', 'i']

over: ['then', 'called', 'however', 'people', 'while', 'years',
'about', 'john']

d: ['back', 'law', 'open', 'curbstone', 'called', 'french', 'bleed',
'UNK']

from: ['occur', 'africa', 'such', 'describe', 'appears', 'island',
'power', 'said']

states: ['time', 'being', 'all', 'early', 'when', 'united', 'like',
'they']

history: ['during', 'often', 'god', 'were', 'terms', 'number',
'being', 'great']

which: ['see', 'to', 'bridging', 'pictured', 'icao', 'largely', 'UNK',
'summit']

united: ['being', 'when', 'british', 'i', 'but', 'were', 'history',
'many']

used: ['many', 'only', 'been', 'into', 'were', 'all', 'they', 'i']

during: ['history', 'often', 'were', 'where', 'u', 'terms', 'no',

'god']
they: ['were', 'all', 'their', 'but', 'i', 'first', 'many', 'used']
him: ['new', 'had', 'may', 'also', 'power', 'system', 'life',
'called']

Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.73it/s, Batch
Loss=0.887, Avg Loss=0.447]

Epoch 5 completed - Average Loss: 0.2930

Checkpoint saved at step 22535 to

./baseline_model_checkpoints_glove//glove_model_step_22535.pt

Printing closest words

but: ['six', 'time', 'its', 'when', 'states', 'being', 'early', 'no']

this: ['have', 'from', 'not', 'another', 'which', 'also', 'even',
'any']

these: ['then', 'however', 'than', 'such', 'there', 'people', 'while',
'about']

war: ['many', 'de', 'other', 'during', 'often', 'i', 'much', 'where']

many: ['other', 'all', 'war', 'after', 'i', 'de', 'often', 'out']

over: ['about', 'however', 'people', 'while', 'called', 'through',
'then', 'known']

d: ['actress', 'n', 'from', 'seven', 'later', 'e', 'four', 'about']

from: ['this', 'four', 'have', 'to', 'also', 'not', 'he', 'any']

states: ['time', 'being', 'but', 'united', 'number', 'early',
'within', 'when']

history: ['other', 'during', 'natural', 'great', 'many', 'often',
'war', 'widespread']

which: ['this', 'they', 'have', 'not', 'but', 'from', 'its', 'six']

united: ['being', 'british', 'number', 'within', 'states', 'any',
'early', 'world']

used: ['state', 'use', 'their', 'became', 'work', 'both', 'has', 'we']

during: ['often', 'god', 'where', 'sometimes', 'england', 'out',
'very', 'u']

they: ['which', 'but', 'she', 'way', 'time', 'when', 'its', 'any']

him: ['area', 'well', 'own', 'said', 'do', 'large', 'australia',
'island']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.44it/s, Batch
Loss=1.99, Avg Loss=0.516]

Epoch 6 completed - Average Loss: 0.3134

Checkpoint saved at step 27042 to

./baseline_model_checkpoints_glove//glove_model_step_27042.pt

Printing closest words

but: ['has', 'provision', 'hideo', 'monica', 'disparity', 'llewellyn',
'dardanelles', 'sovereign']

this: ['to', 'achilles', 'as', 'anchorage', 'still', 'place', 'court',
'sometimes']

```
these: ['their', 'when', 'called', 'second', 'same', 'form',  
'article', 'have']  
war: ['after', 'year', 'would', 'during', 'c', 'i', 'only', 'where']  
many: ['five', 'b', 'i', 'four', 'apollo', 'most', 'eight', 'eardrum']  
over: ['being', 'then', 'about', 'people', 'from', 'four', 'day',  
'states']  
d: ['children', 'five', 'isbn', 'chas', 'four', 'an', 'he', 'had']  
from: ['which', 'his', 'or', 'are', 'first', 'with', 'that', 'being']  
states: ['being', 'day', 'most', 'from', 'm', 'within', 'up', 'by']  
history: ['after', 'her', 'b', 'made', 'based', 'father', 'being',  
'any']  
which: ['from', 'no', 'are', 'then', 'were', 'first', 'not', 'with']  
united: ['them', 'british', 'another', 'human', 'western', 'use',  
'world', 'jewish']  
used: ['has', 'called', 'flawed', 'asteroid', 'however', 'but',  
'transmit', 'imoko']  
during: ['u', 'year', 'after', 'where', 'own', 'often', 'other',  
'france']  
they: ['history', 'his', 'after', 'first', 'from', 'day', 'her',  
'had']  
him: ['new', 'her', 'books', 'own', 'between', 'often', 'more', 'out']  
Starting Epoch 7/10
```

```
Epoch 7/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.84it/s, Batch  
Loss=0.971, Avg Loss=0.416]
```

Epoch 7 completed - Average Loss: 0.2729

Checkpoint saved at step 31549 to

./baseline_model_checkpoints_glove//glove_model_step_31549.pt

Printing closest words

but: ['has', 'used', 'their', 'they', 'while', 'when', 'all', 'see']

this: ['later', 'work', 'who', 'during', 'like', 'several', 'country',
'idea']

these: ['while', 'several', 'however', 'would', 'only', 'has', 'men',
'general']

war: ['during', 'being', 'i', 'still', 'jews', 'water', 'such',
'within']

many: ['there', 'than', 'history', 'under', 'so', 'including', 'well',
'when']

over: ['being', 'such', 'i', 'made', 'then', 'within', 'large',
'official']

d: ['b', 'washington', 'flight', 'king', 'writer', 'imagemap',
'deaths', 'american']

from: ['that', 's', 'his', 'are', 'minutes', 'three', 'eight',
'kharo']

states: ['being', 'day', 'm', 'most', 'made', 'alexander', 'such',
'found']

history: ['so', 'will', 'him', 'under', 'where', 'what', 'after',
'set']

which: ['such', 'not', 'most', 'part', 'states', 'being', 'over',

```
'even']
united: ['same', 'however', 'human', 'ability', 'known', 'various',
'like', 'country']
used: ['called', 'when', 'but', 'their', 'both', 'they', 'may',
'often']
during: ['war', 'i', 'water', 'g', 'idea', 'jews', 'still',
'development']
they: ['used', 'but', 'also', 'had', 'were', 'he', 'may', 'its']
him: ['will', 'where', 'so', 'often', 'between', 'her', 'history',
'after']
```

Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.68it/s, Batch Loss=1.69, Avg Loss=0.462]

Epoch 8 completed - Average Loss: 0.2880

Checkpoint saved at step 36056 to

./baseline_model_checkpoints_glove//glove_model_step_36056.pt

Printing closest words

```
but: ['were', 'first', 'its', 'after', 'had', 'all', 'more', 'while']
this: ['have', 'for', 'that', 'their', 'at', 's', 'was', 'by']
these: ['locrians', 'ecumenical', 'UNK', 'steroids', 'organically',
'UNK', 'morerod', 'verbs']
war: ['i', 'during', 'alexander', 'second', 'still', 'state', 'lost',
'g']
many: ['there', 'also', 'him', 'where', 'may', 'used', 'six', 'some']
over: ['french', 'then', 'time', 'out', 'during', 'i', 'english',
'jews']
d: ['authority', 'apple', 'cars', 'UNK', 'patchily', 'image',
'bubbles', 'ahab']
from: ['had', 'bc', 'society', 'focus', 'claims', 'states', 'UNK',
'enfeoffment']
states: ['sense', 'apple', 'being', 'society', 'play', 'think',
'warhol', 'anarchists']
history: ['may', 'often', 'between', 'well', 'is', 'works', 'own',
'used']
which: ['at', 'this', 'who', 'many', 'was', 'their', 'have', 'that']
united: ['state', 'life', 'war', 'america', 'world', 'until',
'modern', 'became']
used: ['may', 'often', 'him', 'where', 'many', 'some', 'well', 'have']
during: ['part', 'i', 'g', 'war', 'state', 'if', 'time', 'still']
they: ['also', 'it', 'used', 'long', 'as', 'may', 'so', 'have']
him: ['may', 'often', 'between', 'used', 'many', 'have', 'well',
'where']
```

Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 4507/4507 [00:24<00:00, 186.38it/s, Batch Loss=1.02, Avg Loss=0.404]

Epoch 9 completed - Average Loss: 0.2648
Checkpoint saved at step 40563 to
./baseline_model_checkpoints_glove//glove_model_step_40563.pt
Printing closest words
but: ['its', 'first', 'more', 'however', 'would', 'all', 'other',
'when']
this: ['UNK', 'restricting', 'that', 's', 'UNK', 'cooperates', 'UNK',
'wonted']
these: ['each', 'known', 'see', 'used', 'city', 'how', 'book', 'they']
war: ['found', 'part', 'months', 'english', 'countries', 'north',
'tributaries', 'members']
many: ['ctet', 'though', 'add', 'UNK', 'ethica', 'surgit', 'see',
'UNK']
over: ['such', 'who', 'when', 'made', 'however', 'while', 'people',
'would']
d: ['e', 'b', 'however', 'john', 'article', 'day', 'first', 'more']
from: ['with', 'made', 'a', 'to', 'or', 'two', 'day', 'his']
states: ['being', 'without', 'both', 'then', 'than', 'would', 'later',
'like']
history: ['often', 'where', 'used', 'city', 'king', 'book', 'there',
'them']
which: ['it', 'who', 'time', 'or', 'two', 'are', 'other', 'not']
united: ['world', 'system', 'time', 'became', 'modern', 'new', 'life',
'any']
used: ['there', 'often', 'history', 'where', 'book', 'become',
'information', 'support']
during: ['s', 'the', 'work', 'of', 'having', 'UNK', 'fleeced', 'a']
they: ['also', 'used', 'these', 'history', 'king', 'pass', 'god',
'often']
him: ['has', 'satyendra', 'UNK', 'UNK', 'disambiguation', 'logik',
'oresund', 'nemours']
Starting Epoch 10/10

Epoch 10/10: 100%[██████████] 4507/4507 [00:23<00:00, 189.54it/s,
Batch Loss=1.62, Avg Loss=0.434]

Epoch 10 completed - Average Loss: 0.2741
Checkpoint saved at step 45070 to
./baseline_model_checkpoints_glove//glove_model_step_45070.pt
Printing closest words
but: ['two', 'its', 'driscoll', 'weeks', 'sociobiologists', 'name',
'into', 'year']
this: ['is', 'in', 's', 'several', 'energy', 'huston', 'whose',
'that']
these: ['taczanowskii', 'form', 'what', 'apollo', 'last', 'how',
'national', 'UNK']
war: ['state', 'see', 'states', 'known', 'were', 'found', 'had',
'with']
many: ['politician', 'a', 'not', 'five', 'an', 'weisse', 'd', 'eight']
over: ['between', 'early', 'out', 'de', 'western', 'while', 'any',

```
'only']
d: ['five', 'three', 'world', 'however', 'john', 'new', 'with', 'an']
from: ['are', 'president', 'or', 'to', 'two', 'his', 'notched',
'redness']
states: ['state', 'like', 'later', 'see', 'were', 'found', 'had',
'very']
history: ['will', 'story', 'often', 'book', 'articles', 'offered',
'league', 'christian']
which: ['it', 'other', 'father', 'just', 'who', 'jesus', 'such',
'important']
united: ['life', 'well', 'world', 'modern', 'day', 'became', 'new',
'however']
used: ['how', 'where', 'often', 'given', 'government', 'there',
'father', 'god']
during: ['her', 'government', 'number', 'general', 'because', 'based',
'r', 'died']
they: ['also', 'on', 'six', 'later', 'life', 'were', 'five', 'united']
him: ['government', 'how', 'here', 'god', 'just', 'must', 'act',
'given']
Training completed after 10 epochs - Final Average Loss: 0.2741
Created path: ./final_model
Final model saved at ./final_model/glove_final_model.pt
Embeddings saved at ./final_model/glove_embeddings.pkl
```



Training completed in 4.92 minutes

[Reference]: Training completed after 10 epochs - Final Average Loss: 0.2742

Running Experiments with various hyper-parameters for Glove

Make sure you tune the following hyper parameters:

- Learning Rate (Can vary from 0.1 to 10)
- Window size (Can vary from 1 to 10)
- Embedding Size (Can vary from 50 to 300)

For each hyperparameter setting, you will see a plot for each model trained. Please see below for possible values of each hyperparameter that you can experiment with:

- lr - [0.1, 0.5, 0.05, 0.01, 1]
- window_size - [3, 4, 5, 10]
- embedding_size - [50, 100, 200, 300]

Experiment 1

For the first experiment, set window_size = 3. Then, vary the learning rate through all three possible values and train different models for each hyperparameter setting. Use the plotted loss for each model to then select the best value of learning rate to be used going forward. Also make sure to assign appropriate names to the checkpoint_model_path and final_model_path arguments.

Hyperparameter Setting 1 for Experiment 1

```
# TODO(student): start

lr = 0.05 # Learning rate
checkpoint_model_path = "exp_1_setting_1" # Path to save checkpoints
final_model_path = "exp_1_setting_1_final_model" # Path to save the
final model and embeddings

run_training(
    epochs = 10,
    lr=lr, # Learning rate
    checkpoint_model_path=checkpoint_model_path, # Directory to save
model checkpoints
    final_model_path=final_model_path, # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size
    batch_size=512, # Batch size for training
    embedding_size=128, # Embedding size (vector dimension)
    window_size=3, # Context window size for co-occurrence
    checkpoint_step=5000 # Save checkpoints after every 1000 steps
```

```

)

# TODO(student): end

Created path: exp_1_setting_1_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
Co-occurrence matrix generated.
Printing Validation Set Words
Word: world, ID: 66
Word: its, ID: 46
Word: has, ID: 41
Word: he, ID: 32
Word: war, ID: 75
Word: they, ID: 49
Word: the, ID: 1
Word: all, ID: 51
Word: most, ID: 53
Word: would, ID: 77
Word: two, ID: 11
Word: was, ID: 15
Word: over, ID: 88
Word: american, ID: 48
Word: one, ID: 4
Word: b, ID: 59
Device: cuda:0
Total Batches per Epoch: 4507
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 4507/4507 [00:23<00:00, 191.97it/s, Batch
Loss=2.04, Avg Loss=2.18]

Epoch 1 completed - Average Loss: 2.0912
Checkpoint saved at step 4507 to
exp_1_setting_1_glove//glove_model_step_4507.pt
Printing closest words

```


world: ['fact', 'UNK', 'include', 'general', 'pythia', 'UNK', 'chief', 'UNK']
its: ['bourne', 'clayton', 'UNK', 'UNK', 'cape', 'kutha', 'dwellings', 'other']
has: ['elects', 'virginity', 'baryons', 'UNK', 'accents', 'UNK', 'UNK', 'their']
he: ['UNK', 'large', 'view', 'UNK', 'nec', 'never', 'UNK', 'superscript']
war: ['when', 'government', 'did', 'would', 'no', 'of', 'century', 'internet']
they: ['UNK', 'mathrmy', 'UNK', 'sedna', 'places', 'anagrams', 'dufferin', 'UNK']
the: ['obscuring', 'UNK', 'campaign', 'bourgeois', 'UNK', 'esthetic', 'UNK', 'alonzo']
all: ['habor', 'porch', 'UNK', 'silently', 'UNK', 'wizard', 'are', 'UNK']
most: ['used', 'popular', 'can', 'union', 'f', 'what', 'UNK', 'army']
would: ['war', 'main', 'no', 'of', 'strong', 'if', 'north', 'national']
two: ['death', 'u', 'prelude', 'draw', 'UNK', 'biblica', 'sinauer', 'could']
was: ['had', 'esdras', 'reenactment', 'alhazared', 'gianna', 'UNK', 'santamaria', 'UNK']
over: ['years', 'history', 'church', 'more', 'first', 'although', 'family', 'king']
american: ['popcorn', 'consent', 'hurtles', 'scheuch', 'angustiarum', 'UNK', 'elixirs', 'UNK']
one: ['luli', 'berretta', 'ascanio', 'numbers', 'to', 'substrate', 'magician', 'zur']
b: ['state', 'even', 'japanese', 'aq', 'beagle', 'offutt', 'arafura', 'celebrate']

Starting Epoch 2/10

Epoch 2/10: 100%[██████████] 4507/4507 [00:23<00:00, 189.27it/s, Batch Loss=250, Avg Loss=100]

Epoch 2 completed - Average Loss: 58.7100

Checkpoint saved at step 9014 to

exp_1_setting_1_glove/glove_model_step_9014.pt

Printing closest words

world: ['over', 'states', 'each', 'common', 'new', 'different', 'later', 'country']
its: ['UNK', 'oxidative', 'become', 'however', 'huascar', 'hoplites', 'UNK', 'preparing']
has: ['naboth', 'gm', 'these', 'countries', 'hemophiliacs', 'curricula', 'klm', 'same']
he: ['headland', 'UNK', 'halftime', 'josquin', 'crc', 'first', 'partment', 'UNK']
war: ['out', 'would', 'century', 'year', 'through', 'may', 'english', 'since']

they: ['shorts', 'people', 'against', 'anti', 'five', 'usonia',
'retroviral', 'hubert']
the: ['rer', 'UNK', 'scholastic', 'casibari', 'hess',
'bidirectionality', 'UNK', 'hillfoot']
all: ['reconciling', 'baoil', 'carcinoma', 'tenascin', 'aeneid',
'disseisin', 'linked', 'are']
most: ['been', 'states', 'who', 'valuation', 'now', 'nealbutrus',
'monty', 'UNK']
would: ['people', 'year', 'anti', 'right', 'war', 'found', 'where',
'god']
two: ['larder', 'UNK', 'hermias', 'hugues', 'UNK', 'mors', 'mudur',
'glorias']
was: ['UNK', 'mineptah', 'geo', 'exemplary', 'probl', 'noisy',
'andrei', 'mar']
over: ['states', 'science', 'world', 'became', 'made', 'death',
'number', 'years']
american: ['lanoye', 'shemot', 'memorization', 'mock',
'professionalism', 'ark', 'UNK', 'UNK']
one: ['turing', 'recommending', 'incarnation', 'sari', 'stolbova',
'saqqara', 'ded', 'abby']
b: ['irelandseye', 'truda', 'kwango', 'symmetrical', 'him', 'UNK',
'snowfactory', 'modulating']
Starting Epoch 3/10

Epoch 3/10: 100%[██████████] 4507/4507 [00:23<00:00, 189.51it/s, Batch
Loss=933, Avg Loss=337]

Epoch 3 completed - Average Loss: 167.9409

Checkpoint saved at step 13521 to

exp_1_setting_1_glove//glove_model_step_13521.pt

Printing closest words

world: ['feb', 'lum', 'cavalrymen', 'UNK', 'cabannes', 'dicaearchus',
'UNK', 'nolo']
its: ['hevelius', 'have', 'lydon', 'correlation', 'existed', 'skull',
'would', 'UNK']
has: ['recast', 'immortallity', 'featherlite', 'UNK', 'deleted',
'tacklers', 'perales', 'empowers']
he: ['bactrians', 'fixups', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK']
war: ['british', 'brutally', 'amp', 'profane', 'mojadeddi',
'transformation', 'multiparty', 'burlington']
they: ['first', 'UNK', 'UNK', 'raya', 'prantl', 'sireland', 'UNK',
'techniques']
the: ['UNK', 'immoral', 'UNK', 'UNK', 'mujahideen', 'velcro', 'UNK',
'UNK']
all: ['been', 'mononoke', 'folds', 'neutron', 'UNK', 'UNK', 'alembic',
'UNK']
most: ['xii', 'boyle', 'disbeliever', 'things', 'been', 'brewery',
'before', 'kennicott']
would: ['however', 'jost', 'nursery', 'aristotelicum', 'bookshelf',
'cahoots', 'adopt', 'its']

two: ['conceptions', 'UNK', 'curriculum', 'there', 'comparisons',
'operating', 'kingly', 'UNK']
was: ['joop', 'translations', 'UNK', 'hme', 'UNK', 'UNK', 'UNK',
'UNK']
over: ['rwandan', 'mdg', 'retains', 'duodenum', 'UNK', 'UNK',
'recluse', 'natural']
american: ['UNK', 'tractatus', 'UNK', 'UNK', 'spilled', 'UNK',
'sauropsida', 'UNK']
one: ['UNK', 'UNK', 'resorted', 'UNK', 'amusing', 'khufu', 'rotten',
'mynga']
b: ['UNK', 'italian', 'satyrs', 'seii', 'commence', 'UNK',
'antibodies', 'UNK']
Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 4507/4507 [00:24<00:00, 187.02it/s, Batch
Loss=635, Avg Loss=154]

Epoch 4 completed - Average Loss: 95.0727

Checkpoint saved at step 18028 to

exp_1_setting_1_glove//glove_model_step_18028.pt

Printing closest words

world: ['neoptolemus', 'belonged', 'UNK', 'UNK', 'priori', 'UNK',
'party', 'state']

its: ['UNK', 'lengthier', 'renting', 'UNK', 'impart', 'UNK', 'their',
'periods']

has: ['about', 'rather', 'their', 'six', 'some', 'further', 'written',
'napalm']

he: ['headland', 'UNK', 'UNK', 'UNK', 'aran', 'sedimentation',
'stock', 'nondescript']

war: ['city', 'computer', 'so', 'anachronistic', 'flowers', 'UNK',
'made', 'god']

they: ['some', 'vanquishing', 'clumsy', 'warburg', 'sarapo', 'UNK',
'UNK', 'UNK']

the: ['tig', 'drusiana', 'UNK', 'UNK', 'kuwait', 'UNK', 'undeniable',
'unfeasible']

all: ['multiplexed', 'UNK', 'rzburg', 'UNK', 'unrequited', 'UNK',
'greensport', 'UNK']

most: ['UNK', 'lication', 'UNK', 'ergodic', 'photoelectricity',
'mauriya', 'UNK', 'tankian']

would: ['slaves', 'UNK', 'UNK', 'hardenbrook', 'near', 'stark',
'several', 'UNK']

two: ['UNK', 'UNK', 'UNK', 'UNK', 'steamships', 'UNK', 'UNK', 'UNK']
was: ['husbands', 'newschool', 'bibliographer', 'UNK', 'UNK', 'UNK',
'UNK', 'UNK']

over: ['burnett', 'UNK', 'UNK', 'UNK', 'blooded', 'game', 'nio',
'UNK']

american: ['UNK', 'UNK', 'suppressed', 'dihydrogen', 'logical',
'claymation', 'UNK', 'arvid']

one: ['UNK', 'propanol', 'UNK', 'UNK', 'slowly', 'UNK', 'aseev',
'UNK']

b: ['UNK', 'loose', 'norris', 'UNK', 'falklands', 'connexxion',
'chloroplast', 'UNK']
Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.36it/s, Batch
Loss=843, Avg Loss=247]

Epoch 5 completed - Average Loss: 134.4283

Checkpoint saved at step 22535 to

exp_1_setting_1_glove//glove_model_step_22535.pt

Printing closest words

world: ['homooousion', 'UNK', 'melito', 'tib', 'UNK', 'unsuccessful',
'delineations', 'UNK']

its: ['soot', 'wraiths', 'UNK', 'fayette', 'pi', 'hindered', 'koston',
'interceptions']

has: ['six', 'UNK', 'censure', 'UNK', 'their', 'darbari', 'zigzag',
'antonioni']

he: ['UNK', 'bremer', 'UNK', 'scanty', 'UNK', 'platforms', 'seats',
'deserts']

war: ['state', 'UNK', 'century', 'UNK', 'aris', 'samoan',
'lincolnshire', 'UNK']

they: ['some', 'recreational', 'colour', 'glamorous', 'UNK', 'UNK',
'UNK', 'automata']

the: ['UNK', 'listeners', 'UNK', 'UNK', 'UNK', 'degneration', 'UNK',
'prostitution']

all: ['UNK', 'bracketed', 'UNK', 'UNK', 'father', 'UNK', 'hottest',
'labelled']

most: ['UNK', 'curtis', 'UNK', 'attainment', 'UNK', 'UNK', 'cyanide',
'UNK']

would: ['opler', 'between', 'undernourished', 'dormant', 'UNK',
'surpluses', 'have', 'UNK']

two: ['timarchus', 'UNK', 'UNK', 'ardley', 'goodeniaceae', 'UNK',
'UNK', 'arrivals']

was: ['UNK', 'UNK', 'shamanists', 'UNK', 'UNK', 'unmolested', 'UNK',
'UNK']

over: ['icemaking', 'UNK', 'brennan', 'UNK', 'UNK', 'UNK',
'outranked', 'standby']

american: ['davida', 'punted', 'UNK', 'UNK', 'UNK', 'birch', 'UNK',
'dudok']

one: ['UNK', 'nigroaenea', 'jirishanca', 'UNK', 'UNK', 'hiroyuki',
'ncia', 'UNK']

b: ['northern', 'UNK', 'UNK', 'olaf', 'shelter', 'UNK', 'UNK', 'UNK']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.13it/s, Batch
Loss=367, Avg Loss=146]

Epoch 6 completed - Average Loss: 87.1332

Checkpoint saved at step 27042 to

exp_1_setting_1_glove//glove_model_step_27042.pt

Printing closest words

world: ['known', 'him', 'any', 'example', 'since', 'never', 'given', 'times']

its: ['UNK', 'UNK', 'UNK', 'UNK', 'windsor', 'UNK', 'funereus', 'UNK']

has: ['six', 'no', 'siula', 'mwt', 'latchets', 'pinnate', 'UNK', 'UNK']

he: ['had', 'UNK', 'tenedos', 'redeclared', 'leale', 'UNK', 'casually', 'abbacies']

war: ['well', 'e', 'gedanite', 'UNK', 'UNK', 'beasley', 'adonis', 'UNK']

they: ['renegades', 'UNK', 'ratings', 'UNK', 'UNK', 'fukasaku', 'UNK', 'four']

the: ['UNK', 'millions', 'amerikano', 'UNK', 'UNK', 'blankly', 'trucks', 'mutu']

all: ['UNK', 'UNK', 'paths', 'utulei', 'UNK', 'genres', 'been', 'stiftskirche']

most: ['made', 'redlichida', 'nathanael', 'trumble', 'besides', 'somers', 'UNK', 'effeminate']

would: ['michiel', 'dekhoda', 'UNK', 'UNK', 'UNK', 'mislead', 'inversion', 'leper']

two: ['jos', 'pallida', 'UNK', 'transducer', 'illimitable', 'UNK', 'UNK', 'UNK']

was: ['UNK', 'UNK', 'facilitate', 'grandfathering', 'UNK', 'disappointed', 'ivoire', 'talmudic']

over: ['him', 'recognized', 'filaments', 'UNK', 'UNK', 'giggles', 'UNK', 'saviour']

american: ['UNK', 'UNK', 'UNK', 'gamow', 'keenly', 'telfair', 'UNK', 'UNK']

one: ['wine', 'downloaded', 'chi', 'viet', 'bryophytes', 'reign', 'renal', 'UNK']

b: ['australia', 'bilingualism', 'UNK', 'UNK', 'eclipse', 'than', 'UNK', 'UNK']

Starting Epoch 7/10

Epoch 7/10: 100%[██████████] 4507/4507 [00:23<00:00, 190.35it/s, Batch Loss=739, Avg Loss=222]

Epoch 7 completed - Average Loss: 120.4184

Checkpoint saved at step 31549 to

exp_1_setting_1_glove/glove_model_step_31549.pt

Printing closest words

world: ['any', 'united', 'vegetarianism', 'seven', 'french', 'UNK', 'estuary', 'example']

its: ['UNK', 'anniversarial', 'UNK', 'UNK', 'cartleyhole', 'conquista', 'UNK', 'UNK']

has: ['six', 'UNK', 'not', 'UNK', 'UNK', 'UNK', 'khans', 'slayer']

he: ['had', 'associate', 'UNK', 'after', 'enclosure', 'but', 'UNK', 'UNK']

war: ['de', 'can', 'leloaloa', 'UNK', 'letelier', 'UNK', 'hexaminecopper', 'bight']

they: ['poached', 'ment', 'elope', 'sergio', 'poaching', 'multiparty',
'UNK', 'wayassa']
the: ['UNK', 'UNK', 'revoluta', 'UNK', 'roadblock', 'UNK',
'councilor', 'penny']
all: ['vee', 'UNK', 'datalink', 'UNK', 'UNK', 'spiced', 'behaves',
'akhaltzikke']
most: ['span', 'UNK', 'UNK', 'synthesizing', 'cone', 'alphonse',
'gastritis', 'UNK']
would: ['wolcott', 'competing', 'UNK', 'denaturing', 'UNK',
'synthasizer', 'UNK', 'states']
two: ['UNK', 'iec', 'muddy', 'UNK', 'UNK', 'UNK', 'by', 'UNK']
was: ['cholesterol', 'amo', 'UNK', 'UNK', 'sadachbia', 'UNK', 'UNK',
'UNK']
over: ['UNK', 'cloaking', 'stupid', 'UNK', 'UNK', 'entertained',
'lili', 'loder']
american: ['UNK', 'UNK', 'UNK', 'blooded', 'cyberpunk', 'UNK', 'fisk',
'tordylum']
one: ['thar', 'UNK', 'businesses', 'UNK', 'complaint', 'UNK', 'hcl',
'dingo']
b: ['UNK', 'polynesia', 'theatrical', 'UNK', 'mnras', 'system', 'UNK',
'imps']
Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.29it/s, Batch
Loss=321, Avg Loss=151]

Epoch 8 completed - Average Loss: 88.3898

Checkpoint saved at step 36056 to

exp_1_setting_1_glove//glove_model_step_36056.pt

Printing closest words

world: ['seven', 'union', 'plunderers', 'offensives', 'UNK', 'wove',
'philately', 'intra']

its: ['UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'first', 'UNK']

has: ['six', 'sprint', 'UNK', 'UNK', 'UNK', 'ramifications',
'amoebzoa', 'UNK']

he: ['had', 'but', 'UNK', 'UNK', 'UNK', 'squandering', 'susitna',
'UNK']

war: ['UNK', 'UNK', 'UNK', 'longtime', 'liquefy', 'independance',
'detest', 'UNK']

they: ['organize', 'UNK', 'stalinist', 'distraught', 'business',
'UNK', 'brugsch', 'cesstion']

the: ['grandnephew', 'UNK', 'valentinians', 'producing', 'UNK', 'UNK',
'deinosuchus', 'accumulator']

all: ['used', 'meditative', 'tallapoosa', 'billionaire', 'UNK', 'UNK',
'karting', 'either']

most: ['orients', 'bransfield', 'UNK', 'goal', 'UNK', 'lettres',
'UNK', 'castra']

would: ['sociolinguistic', 'UNK', 'UNK', 'distinguishable',
'confessors', 'infraclass', 'fargas', 'UNK']

two: ['anagramming', 'equitable', 'UNK', 'gasperi', 'UNK', 'UNK',

'aerodrome', 'UNK']
was: ['matching', 'emerson', 'UNK', 'UNK', 'grapple', 'biofacts',
'UNK', 'UNK']
over: ['UNK', 'pacifists', 'sacrobosco', 'researcher', 'corinthios',
'ectoderm', 'topalov', 'UNK']
american: ['hired', 'usian', 'symbolist', 'UNK', 'cane', 'lagoons',
'UNK', 'darjeeling']
one: ['UNK', 'arrogant', 'reggiani', 'UNK', 'UNK', 'UNK', 'gebavult',
'metates']
b: ['promontorii', 'UNK', 'hengen', 'UNK', 'choi', 'savimbi',
'tofana', 'UNK']
Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 4507/4507 [00:24<00:00, 187.22it/s, Batch
Loss=608, Avg Loss=208]

Epoch 9 completed - Average Loss: 112.7560
Checkpoint saved at step 40563 to
exp_1_setting_1_glove/glove_model_step_40563.pt
Printing closest words
world: ['seven', 'greats', 'UNK', 'phagos', 'denude', 'inherits',
'equates', 'norwich']
its: ['UNK', 'UNK', 'ogilvie', 'returning', 'umm', 'stargate', 'UNK',
'UNK']
has: ['samothrace', 'UNK', 'UNK', 'UNK', 'questr', 'UNK', 'UNK',
'locations']
he: ['but', 'vera', 'riley', 'UNK', 'sits', 'peer', 'stepanov', 'UNK']
war: ['inaction', 'UNK', 'UNK', 'UNK', 'UNK', 'theotokos', 'UNK',
'god']
they: ['UNK', 'grazing', 'UNK', 'aper', 'aeschylus', 'UNK', 'gains',
'acetylates']
the: ['UNK', 'sepsis', 'radegund', 'UNK', 'walid', 'UNK',
'segregation', 'collision']
all: ['UNK', 'urgent', 'UNK', 'anagyros', 'UNK', 'homogeneity',
'filipino', 'inflexion']
most: ['UNK', 'UNK', 'UNK', 'UNK', 'ingelham', 'UNK', 'spilled',
'UNK']
would: ['being', 'united', 'them', 'bijective', 'antoine', 'UNK',
'only', 'UNK']
two: ['UNK', 'stunting', 'prynne', 'vhdl', 'UNK', 'oreca', 'society',
'citation']
was: ['UNK', 'snorkel', 'repealed', 'UNK', 'attractive', 'UNK', 'UNK',
'virginity']
over: ['UNK', 'UNK', 'number', 'several', 'being', 'united', 'south',
'time']
american: ['UNK', 'shenderovich', 'citium', 'twists', 'sediments',
'oakland', 'hydroxide', 'tengmalm']
one: ['UNK', 'pitts', 'UNK', 'UNK', 'UNK', 'UNK', 'moralist',
'rivalries']

b: ['time', 'them', 'UNK', 'both', 'UNK', 'UNK', 'many', 'difference']
Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 4507/4507 [00:23<00:00, 187.98it/s,
Batch Loss=340, Avg Loss=160]

Epoch 10 completed - Average Loss: 91.7088

Checkpoint saved at step 45070 to

exp_1_setting_1_glove//glove_model_step_45070.pt

Printing closest words

world: ['seven', 'videla', 'xn', 'tiresias', 'ramos', 'hayq', 'UNK',
'UNK']

its: ['corrupts', 'permanganate', 'UNK', 'UNK', 'UNK', 'UNK',
'narrowness', 'UNK']

has: ['UNK', 'antennae', 'averroes', 'richter', 'UNK', 'UNK', 'UNK',
'malebranche']

he: ['but', 'cummin', 'UNK', 'UNK', 'ambroid', 'UNK', 'embeded',
'gatsby']

war: ['would', 'about', 'any', 'cushite', 'indented', 'then', 'UNK',
'UNK']

they: ['abs', 'UNK', 'submissions', 'utero', 'UNK', 'six', 'UNK',
'adjusted']

the: ['sofia', 'scattered', 'colonizer', 'UNK', 'shibasaburo',
'lxiii', 'inert', 'UNK']

all: ['UNK', 'telephony', 'UNK', 'UNK', 'UNK', 'UNK', 'cafe', 'UNK']

most: ['curiously', 'catwalk', 'UNK', 'asynchronous', 'UNK', 'mitaka',
'resold', 'schumpeter']

would: ['war', 'clay', 'UNK', 'these', 'UNK', 'may', 'sativum',
'smiths']

two: ['saltation', 'UNK', 'precept', 'UNK', 'UNK', 'UNK', 'UNK',
'UNK']

was: ['UNK', 'UNK', 'UNK', 'UNK', 'rohan', 'punks', 'warcraft',
'indulgences']

over: ['UNK', 'teau', 'UNK', 'UNK', 'UNK', 'seamounts', 'fuscus',
'pleasing']

american: ['volupta', 'rijkaard', 'authorizes', 'sea', 'UNK', 'UNK',
'perpetrated', 'UNK']

one: ['UNK', 'cysteine', 'UNK', 'UNK', 'UNK', 'UNK', 'goldstone',
'wei']

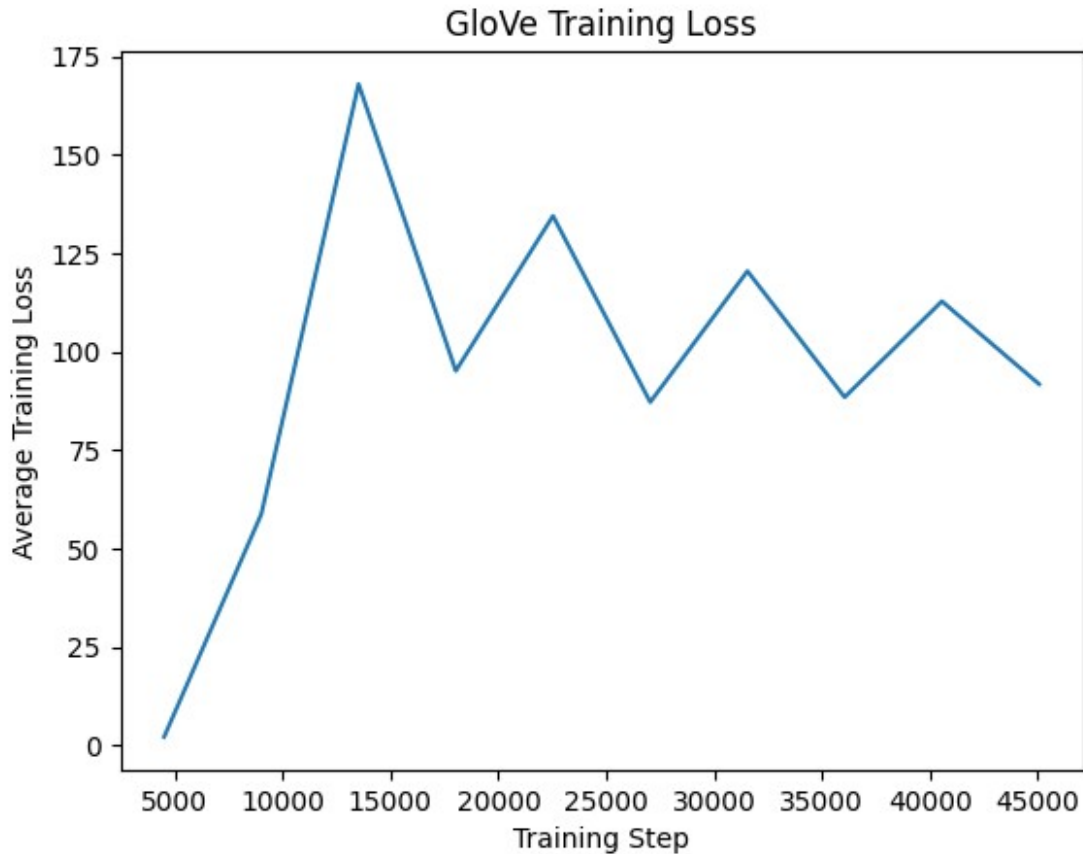
b: ['very', 'holzerni', 'england', 'UNK', 'term', 'UNK', 'UNK', 'UNK']

Training completed after 10 epochs - Final Average Loss: 91.7088

Created path: exp_1_setting_1_final_model

Final model saved at exp_1_setting_1_final_model/glove_final_model.pt

Embeddings saved at exp_1_setting_1_final_model/glove_embeddings.pkl



Hyperparameter Setting 2 for Experiment 1

```
# TODO(student): start
```

```
lr = 0.1 # Learning rate
checkpoint_model_path = "exp_1_setting_2" # Path to save checkpoints
final_model_path = "exp_1_setting_2_final_model" # Path to save the
final model and embeddings

run_training(
    epochs = 10,
    lr=lr, # Learning rate
    checkpoint_model_path=checkpoint_model_path, # Directory to save
model checkpoints
    final_model_path=final_model_path, # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size
    batch_size=512, # Batch size for training
    embedding_size=128, # Embedding size (vector dimension)
    window_size=3, # Context window size for co-occurrence
    checkpoint_step=5000 # Save checkpoints after every 1000 steps
)
```

```
# TODO(student): end
```

```
Created path: exp_1_setting_2_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
Co-occurrence matrix generated.
Printing Validation Set Words
Word: only, ID: 74
Word: other, ID: 43
Word: see, ID: 71
Word: for, ID: 20
Word: had, ID: 44
Word: also, ID: 38
Word: by, ID: 16
Word: that, ID: 19
Word: who, ID: 54
Word: states, ID: 76
Word: eight, ID: 13
Word: four, ID: 21
Word: when, ID: 67
Word: than, ID: 70
Word: from, ID: 28
Word: zero, ID: 8
Device: cuda:0
Total Batches per Epoch: 4507
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.68it/s, Batch
Loss=12.3, Avg Loss=7.94]

Epoch 1 completed - Average Loss: 8.6502
Checkpoint saved at step 4507 to
exp_1_setting_2_glove//glove_model_step_4507.pt
Printing closest words
only: ['convicted', 'south', 'UNK', 'auxiliary', 'betrothed',
'recruits', 'UNK', 'reading']
other: ['this', 'andurrial', 'james', 'galilean', 'cypherpunk', 'UNK',
'similarity', 'UNK']
```

see: ['barallete', 'secundus', 'bellingham', 'fesas', 'cavefish',
'deeps', 'semantics', 'anchorage']
for: ['flossing', 'japonic', 'three', 'derick', 'noor', 'vopo',
'appreciative', 'fancy']
had: ['but', 'be', 'basilica', 'fernbank', 'UNK', 'unflown', 'jiang',
'alger']
also: ['arsenal', 'carfree', 'bron', 'UNK', 'have', 'hologram',
'cutters', 'anthropology']
by: ['sobrero', 'mixtec', 'veerle', 'tarascan', 'redman', 'UNK',
'colonialism', 'domination']
that: ['tofana', 'the', 'referendum', 'second', 'aiwaz', 'chiba',
'he', 'UNK']
who: ['when', 'coves', 'arthrospira', 'lit', 'UNK', 'advent', 'UNK',
'recently']
states: ['organic', 'back', 'UNK', 'over', 'westhill', 'times',
'used', 'UNK']
eight: ['moriyama', 'feore', 'lotharingia', 'digard', 'kenyatta',
'UNK', 'barnet', 'sursok']
four: ['tractat', 'biblica', 'postiga', 'UNK', 'xxi', 'mumy',
'cataldi', 'andersen']
when: ['bergstr', 'who', 'conditioned', 'culture', 'similar',
'torque', 'areas', 'metallurgic']
than: ['control', 'robert', 'topics', 'yielded', 'mills', 'UNK',
'kazimierz', 'cross']
from: ['anatemnein', 'xx', 'synthesizers', 'UNK', 'biodiesel',
'aeons', 'pamir', 'knots']
zero: ['second', 'helots', 'montanari', 'huayna', 'clear', 'rupee',
'kochubey', 'landscaping']
Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.70it/s, Batch
Loss=2.79e+3, Avg Loss=1.22e+3]

Epoch 2 completed - Average Loss: 832.7566

Checkpoint saved at step 9014 to

exp_1_setting_2_glove//glove_model_step_9014.pt

Printing closest words

only: ['well', 'UNK', 'world', 'UNK', 'up', 'we', 'vil', 'would']

other: ['there', 'new', 'UNK', 'robinia', 'english', 'UNK',
'greviously', 'easily']

see: ['UNK', 'UNK', 'grewingk', 'cel', 'naguals', 'kan', 'fries',
'UNK']

for: ['newington', 'dhul', 'tiebreak', 'responds', 'UNK', 'UNK',
'teetotalers', 'demyelination']

had: ['their', 'also', 'internetwork', 'ultraviolet', 'milankovi',
'UNK', 'shirts', 'UNK']

also: ['their', 'helped', 'had', 'hazardous', 'UNK', 'lofts', 'UNK',
'heracleum']

by: ['negative', 'cheops', 'laudation', 'upanishads', 'striving',
'semin', 'sch', 'torta']

that: ['abuses', 'missourians', 'girdle', 'preternatural', 'lent',
'reflexive', 'systemically', 'history']
who: ['sure', 'aegae', 'laurinaitis', 'UNK', 'cleaners', 'UNK',
'thug', 'turner']
states: ['often', 'like', 'said', 'do', 'world', 'them', 'word',
'century']
eight: ['christy', 'vicente', 'karlheinz', 'orsi', 'UNK', 'fortunes',
'mix', 'UNK']
four: ['keating', 'strabo', 'UNK', 'bonnaf', 'UNK', 'cavefish', 'UNK',
'polytechnique']
when: ['city', 'world', 'any', 'under', 'against', 'form',
'generally', 'will']
than: ['where', 'called', 'new', 'many', 'own', 'about', 'government',
'god']
from: ['arevac', 'glorifying', 'natufian', 'phibunsongkhram',
'bayram', 'slates', 'UNK', 'database']
zero: ['shamir', 'rashomon', 'been', 'colophony', 'stapledon',
'convicts', 'subordinates', 'neodamodes']
Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 4507/4507 [00:23<00:00, 190.11it/s, Batch
Loss=2.03e+4, Avg Loss=6.38e+3]

Epoch 3 completed - Average Loss: 3166.0162

Checkpoint saved at step 13521 to

exp_1_setting_2_glove//glove_model_step_13521.pt

Printing closest words

only: ['founded', 'cove', 'UNK', 'investigates', 'UNK', 'verse',
'UNK', 'huambo']

other: ['UNK', 'topsides', 'ial', 'first', 'typological', 'dope',
'militias', 'greedy']

see: ['UNK', 'UNK', 'pygmalion', 'minute', 'tsurai', 'recalls', 'UNK',
'UNK']

for: ['UNK', 'hemichordata', 'UNK', 'laici', 'rituals', 'cardinals',
'revealing', 'leadership']

had: ['heffe', 'found', 'lavishly', 'UNK', 'UNK', 'csascii',
'webcasts', 'orient']

also: ['incontinent', 'nonpolar', 'UNK', 'reassured', 'UNK', 'cnet',
'fifth', 'blockade']

by: ['scintillating', 'aramata', 'britannia', 'hakama', 'propellers',
'UNK', 'UNK', 'pari']

that: ['nucleosynthesis', 'UNK', 'tropics', 'construction', 'UNK',
'UNK', 'pyramidalis', 'agnostid']

who: ['infatuation', 'hpo', 'supposes', 'someset', 'she', 'rann',
'ostariophysi', 'obi']

states: ['UNK', 'carnosaurs', 'feasibility', 'ci', 'semiofficial',
'UNK', 'UNK', 'UNK']

eight: ['ile', 'pentecostal', 'passport', 'UNK', 'UNK', 'hobart',
'UNK', 'roman']

four: ['UNK', 'UNK', 'jbjerg', 'match', 'modulated', 'UNK', 'chives',

```
'UNK']
when: ['language', 'time', 'UNK', 'thriller', 'UNK', 'enforced',
'UNK', 'plekhanov']
than: ['pavilion', 'twentieth', 'rebuilding', 'loyalist', 'UNK',
'UNK', 'UNK', 'pulse']
from: ['UNK', 'UNK', 'uneasy', 'chella', 'UNK', 'UNK', 'reforming',
'hyper']
zero: ['stuffing', 'polmont', 'cloomber', 'UNK', 'UNK', 'UNK', 'UNK',
'UNK']
Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 4507/4507 [00:23<00:00, 190.63it/s, Batch
Loss=3.41e+3, Avg Loss=1.96e+3]

Epoch 4 completed - Average Loss: 1371.6389
Checkpoint saved at step 18028 to
exp_1_setting_2_glove/glove_model_step_18028.pt
Printing closest words
only: ['UNK', 'UNK', 'against', 'union', 'egypt', 'footage', 'story',
'khad']
other: ['had', 'its', 'UNK', 'livius', 'anallajsi', 'relieve', 'UNK',
'sajama']
see: ['peste', 'UNK', 'lzig', 'UNK', 'UNK', 'unlawful', 'UNK',
'obedientiam']
for: ['movers', 'naoroji', 'panther', 'fundamental', 'UNK',
'stoddart', 'UNK', 'speea']
had: ['other', 'heuvelmans', 'defector', 'alchemist', 'sumter',
'devotees', 'weinberger', 'tetrameter']
also: ['laurencia', 'coucy', 'search', 'deliverer', 'tuvalu', 'nyse',
'UNK', 'carfree']
by: ['UNK', 'UNK', 'pratti', 'UNK', 'UNK', 'assunta', 'UNK', 'exuere']
that: ['bowdoin', 'commonplace', 'ingens', 'UNK', 'pelethites', 'UNK',
'UNK', 'UNK']
who: ['pelopidas', 'fleeting', 'perverse', 'UNK', 'lip', 'pedophilia',
'UNK', 'acipenseriformes']
states: ['them', 'will', 'did', 'what', 'sea', 'area', 'small', 'few']
eight: ['was', 'bourdieu', 'pension', 'wires', 'limits', 'UNK',
'unmask', 'UNK']
four: ['UNK', 'UNK', 'borg', 'UNK', 'outgoing', 'UNK', 'vailoatai',
'UNK']
when: ['name', 'usually', 'out', 'than', 'UNK', 'made', 'every',
'left']
than: ['time', 'name', 'made', 'left', 'when', 'known', 'there',
'original']
from: ['gangsters', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'paenitentia',
'nongovernmental']
zero: ['UNK', 'UNK', 'glossed', 'UNK', 'indifferent', 'UNK', 'medic',
'UNK']
Starting Epoch 5/10
```

Epoch 5/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.10it/s, Batch Loss=1.47e+4, Avg Loss=4.05e+3]

Epoch 5 completed - Average Loss: 2267.6647

Checkpoint saved at step 22535 to

exp_1_setting_2_glove//glove_model_step_22535.pt

Printing closest words

only: ['way', 'set', 'john', 'alexander', 'early', 'work', 'g', 'system']

other: ['had', 'UNK', 'political', 'term', 'its', 'considered', 'UNK', 'into']

see: ['time', 'those', 'state', 'where', 'history', 'set', 'achaeans', 'left']

for: ['UNK', 'UNK', 'patribus', 'UNK', 'UNK', 'UNK', 'brakeman', 'UNK']

had: ['her', 'any', 'other', 'years', 'still', 'rabbi', 'UNK', 'UNK']

also: ['ipv', 'UNK', 'carron', 'UNK', 'norteamericano', 'cryptographer', 'UNK', 'duprez']

by: ['arta', 'UNK', 'affirms', 'UNK', 'UNK', 'UNK', 'UNK', 'consonantary']

that: ['UNK', 'UNK', 'claudes', 'UNK', 'UNK', 'pronunciation', 'murphys', 'ecclesiastes']

who: ['UNK', 'UNK', 'aga', 'zayid', 'lymphocytes', 'UNK', 'serpents', 'innuendo']

states: ['into', 'them', 'kung', 'king', 'UNK', 'origenes', 'modern', 'UNK']

eight: ['transcendentalist', 'wotton', 'insinuating', 'UNK', 'thyestes', 'transesterification', 'UNK', 'mandaeans']

four: ['perlecan', 'redgrave', 'fineness', 'infer', 'mazzocchi', 'UNK', 'UNK', 'UNK']

when: ['colonel', 'gilroy', 'satyrs', 'UNK', 'own', 'good', 'take', 'pipelines']

than: ['UNK', 'soaked', 'where', 'union', 'these', 'sa', 'time', 'said']

from: ['UNK', 'sigmoid', 'poses', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK']

zero: ['savanna', 'succeeded', 'sassanid', 'UNK', 'tamim', 'UNK', 'UNK', 'monsters']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 4507/4507 [00:24<00:00, 186.60it/s, Batch Loss=3.02e+3, Avg Loss=1.93e+3]

Epoch 6 completed - Average Loss: 1202.2810

Checkpoint saved at step 27042 to

exp_1_setting_2_glove//glove_model_step_27042.pt

Printing closest words

only: ['thought', 'UNK', 'UNK', 'misidentification', 'aschaffenburg', 'situational', 'UNK', 'UNK']

other: ['went', 'coughlin', 'UNK', 'UNK', 'roshanak', 'ornamented', 'auerbach', 'UNK']

see: ['spoils', 'UNK', 'UNK', 'time', 'constants', 'UNK', 'aarhus', 'depopulation']
for: ['UNK', 'UNK', 'UNK', 'webblethorpe', 'UNK', 'outwards', 'UNK', 'prostitution']
had: ['set', 'UNK', 'acutilobius', 'maharaja', 'yih', 'pysharmonika', 'UNK', 'transponders']
also: ['UNK', 'UNK', 'all', 'UNK', 'UNK', 'hyksos', 'UNK', 'tageblatt']
by: ['aisle', 'UNK', 'lynne', 'polio', 'ruhub', 'solipsist', 'distilling', 'UNK']
that: ['singularly', 'UNK', 'survive', 'UNK', 'interacting', 'retorted', 'lankabhimanya', 'poison']
who: ['UNK', 'UNK', 'UNK', 'pluricentric', 'goebel', 'UNK', 'atn', 'simintov']
states: ['UNK', 'UNK', 'UNK', 'michoac', 'UNK', 'gouverneur', 'UNK', 'plateaued']
eight: ['three', 'descensus', 'UNK', 'carcosa', 'UNK', 'origenes', 'shallowest', 'UNK']
four: ['UNK', 'UNK', 'coastline', 'stool', 'straits', 'violet', 'UNK', 'UNK']
when: ['sweet', 'teenagers', 'dominic', 'butteries', 'amplitude', 'pasa', 'UNK', 'UNK']
than: ['born', 'UNK', 'UNK', 'kearny', 'eagle', 'UNK', 'UNK', 'bonesy']
from: ['oxidizer', 'pentanol', 'UNK', 'UNK', 'UNK', 'sarx', 'UNK', 'orthogenysuchus']
zero: ['UNK', 'refutations', 'posner', 'enfranchized', 'UNK', 'UNK', 'UNK', 'dissatisfaction']
Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.96it/s, Batch Loss=1.21e+4, Avg Loss=3.66e+3]

Epoch 7 completed - Average Loss: 1992.1390

Checkpoint saved at step 31549 to

exp_1_setting_2_glove//glove_model_step_31549.pt

Printing closest words

only: ['unbounded', 'offences', 'extraperitoneal', 'year', 'mixes', 'jenk', 'egress', 'UNK']

other: ['cq', 'UNK', 'valued', 'quarterfinal', 'UNK', 'synelcosciadium', 'distracted', 'UNK']

see: ['occupies', 'candidacy', 'cannibalistic', 'illa', 'window', 'croatia', 'plebeians', 'both']

for: ['UNK', 'UNK', 'sunnism', 'UNK', 'UNK', 'unverifiable', 'messaging', 'superintendence']

had: ['assises', 'olbracht', 'UNK', 'keanu', 'UNK', 'tournaments', 'UNK', 'hoffmann']

also: ['mclaughlin', 'UNK', 'he', 'UNK', 'UNK', 'UNK', 'exits', 'UNK']

by: ['two', 'UNK', 'UNK', 'UNK', 'UNK', 'physiology', 'franconian', 'bubbles']

that: ['UNK', 'pathetic', 'UNK', 'UNK', 'UNK', 'bus', 'lodging',
'broderick']
who: ['UNK', 'UNK', 'committee', 'sweyn', 'term', 'UNK', 'UNK', 'UNK']
states: ['people', 'UNK', 'being', 'language', 'great', 'but',
'still', 'known']
eight: ['was', 'UNK', 'parage', 'veloqx', 'UNK', 'UNK', 'campaigning',
'al']
four: ['agendas', 'hopefully', 'UNK', 'bridges', 'UNK', 'chloroform',
'UNK', 'shoal']
when: ['time', 'during', 'war', 'b', 'they', 'both', 'early', 'thus']
than: ['war', 'both', 'number', 'own', 'sometimes', 'so', 'south',
'east']
from: ['coloma', 'percentage', 'commonplace', 'baptiste',
'omnivorous', 'nsdap', 'UNK', 'postulation']
zero: ['interrelationships', 'wawel', 'UNK', 'UNK', 'UNK', 'vastness',
'becanfield', 'eton']
Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.35it/s, Batch
Loss=5.77e+3, Avg Loss=2.09e+3]

Epoch 8 completed - Average Loss: 1272.8285

Checkpoint saved at step 36056 to

exp_1_setting_2_glove//glove_model_step_36056.pt

Printing closest words

only: ['pelican', 'UNK', 'proline', 'ebook', 'hoping', 'atra',
'interconversion', 'UNK']

other: ['liar', 'straits', 'zooxanthellae', 'exuere', 'ingeborg',
'klondike', 'fledge', 'UNK']

see: ['UNK', 'part', 'frigidarium', 'resonator', 'UNK', 'syllabics',
'UNK', 'quite']

for: ['drinkers', 'output', 'UNK', 'poullain', 'hurriedly', 'UNK',
'dionysos', 'unanimous']

had: ['sq', 'has', 'UNK', 'UNK', 'acacio', 'issue', 'scenes',
'lomonosov']

also: ['UNK', 'UNK', 'demigod', 'eedles', 'UNK', 'toleration',
'jennifer', 'UNK']

by: ['geoworks', 'UNK', 'UNK', 'hurteri', 'responds', 'tories', 'UNK',
'syllogisms']

that: ['UNK', 'portrayed', 'UNK', 'UNK', 'UNK', 'UNK', 'shepherd',
'UNK']

who: ['UNK', 'UNK', 'vauvenargues', 'fissures', 'sakuma', 'rainwater',
'hinders', 'compliant']

states: ['world', 'between', 'may', 'them', 'agrarianism', 'UNK',
'much', 'UNK']

eight: ['sported', 'johnnie', 'unnoticed', 'UNK', 'smithery', 'UNK',
'apprenticeship', 'forts']

four: ['UNK', 'venserpolder', 'centennial', 'humid', 'UNK', 'UNK',
'martha', 'UNK']

when: ['unscrupulous', 'UNK', 'ulcer', 'horus', 'egg', 'polished',

'UNK', 'UNK']
than: ['so', 'both', 'UNK', 'about', 'same', 'found', 'UNK', 'UNK']
from: ['UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'leveller', 'supernova',
'gza']
zero: ['UNK', 'refinements', 'deluxe', 'UNK', 'UNK', 'UNK',
'retaliation', 'toitsu']
Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 4507/4507 [00:23<00:00, 187.92it/s, Batch
Loss=1.01e+4, Avg Loss=3.42e+3]

Epoch 9 completed - Average Loss: 1839.8145

Checkpoint saved at step 40563 to

exp_1_setting_2_glove//glove_model_step_40563.pt

Printing closest words

only: ['UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK']

other: ['melts', 'UNK', 'representation', 'morphogenesis',
'adocracies', 'UNK', 'not', 'UNK']

see: ['UNK', 'leaping', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK']

for: ['UNK', 'precaution', 'mediate', 'levelled', 'UNK', 'UNK', 'UNK',
'shr']

had: ['landscaped', 'zaro', 'agattu', 'UNK', 'UNK', 'belissen',
'noblewomen', 'UNK']

also: ['hypothetical', 'hoover', 'UNK', 'UNK', 'tsarina', 'estimate',
'arsenopyrite', 'UNK']

by: ['UNK', 'UNK', 'legal', 'magnesia', 'directing', 'UNK', 'gayet',
'mvd']

that: ['altay', 'anarchoblogs', 'imply', 'inkas', 'UNK', 'lfa',
'iuventutis', 'UNK']

who: ['UNK', 'UNK', 'writes', 'stupa', 'iy', 'UNK', 'natur',
'resolutely']

states: ['UNK', 'tropics', 'UNK', 'electorates', 'UNK', 'UNK', 'UNK',
'morse']

eight: ['gravitation', 'pawns', 'UNK', 'UNK', 'UNK', 'caption',
'criticizing', 'updates']

four: ['bow', 'UNK', 'UNK', 'UNK', 'UNK', 'magnum', 'UNK',
'roundabout']

when: ['UNK', 'qb', 'been', 'solvay', 'framtida', 'beloved',
'eugenius', 'UNK']

than: ['greer', 'troll', 'sz', 'deutschmarks', 'UNK', 'UNK',
'maryborough', 'UNK']

from: ['UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'sport', 'UNK',
'hypothesis']

zero: ['UNK', 'fletcher', 'posner', 'griechischen', 'irrigation',
'permeated', 'respondents', 'UNK']

Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 4507/4507 [00:24<00:00, 187.36it/s,
Batch Loss=6.3e+3, Avg Loss=2.28e+3]

Epoch 10 completed - Average Loss: 1344.1214
Checkpoint saved at step 45070 to
exp_1_setting_2_glove//glove_model_step_45070.pt
Printing closest words
only: ['laboratories', 'will', 'these', 'zapotec', 'UNK', 'wara',
'iceland', 'explicit']
other: ['protodynastic', 'affectional', 'UNK', 'inexorably', 'nedao',
'electrodynamic', 'UNK', 'looking']
see: ['actaeon', 'ups', 'UNK', 'emboldened', 'laptop', 'opposition',
'ethnological', 'ionianism']
for: ['UNK', 'novelists', 'UNK', 'UNK', 'UNK', 'april', 'goodrich',
'nazism']
had: ['excessive', 'theodefrid', 'heralds', 'UNK', 'stocks', 'UNK',
'kawm', 'verbally']
also: ['UNK', 'UNK', 'marseillaise', 'UNK', 'minimize', 'UNK',
'versified', 'be']
by: ['leftarrow', 'UNK', 'UNK', 'wrtn', 'UNK', 'UNK', 'mapusagafou',
'basketmaker']
that: ['bothering', 'UNK', 'pliocene', 'UNK', 'UNK', 'UNK', 'UNK',
'childish']
who: ['douala', 'UNK', 'UNK', 'gdansk', 'schemes', 'patchy',
'foremost', 'UNK']
states: ['war', 'however', 'world', 'UNK', 'UNK', 'now', 'UNK',
'platonisme']
eight: ['iee', 'threshold', 'three', 'haliartus', 'jurisdictional',
'praxeological', 'manual', 'numerus']
four: ['thelema', 'UNK', 'jasper', 'spoken', 'UNK', 'bligh',
'lambeth', 'gratianum']
when: ['UNK', 'has', 'been', 'UNK', 'garth', 'UNK', 'UNK',
'perisapora']
than: ['ntlichen', 'entrusted', 'UNK', 'geneticist', 'UNK', 'UNK',
'UNK', 'chinatowns']
from: ['mistakenly', 'UNK', 'UNK', 'conception', 'recounting',
'seconded', 'melbourne', 'UNK']
zero: ['winches', 'UNK', 'UNK', 'unanticipated', 'vercellenses',
'UNK', 'UNK', 'UNK']
Training completed after 10 epochs - Final Average Loss: 1344.1214
Created path: exp_1_setting_2_final_model
Final model saved at exp_1_setting_2_final_model/glove_final_model.pt
Embeddings saved at exp_1_setting_2_final_model/glove_embeddings.pkl



Hyperparameter Setting 3 for Experiment 1

```
# TODO(student): start
```

```
lr = 0.5
```

```
checkpoint_model_path = "exp_1_setting_3"
```

```
final_model_path = "exp_1_setting_3_final_model"
```

```
run_training(
```

```
    epochs = 10,
```

```
    lr=lr, # Learning rate
```

```
    checkpoint_model_path=checkpoint_model_path, # Directory to save  
    model checkpoints
```

```
    final_model_path=final_model_path, # Directory to save the final  
    model and embeddings
```

```
    vocab_size=int(1e5), # Vocabulary size
```

```
    batch_size=512, # Batch size for training
```

```
    embedding_size=128, # Embedding size (vector dimension)
```

```
    window_size=3, # Context window size for co-occurrence
```

```
    checkpoint_step=5000 # Save checkpoints after every 1000 steps
```

```
)
```

```
# TODO(student): end
```

```
Created path: exp_1_setting_3_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
Co-occurrence matrix generated.
Printing Validation Set Words
Word: them, ID: 96
Word: four, ID: 21
Word: it, ID: 30
Word: history, ID: 83
Word: years, ID: 86
Word: its, ID: 46
Word: states, ID: 76
Word: people, ID: 79
Word: no, ID: 80
Word: or, ID: 29
Word: these, ID: 65
Word: be, ID: 33
Word: from, ID: 28
Word: world, ID: 66
Word: zero, ID: 8
Word: nine, ID: 9
Device: cuda:0
Total Batches per Epoch: 4507
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 4507/4507 [00:23<00:00, 190.89it/s, Batch
Loss=180, Avg Loss=165]

Epoch 1 completed - Average Loss: 907.7487
Checkpoint saved at step 4507 to
exp_1_setting_3_glove//glove_model_step_4507.pt
Printing closest words
them: ['met', 'easier', 'debates', 'lastly', 'hampers', 'guthrie',
'bringers', 'mouse']
four: ['grosjean', 'tamerlano', 'sebastien', 'bykovsky', 'blewski',
'beretta', 'amin', 'convent']
```

it: ['hurled', 'intentional', 'juxtaposition', 'rasav', 'chun',
'reevaulation', 'surely', 'dilemmas']
history: ['executing', 'british', 'production', 'sensual',
'nicotinic', 'acetaminophenol', 'fragile', 'avshalom']
years: ['wales', 'wishes', 'review', 'state', 'pinnacle', 'walls',
'nicotinic', 'set']
its: ['fillm', 'faith', 'government', 'bandage', 'trummet', 'poised',
'robby', 'epiph']
states: ['symbiotic', 'radar', 'despite', 'legalized', 'UNK',
'altruists', 'waves', 'michael']
people: ['damned', 'ayo', 'grabs', 'unawatuna', 'larson', 'stimming',
'salat', 'housman']
no: ['dismissing', 'may', 'force', 'taper', 'imsa', 'muttered',
'cassius', 'flourish']
or: ['elfhame', 'cavalrymen', 'ero', 'politeia', 'lavender',
'folkrace', 'alvan', 'isopropyl']
these: ['tenascin', 'clearest', 'confidently', 'UNK', 'sofia',
'ballston', 'mortal', 'repealed']
be: ['escalating', 'malfeasance', 'stage', 'splendour', 'randomise',
'crimea', 'commonplace', 'zuiden']
from: ['heraclea', 'paestum', 'pireas', 'stati', 'summon', 'belgrade',
'thane', 'cafe']
world: ['than', 'immunosuppression', 'UNK', 'complications', 'larynx',
'hanlon', 'demanding', 'shellfish']
zero: ['doku', 'dinar', 'rickety', 'rupee', 'setanta', 'lorraine',
'main', 'trillion']
nine: ['welty', 'dec', 'schwantes', 'cas', 'piccadilly',
'leoncavallo', 'priesthood', 'rupes']
Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 4507/4507 [00:23<00:00, 187.90it/s, Batch
Loss=6.55e+5, Avg Loss=5.37e+5]

Epoch 2 completed - Average Loss: 413914.3055

Checkpoint saved at step 9014 to

exp_1_setting_3_glove//glove_model_step_9014.pt

Printing closest words

them: ['through', 'during', 'physical', 'down', 'alexander', 'player',
'german', 'will']

four: ['eccentric', 'rhinoceroses', 'helots', 'vartan', 'donate',
'educators', 'digeridoo', 'ascorbic']

it: ['succinifera', 'excuse', 'wenthal', 'develop', 'house',
'moralistic', 'UNK', 'philological']

history: ['i', 'films', 'non', 'includes', 'its', 'free',
'references', 'made']

years: ['national', 'philosophy', 'between', 'set', 'state', 'e',
'without', 'absolute']

its: ['her', 'history', 'not', 'fortunate', 'known', 'jews', 'they',
'version']

states: ['life', 'city', 'b', 'those', 'film', 'although', 'people',

'following']
people: ['because', 'film', 'first', 'although', 'states', 'words', 'good', 'she']
no: ['world', 'south', 'day', 'UNK', 'come', 'up', 'their', 'century']
or: ['nontheism', 'UNK', 'perisapora', 'crafoord', 'haloalkane', 'mefferdi', 'markings', 'ite']
these: ['european', 'however', 'year', 'bc', 'published', 'up', 'against', 'th']
be: ['at', 'cossack', 'autonoe', 'may', 'well', 'visa', 'osculating', 'kentuckians']
from: ['sterility', 'abdication', 'lito', 'municipality', 'hawaiian', 'swabia', 'UNK', 'organ']
world: ['war', 'form', 'english', 'under', 'century', 'government', 'what', 'm']
zero: ['handcuffs', 'leviticus', 'jobs', 'after', 'marnie', 'prestige', 'allow', 'dorilla']
nine: ['is', 'ventotene', 'same', 'morrocan', 'galatian', 'abusina', 'nirad', 'soviet']
Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 4507/4507 [00:24<00:00, 186.89it/s, Batch Loss=1.53e+7, Avg Loss=3.43e+6]

Epoch 3 completed - Average Loss: 1829804.1007

Checkpoint saved at step 13521 to

exp_1_setting_3_glove//glove_model_step_13521.pt

Printing closest words

them: ['him', 'UNK', 'akin', 'less', 'before', 'nightclubs', 'UNK', 'population']

four: ['narrowest', 'nurses', 'sigindunum', 'UNK', 'paleontologist', 'lucanians', 'UNK', 'goff']

it: ['several', 'velocites', 'UNK', 'radio', 'td', 'anguish', 'trouble', 'drunkenness']

history: ['niosh', 'use', 'all', 'shrunk', 'part', 'jiang', 'UNK', 'endure']

years: ['bombay', 'gust', 'dealings', 'congressional', 'use', 'between', 'other', 'niosh']

its: ['toil', 'they', 'each', 'nine', 'railway', 'another', 'mutinous', 'homophobia']

states: ['finely', 'efefef', 'discontinues', 'UNK', 'asbury', 'pittsburg', 'UNK', 'UNK']

people: ['about', 'only', 'much', 'greek', 'apollo', 'may', 'british', 'slobodan']

no: ['although', 'become', 'long', 'like', 'known', 'called', 'book', 'arches']

or: ['sarcodina', 'denouement', 'cap', 'giraud', 'brasserie', 'UNK', 'nishapur', 'rashly']

these: ['other', 'between', 'more', 'all', 'augusta', 'use', 'eber', 'henry']

be: ['atcm', 'virtuous', 'kunstmuseum', 'wishful', 'braingames', 'at',

```
'also', 'whitesides']
from: ['UNK', 'aware', 'UNK', 'laffer', 'UNK', 'anguillita',
'theatres', 'pads']
world: ['name', 'end', 'laharpe', 'between', 'anion', 'splashes',
'irrational', 'outvoted']
zero: ['warmth', 'more', 'pacifistic', 'UNK', 'enthusiastically',
'use', 'twain', 'ghazni']
nine: ['only', 'after', 'another', 'turing', 'lobed', 'poisoning',
'toil', 'profound']
Starting Epoch 4/10
```

```
Epoch 4/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.40it/s, Batch
Loss=1.49e+6, Avg Loss=1.06e+6]
```

```
Epoch 4 completed - Average Loss: 801993.2586
```

```
Checkpoint saved at step 18028 to
```

```
exp_1_setting_3_glove//glove_model_step_18028.pt
```

```
Printing closest words
```

```
them: ['has', 'spoto', 'thou', 'bomb', 'indecisive', 'outpost',
'jazaka', 'granus']
four: ['geopolitically', 'willi', 'conspiracy', 'warne', 'tubular',
'danorum', 'prostrate', 'UNK']
it: ['be', 'booty', 'several', 'extortionate', 'ers', 'closer',
'keying', 'containment']
history: ['large', 'more', 'over', 'battle', 'including', 'until',
'do', 'between']
years: ['jimenez', 'avenging', 'UNK', 'abbey', 'UNK', 'divergently',
'biopharmaceuticals', 'UNK']
its: ['they', 'anachronistic', 'pictured', 'osbert', 'applause',
'brakeman', 'molten', 'boost']
states: ['use', 'about', 'UNK', 'her', 'benefiting', 'currents',
'more', 'name']
people: ['president', 'since', 'science', 'aryans', 'personation',
'swarmed', 'city', 'indispensable']
no: ['geelong', 'moderation', 'UNK', 'aloft', 'firestone', 'UNK',
'tubenoses', 'aditi']
or: ['ravages', 'codebreaking', 'rallycross', 'UNK', 'term', 'any',
'pacifists', 'unsupported']
these: ['other', 'between', 'including', 'miscoordination', 'but',
'son', 'more', 'slobodan']
be: ['it', 'has', 'UNK', 'impeachment', 'rightfully', 'direst', 'UNK',
'had']
from: ['fond', 'keuper', 'ifaf', 'peaceable', 'UNK', 'UNK',
'chiquitita', 'forested']
world: ['use', 'than', 'until', 'over', 'when', 'however', 'so',
'used']
zero: ['symptoms', 'UNK', 'righteousness', 'larissa', 'graceful',
'disobedience', 'dockings', 'serotonin']
nine: ['line', 'UNK', 'shomenuchi', 'thallus', 'megalomania',
```

'kocharian', 'holten', 'homophobia']

Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 4507/4507 [00:23<00:00, 190.28it/s, Batch Loss=8.15e+6, Avg Loss=2.04e+6]

Epoch 5 completed - Average Loss: 1160869.0102

Checkpoint saved at step 22535 to

exp_1_setting_3_glove//glove_model_step_22535.pt

Printing closest words

them: ['up', 'these', 'states', 'called', 'where', 'science', 'later', 'most']

four: ['madagascar', 'UNK', 'UNK', 'kbit', 'treble', 'amado', 'aufzug', 'pedrosa']

it: ['be', 'mediate', 'time', 'lobelia', 'credibility', 'karatantcheva', 'abrasax', 'counterexamples']

history: ['being', 'up', 'called', 'states', 'these', 'about', 'life', 'most']

years: ['UNK', 'UNK', 'acyl', 'UNK', 'panes', 'UNK', 'UNK', 'ichthyostega']

its: ['giamatti', 'after', 'trotskyite', 'UNK', 'miscoordination', 'inflame', 'rejection', 'illusionist']

states: ['history', 'up', 'being', 'most', 'known', 'where', 'called', 'them']

people: ['and', 'UNK', 'only', 'heuristics', 'hypaspists', 'afghani', 'after', 'seydlitz']

no: ['known', 'audible', 'like', 'most', 'her', 'kleinwalsertal', 'UNK', 'anarcha']

or: ['glace', 'other', 'judit', 'perfumed', 'meteosat', 'obstinately', 'made', 'UNK']

these: ['history', 'including', 'them', 'life', 'supremely', 'languages', 'being', 'idealism']

be: ['it', 'time', 'world', 'several', 'UNK', 'i', 'diluted', 'UNK']

from: ['oskar', 'giorgio', 'telecommunications', 'issue', 'UNK', 'UNK', 'soundtrackinfo', 'peleg']

world: ['only', 'she', 'government', 'be', 'american', 'much', 'day', 'freakonomics']

zero: ['reichswehr', 'UNK', 'ktzn', 'accredited', 'ori', 'asli', 'spectrochemical', 'pastiche']

nine: ['neatly', 'UNK', 'UNK', 'rotunda', 'multiplicity', 'UNK', 'UNK', 'UNK']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.59it/s, Batch Loss=2.81e+6, Avg Loss=1.06e+6]

Epoch 6 completed - Average Loss: 677303.5065

Checkpoint saved at step 27042 to

exp_1_setting_3_glove//glove_model_step_27042.pt

Printing closest words

them: ['addictiveness', 'UNK', 'changes', 'unilateralism', 'UNK', 'UNK', 'liliales', 'UNK']
four: ['updates', 'was', 'akira', 'miki', 'arachosia', 'choueiri', 'subhanahu', 'UNK']
it: ['be', 'merican', 'intelligible', 'bandits', 'closing', 'barrias', 'ref', 'revises']
history: ['states', 'over', 'there', 'being', 'if', 'united', 'only', 'up']
years: ['staley', 'UNK', 'leftpondian', 'UNK', 'tilkin', 'trope', 'UNK', 'UNK']
its: ['up', 'without', 'united', 'deceive', 'algorism', 'heath', 'set', 'powerpc']
states: ['history', 'only', 'being', 'united', 'over', 'if', 'there', 'part']
people: ['fairhair', 'uru', 'languages', 'revises', 'UNK', 'panavia', 'plebiscite', 'farnell']
no: ['UNK', 'breakdown', 'UNK', 'soc', 'UNK', 'custom', 'UNK', 'UNK']
or: ['reds', 'scriptural', 'household', 'psychoanalytic', 'UNK', 'deaflibrary', 'apostol', 'heldenplatz']
these: ['UNK', 'livy', 'shuffling', 'UNK', 'UNK', 'marooned', 'charinus', 'other']
be: ['it', 'captured', 'hotels', 'UNK', 'locrians', 'ancile', 'iota', 'UNK']
from: ['yaqub', 'bravas', 'UNK', 'tromso', 'UNK', 'perplex', 'UNK', 'UNK']
world: ['known', 'general', 'past', 'UNK', 'him', 'well', 'have', 'UNK']
zero: ['planisphere', 'inertness', 'UNK', 'UNK', 'UNK', 'transgressive', 'advised', 'chlorpromazine']
nine: ['patilla', 'openness', 'painstakingly', 'UNK', 'UNK', 'UNK', 'UNK', 'stared']
Starting Epoch 7/10

Epoch 7/10: 100%[██████████] 4507/4507 [00:24<00:00, 187.31it/s, Batch Loss=7.32e+6, Avg Loss=2.01e+6]

Epoch 7 completed - Average Loss: 1090455.2835

Checkpoint saved at step 31549 to

exp_1_setting_3_glove//glove_model_step_31549.pt

Printing closest words

them: ['redressed', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'nearchus', 'UNK']

four: ['acceding', 'sozomen', 'void', 'robber', 'units', 'cubic', 'UNK', 'UNK']

it: ['be', 'prosody', 'UNK', 'UNK', 'carburettor', 'UNK', 'UNK', 'UNK']

history: ['often', 'general', 'there', 'such', 'so', 'number', 'country', 'nadal']

years: ['death', 'undiscernable', 'UNK', 'she', 'may', 'UNK', 'there', 'kfqd']

its: ['irrational', 'UNK', 'bouillier', 'giamatti', 'owusu', 'UNK', 'despotic', 'fees']
states: ['UNK', 'UNK', 'UNK', 'dir', 'UNK', 'UNK', 'causative', 'wwi']
people: ['back', 'still', 'farce', 'film', 'm', 'UNK', 'deledda', 'until']
no: ['wakes', 'sakhalin', 'testimony', 'eventually', 'religious', 'newtons', 'UNK', 'plutocracy']
or: ['UNK', 'ernighan', 'nigga', 'frameworks', 'nietzsche', 'optimally', 'catastrophic', 'earthenware']
these: ['other', 'often', 'between', 'brute', 'connivance', 'dicotyledonous', 'montessori', 'naumburg']
be: ['it', 'also', 'UNK', 'shank', 'UNK', 'UNK', 'UNK', 'rafted']
from: ['UNK', 'UNK', 'UNK', 'UNK', 'vivendi', 'UNK', 'sergio', 'UNK']
world: ['UNK', 'altair', 'abdication', 'guzm', 'UNK', 'crypts', 'svanhildr', 'patonii']
zero: ['aqiba', 'rickety', 'woodward', 'harpsichordist', 'UNK', 'UNK', 'phrygius', 'predictably']
nine: ['l', 'strait', 'hayes', 'updates', 'UNK', 'gevers', 'UNK', 'dry']

Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.44it/s, Batch Loss=3.47e+6, Avg Loss=1.22e+6]

Epoch 8 completed - Average Loss: 735228.0957

Checkpoint saved at step 36056 to

exp_1_setting_3_glove//glove_model_step_36056.pt

Printing closest words

them: ['UNK', 'alludes', 'since', 'UNK', 'counting', 'UNK', 'UNK', 'capcoms']

four: ['credo', 'UNK', 'UNK', 'UNK', 'giese', 'machining', 'asher', 'dissent']

it: ['UNK', 'fashions', 'UNK', 'UNK', 'scrutinised', 'western', 'UNK', 'UNK']

history: ['transfinite', 'UNK', 'diaries', 'UNK', 'transaction', 'gaia', 'dumasia', 'simplicity']

years: ['proviso', 'kingssonar', 'wr', 'demarcation', 'gestis', 'UNK', 'amali', 'UNK']

its: ['UNK', 'dismissing', 'such', 'subdititos', 'muster', 'bewildering', 'kalbbia', 'UNK']

states: ['world', 'UNK', 'UNK', 'UNK', 'bokuto', 'UNK', 'UNK', 'UNK']

people: ['hieroglyph', 'confederal', 'capitalised', 'war', 'what', 'known', 'scilly', 'hallowed']

no: ['kushi', 'UNK', 'drupe', 'smyrnaeus', 'nearchus', 'bonito', 'UNK', 'heracles']

or: ['UNK', 'energetic', 'UNK', 'occultists', 'grizzard', 'pentanol', 'entropyandtemp', 'repository']

these: ['webcam', 'what', 'bleek', 'UNK', 'huddling', 'fat', 'tribunal', 'UNK']

be: ['also', 'lorikeet', 'UNK', 'UNK', 'UNK', 'UNK', 'streeton',

'UNK']
from: ['buffs', 'subcostal', 'arus', 'realien', 'crassipes', 'UNK',
'diatom', 'UNK']
world: ['UNK', 'UNK', 'states', 'estranged', 'datapoint', 'UNK',
'UNK', 'moorcroft']
zero: ['reese', 'sleight', 'resolve', 'UNK', 'UNK', 'chancery',
'chemische', 'reubens']
nine: ['georgi', 'charismatics', 'UNK', 'UNK', 'UNK', 'departments',
'antiquarian', 'aboard']
Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 4507/4507 [00:23<00:00, 189.12it/s, Batch
Loss=5.9e+6, Avg Loss=1.95e+6]

Epoch 9 completed - Average Loss: 1063249.1910

Checkpoint saved at step 40563 to

exp_1_setting_3_glove//glove_model_step_40563.pt

Printing closest words

them: ['survivor', 'UNK', 'tradewinds', 'concubines', 'UNK',
'innuendo', 'franck', 'dashing']

four: ['sozomen', 'millimetres', 'everitt', 'UNK', 'squat',
'obstruction', 'outta', 'UNK']

it: ['portmanteau', 'UNK', 'flexispina', 'UNK', 'nc', 'observances',
'UNK', 'monomaniacal']

history: ['century', 'out', 'film', 'same', 'vedrosha', 'did', 'day',
'year']

years: ['century', 'book', 'UNK', 'hout', 'UNK', 'beatified', 'dan',
'UNK']

its: ['proslavery', 'faculties', 'UNK', 'mendel', 'UNK', 'ner', 'UNK',
'UNK']

states: ['careers', 'UNK', 'UNK', 'recoverable', 'korbinian', 'known',
'UNK', 'however']

people: ['UNK', 'throat', 'UNK', 'UNK', 'UNK', 'sweeteners', 'UNK',
'rocketry']

no: ['UNK', 'UNK', 'sadal melik', 'anubis', 'UNK', 'iiaa', 'UNK',
'UNK']

or: ['mitra', 'lens', 'tkabir', 'UNK', 'column', 'UNK', 'togarmah',
'inflict']

these: ['UNK', 'UNK', 'difference', 'UNK', 'tcsec', 'nizar', 'eight',
'valued']

be: ['also', 'disarray', 'UNK', 'azincourt', 'UNK', 'whizzer', 'UNK',
'savages']

from: ['turkestan', 'UNK', 'alta', 'oxidises', 'UNK', 'prolong',
'basketball', 'UNK']

world: ['than', 'electromotive', 'buddhism', 'UNK', 'UNK', 'stettin',
'UNK', 'UNK']

zero: ['bishops', 'UNK', 'intently', 'UNK', 'UNK', 'impression',
'betts', 'UNK']

nine: ['refrigerant', 'directv', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK',

'UNK']

Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 4507/4507 [00:23<00:00, 188.30it/s,
Batch Loss=3.74e+6, Avg Loss=1.35e+6]

Epoch 10 completed - Average Loss: 801748.5250

Checkpoint saved at step 45070 to

exp_1_setting_3_glove//glove_model_step_45070.pt

Printing closest words

them: ['kuwaiti', 'UNK', 'tantra', 'UNK', 'coelom', 'shaoyin',
'apology', 'bletchley']

four: ['credo', 'penrose', 'was', 'pirze', 'updates', 'pawns', 'UNK',
'cma']

it: ['UNK', 'UNK', 'jiggled', 'helmut', 'UNK', 'hecuba', 'altruistic',
'trakl']

history: ['UNK', 'UNK', 'seyss', 'caudatus', 'marred', 'sergeant',
'UNK', 'cyprus']

years: ['then', 'yourdon', 'UNK', 'tenor', 'tiled', 'offset',
'nicklaus', 'dositheus']

its: ['muster', 'tereshkova', 'UNK', 'UNK', 'drumheller', 'elapidae',
'exchristian', 'UNK']

states: ['UNK', 'time', 'UNK', 'habitable', 'unilateralis',
'camarasaurus', 'UNK', 'UNK']

people: ['lorne', 'fitzjames', 'macleod', 'UNK', 'UNK', 'UNK', 'UNK',
'UNK']

no: ['UNK', 'UNK', 'UNK', 'UNK', 'carborundum', 'UNK',
'sentimentality', 'UNK']

or: ['UNK', 'UNK', 'meal', 'UNK', 'epub', 'zermelo', 'UNK', 'khene']

these: ['bourg', 'hundreds', 'UNK', 'marple', 'previews',
'syncretistic', 'UNK', 'reenacted']

be: ['also', 'not', 'encarta', 'barrett', 'UNK', 'incurred', 'aye',
'slievemore']

from: ['UNK', 'UNK', 'able', 'UNK', 'advance', 'UNK', 'UNK', 'UNK']

world: ['work', 'up', 'UNK', 'aesthetics', 'UNK', 'embassies',
'ratnadh', 'UNK']

zero: ['ktzn', 'UNK', 'gevers', 'heuvelmans', 'UNK', 'UNK',
'postcard', 'UNK']

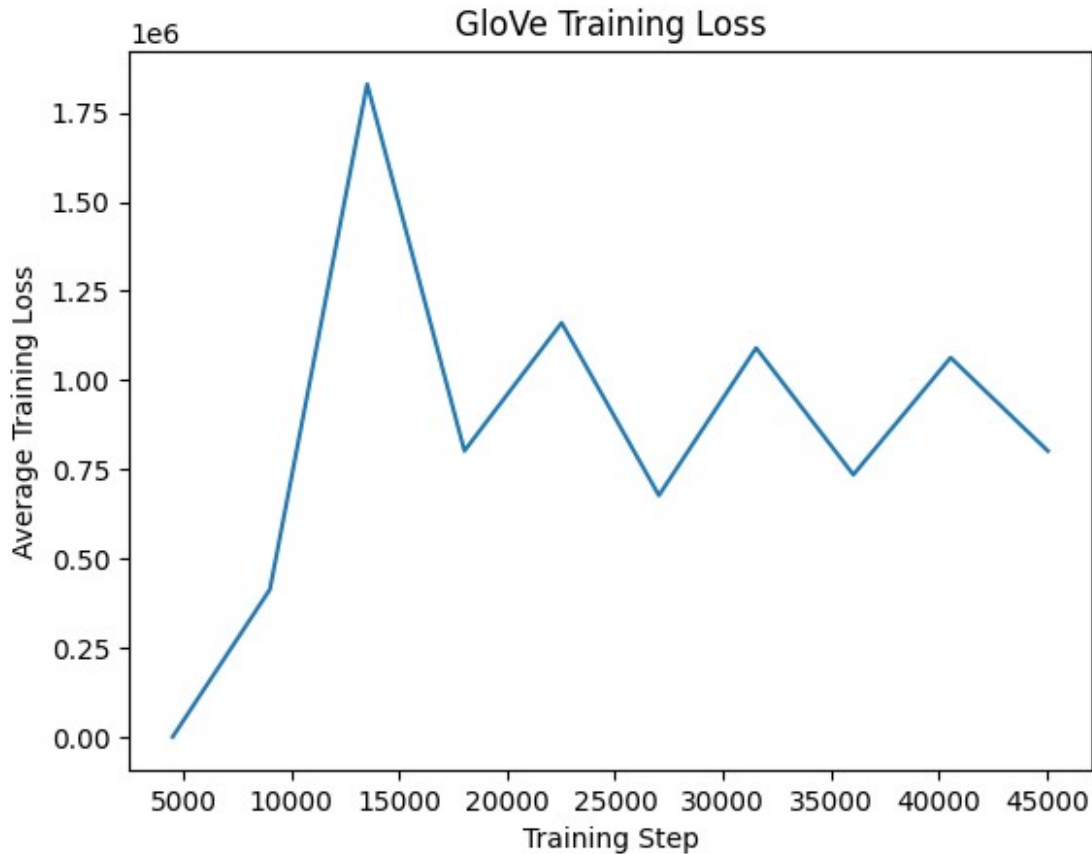
nine: ['UNK', 'transaction', 'UNK', 'UNK', 'UNK', 'UNK', 'parini',
'garson']

Training completed after 10 epochs - Final Average Loss: 801748.5250

Created path: exp_1_setting_3_final_model

Final model saved at exp_1_setting_3_final_model/glove_final_model.pt

Embeddings saved at exp_1_setting_3_final_model/glove_embeddings.pkl



What is the best learning rate for GloVe model and why? What did you learn from the plots?

TODO(students): start

Best learning rate for above Glove model is 0.01. From above experiments, as learning rate increases, the avg loss for the whole experiment is going to increase a lot. So, learning rate should be as less as possible and also not too small. I think 0.01 is good enough to train the glove model.

Learning from Plots:

1. Avg loss increases initially. And then decreases to a saturation point in each epoch.
2. as learning rate increases from 0.01, the avg loss of model increases drastically.
3. for $lr=0.01$, loss is in order of 0.1
4. for $lr=0.05$, loss is in order of 10^2 (drastical increase)
5. for $lr=0.1$, loss is in order of 10^6 .

TODO(students): end

```
## TODO(students): start
```

```
best_lr = 0.01
```

```
## TODO(students): end
```

Experiment 2

Now, going forward, set the value of the learning rate to the best lr for that model type. For the second experiment, we will focus on varying the window size while keeping other parameters like the learning rate, embedding size, and batch size constant. The window size controls how many words before and after the target word are considered when building the co-occurrence matrix. This can have a significant effect on the quality of the learned embeddings.

Hyperparameter Setting 1 for Experiment 2

```
# TODO(student): start

lr = best_lr # Learning rate (fixed) - from experiment 1
checkpoint_model_path = "exp_2_setting_1" # Path to save checkpoints
final_model_path = "exp_2_setting_1_final_model" # Path to save the
final model and embeddings

# Experiment 2: Vary the window size
# You can run this with different window sizes (e.g., 2, 3, 5, 10)
window_size = 2

run_training(
    epochs=10, # Number of epochs to train
    lr=lr, # Learning rate
    checkpoint_model_path=checkpoint_model_path, # Directory to save
model checkpoints
    final_model_path=final_model_path, # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size
    batch_size=512, # Batch size for training
    embedding_size=128, # Embedding size (vector dimension)
    window_size=window_size, # Context window size for co-occurrence
    checkpoint_step=5000, # Save checkpoints after every 2000 steps
)

# TODO(student): end

Created path: exp_2_setting_1_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
```

Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000

Co-occurrence matrix generated.

Printing Validation Set Words

Word: new, ID: 68
Word: were, ID: 42
Word: that, ID: 19
Word: called, ID: 89
Word: b, ID: 59
Word: which, ID: 34
Word: some, ID: 50
Word: these, ID: 65
Word: from, ID: 28
Word: they, ID: 49
Word: not, ID: 37
Word: had, ID: 44
Word: at, ID: 35
Word: world, ID: 66
Word: time, ID: 69
Word: only, ID: 74

Device: cuda:0

Total Batches per Epoch: 3116

Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 3116/3116 [00:16<00:00, 191.19it/s, Batch Loss=0.268, Avg Loss=0.432]

Epoch 1 completed - Average Loss: 0.7658

Checkpoint saved at step 3116 to

exp_2_setting_1_glove//glove_model_step_3116.pt

Printing closest words

new: ['near', 'bourses', 'of', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK']

were: ['UNK', 'milne', 'UNK', 'incoming', 'llamas', 'which', 'bam', 'UNK']

that: ['UNK', 'UNK', 'UNK', 'main', 'anniversary', 'orb', 'united', 'UNK']

called: ['UNK', 'rustica', 'UNK', 'UNK', 'UNK', 'amory', 'UNK', 'UNK']

b: ['UNK', 'afield', 'UNK', 'UNK', 'UNK', 'mkb', 'diamonds', 'nine']

which: ['predated', 'black', 'terms', 'UNK', 'UNK', 'hinged', 'UNK', 'ubiquitous']

some: ['UNK', 'trained', 'at', 'parliamentarian', 'aeroplanes', 'UNK', 'UNK', 'quotes']

these: ['programming', 'behaviors', 'barbadensis', 'UNK', 'dachau', 'emerging', 'alfrun', 'ned']

from: ['UNK', 'UNK', 's', 'yeshiva', 'UNK', 'UNK', 'cia', 'UNK']

they: ['UNK', 'o', 'themselves', 'doujinshi', 'inspectors', 'UNK', 'UNK', 'afyon']

not: ['splendor', 'UNK', 'thorough', 'UNK', 'aasim', 'naylan', 'UNK', 'record']

had: ['UNK', 'declass', 'morea', 'hematoma', 'alpha', 'freckles',
'clotaire', 'reciprocum']
at: ['gchq', 'UNK', 'UNK', 'UNK', 'following', 'UNK', 'some', 'UNK']
world: ['vossische', 'acupuncturists', 'mccclxxii', 'UNK', 'UNK',
'dyes', 'UNK', 'UNK']
time: ['manuals', 'UNK', 'UNK', 'formulas', 'UNK', 'UNK', 'UNK',
'poe']
only: ['aim', 'ardor', 'write', 'pose', 'UNK', 'god', 'miles',
'sulla']

Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 3116/3116 [00:16<00:00, 189.01it/s, Batch
Loss=3.6, Avg Loss=1.22]

Epoch 2 completed - Average Loss: 0.9037

Checkpoint saved at step 6232 to

exp_2_setting_1_glove//glove_model_step_6232.pt

Printing closest words

new: ['on', 'own', 'her', 'UNK', 'where', 'national', 'beijing',
'into']

were: ['this', 'which', 'its', 'had', 'where', 'some', 'on', 'to']

that: ['for', 'which', 'its', 'this', 'it', 'at', 'above', 'not']

called: ['gis', 'known', 'made', 'where', 'way', 'glotz', 'ft',
'webobjects']

b: ['which', 'originally', 'UNK', 'residual', 'although', 'against',
'lanes', 'it']

which: ['this', 'its', 'for', 'also', 'where', 'were', 'had', 'it']

some: ['its', 'were', 'after', 'this', 'into', 'to', 'which', 'it']

these: ['such', 'less', 'UNK', 'submachine', 'given', 'children',
'play', 'shimura']

from: ['were', 'had', 'its', 'of', 'by', 'this', 'which', 'for']

they: ['into', 'this', 'made', 'which', 'where', 'own', 'for',
'philosophy']

not: ['after', 'also', 'this', 'were', 'which', 'it', 'most', 'being']

had: ['which', 'were', 'its', 'for', 'this', 'where', 'from', 'when']

at: ['this', 'its', 'which', 'on', 'for', 'were', 'also', 'who']

world: ['UNK', 'UNK', 'alvarez', 'buds', 'UNK', 'UNK', 'urals', 'UNK']

time: ['another', 'states', 'than', 'among', 'have', 'things', 'fact',
'home']

only: ['such', 'he', 'so', 'nobuyoshi', 'around', 'various', 'UNK',
'demand']

Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 3116/3116 [00:16<00:00, 189.01it/s, Batch
Loss=1.12, Avg Loss=0.603]

Epoch 3 completed - Average Loss: 0.4892

Checkpoint saved at step 9348 to

exp_2_setting_1_glove//glove_model_step_9348.pt

Printing closest words

new: ['during', 'through', 'other', 'called', 'against', 'english', 'these', 'be']
were: ['had', 'when', 'his', 'for', 'that', 'not', 'their', 'who']
that: ['for', 'is', 'his', 's', 'zero', 'with', 'to', 'when']
called: ['through', 'however', 'will', 'be', 'been', 'they', 'up', 'where']
b: ['six', 'but', 'eight', 'about', 'english', 'other', 'd', 'm']
which: ['this', 'on', 'at', 'for', 'to', 'that', 'is', 'in']
some: ['blistering', 'manship', 'one', 'in', 'UNK', 'smothers', 'lookout', 'mathematician']
these: ['other', 'called', 'however', 'new', 'be', 'but', 'been', 'six']
from: ['it', 'seven', 'which', 'on', 'by', 'are', 'in', 'itself']
they: ['be', 'will', 'called', 'been', 'however', 'through', 'him', 'but']
not: ['but', 'be', 'other', 'had', 'been', 'were', 'when', 'who']
had: ['when', 'were', 'their', 'his', 'during', 'through', 'not', 'been']
at: ['which', 'on', 'in', 'for', 'is', 'a', 'to', 'this']
world: ['both', 'about', 'four', 'u', 'b', 'france', 'six', 'UNK']
time: ['then', 'did', 'states', 'small', 'jews', 'life', 'death', 'population']
only: ['no', 'because', 'so', 'use', 'since', 'include', 'australian', 'has']

Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 3116/3116 [00:16<00:00, 189.85it/s, Batch Loss=1.44, Avg Loss=0.537]

Epoch 4 completed - Average Loss: 0.4133

Checkpoint saved at step 12464 to

exp_2_setting_1_glove//glove_model_step_12464.pt

Printing closest words

new: ['natural', 'theory', 'sometimes', 'called', 'jews', 'alexander', 'up', 'men']
were: ['had', 'but', 'can', 'see', 'be', 'UNK', 'other', 'american']
that: ['for', 'to', 'is', 'never', 'vermiform', 'UNK', 'UNK', 'pueblo']
called: ['under', 'generally', 'those', 'theory', 'built', 'national', 'when', 'groups']
b: ['arabic', 'called', 'force', 'groups', 'methods', 'architecture', 'development', 'national']
which: ['for', 'to', 'one', 'stdma', 'originally', 'that', 'conquering', 'tulli']
some: ['after', 'it', 'into', 'all', 'they', 'would', 'its', 'by']
these: ['history', 'about', 'during', 'even', 'i', 'government', 'can', 'name']
from: ['UNK', 'joys', 'pigeons', 'affirm', 'UNK', 'obscured', 'shredded', 'democracies']
they: ['however', 'about', 'him', 'would', 'be', 'can', 'but', 'some']

not: ['said', 'UNK', 'UNK', 'makes', 'believe', 'considered', 'does', 'created']
had: ['other', 'but', 'into', 'be', 'after', 'can', 'also', 'they']
at: ['an', 'states', 'party', 'built', 'economy', 'national', 'sacrosanct', 'up']
world: ['can', 'american', 'by', 'after', 'any', 'some', 'would', 'even']
time: ['states', 'human', 'development', 'force', 'until', 'down', 'death', 'jews']
only: ['such', 'use', 'no', 'between', 'so', 'many', 'first', 'more']
Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 3116/3116 [00:16<00:00, 190.29it/s, Batch Loss=1.1, Avg Loss=0.449]

Epoch 5 completed - Average Loss: 0.3529

Checkpoint saved at step 15580 to

exp_2_setting_1_glove//glove_model_step_15580.pt

Printing closest words

new: ['when', 'their', 'control', 'up', 'have', 'named', 'called', 'an']

were: ['had', 'where', 'most', 'into', 'including', 'by', 'three', 'could']

that: ['is', 'his', 'only', 'not', 'in', 'zero', 'eight', 'been']

called: ['up', 'when', 'those', 'very', 'sometimes', 'under', 'time', 'well']

b: ['m', 'when', 'called', 'under', 'c', 'only', 'time', 'six']

which: ['UNK', 'southbound', 'UNK', 'thrombocyte', 'UNK', 'state', 'augusto', 'case']

some: ['political', 'other', 'christian', 'south', 'earlier', 'systems', 'commuter', 'th']

these: ['UNK', 'UNK', 'carts', 'miya', 'UNK', 'UNK', 'noronha', 'UNK']

from: ['six', 'an', 'are', 'eight', 'apollo', 'his', 'have', 'zero']

they: ['it', 'not', 'been', 'more', 'would', 'its', 'may', 'be']

not: ['only', 'later', 'it', 'if', 'at', 'may', 'they', 'more']

had: ['into', 'most', 'three', 'has', 'by', 'against', 'where', 'were']

at: ['not', 'an', 'only', 'have', 'six', 'zero', 'than', 'while']

world: ['see', 'her', 'city', 'against', 'who', 'nine', 'any', 'most']

time: ['up', 'called', 'those', 'under', 'when', 'both', 'than', 'others']

only: ['may', 'later', 'since', 'while', 'i', 'over', 'not', 'its']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 3116/3116 [00:16<00:00, 189.06it/s, Batch Loss=1.19, Avg Loss=0.422]

Epoch 6 completed - Average Loss: 0.3296

Checkpoint saved at step 18696 to

exp_2_setting_1_glove//glove_model_step_18696.pt

Printing closest words

new: ['bessus', 'clements', 'UNK', 'UNK', 'rapprochement', 'UNK', 'UNK', 'happens']

were: ['first', 'so', 'this', 'there', 'some', 'after', 'such', 'him']

that: ['for', 'he', 'his', 's', 'eight', 'with', 'have', 'are']

called: ['when', 'well', 'found', 'under', 'since', 'those', 'used', 'very']

b: ['nine', 'their', 's', 'are', 'french', 'late', 'four', 'when']

which: ['after', 'when', 'are', 'e', 'times', 'four', 'such', 'well']

some: ['such', 'after', 'many', 'were', 'there', 'first', 'this', 'also']

these: ['there', 'this', 'his', 'many', 'used', 'him', 'no', 'he']

from: ['great', 'westview', 'coibae', 'solution', 'UNK', 'palaeocontact', 'UNK', 'tradition']

they: ['all', 'during', 'i', 'may', 'state', 'its', 'it', 'be']

not: ['who', 'acts', 'word', 'although', 'nine', 'their', 'made', 'should']

had: ['group', 'see', 'government', 'major', 'due', 'her', 'said', 'title']

at: ['became', 'september', 'silv', 'south', 'who', 'them', 'deaths', 'u']

world: ['her', 'against', 'east', 'government', 'applied', 'any', 'abraham', 'atom']

time: ['up', 'i', 'during', 'only', 'they', 'while', 'may', 'than']

only: ['i', 'may', 'while', 'general', 'during', 'modern', 'state', 'film']

Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 3116/3116 [00:16<00:00, 188.33it/s, Batch Loss=1.02, Avg Loss=0.381]

Epoch 7 completed - Average Loss: 0.3068

Checkpoint saved at step 21812 to

exp_2_setting_1_glove//glove_model_step_21812.pt

Printing closest words

new: ['UNK', 'UNK', 'aldine', 'vehicle', 'distributors', 'UNK', 'rastenni', 'sapiens']

were: ['so', 'also', 'use', 'him', 'most', 'he', 'luddites', 'work']

that: ['eight', 'is', 'was', 'these', 'will', 'five', 'if', 'no']

called: ['since', 'often', 'over', 'although', 'like', 'though', 'then', 'word']

b: ['who', 'king', 'french', 'over', 'characters', 'christian', 'n', 'languages']

which: ['it', 'be', 'after', 'this', 'another', 'to', 'also', 'until']

some: ['many', 'such', 'after', 'can', 'this', 'states', 'which', 'very']

these: ['more', 'god', 'through', 'people', 'should', 'since', 'them', 'usually']

from: ['seven', 'no', 'william', 'if', 'year', 'used', 'herbie', 'between']

they: ['to', 'panepistimiou', 'abdicates', 'cautioned', 'UNK',
'umbilicus', 'it', 'hinz']
not: ['although', 'contains', 'true', 'eight', 'more', 'five', 'been',
'recorded']
had: ['into', 'other', 'all', 'than', 'being', 'can', 'work', 'what']
at: ['savage', 'UNK', 'UNK', 'unless', 'emissions', 'pippin',
'religionists', 'UNK']
world: ['than', 'all', 'd', 'what', 'major', 'had', 'work', 'form']
time: ['being', 'than', 'large', 'man', 'way', 'born', 'political',
'common']
only: ['which', 'now', 'all', 'world', 'other', 'd', 'than', 'it']
Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 3116/3116 [00:16<00:00, 187.25it/s, Batch
Loss=1.06, Avg Loss=0.365]

Epoch 8 completed - Average Loss: 0.2915

Checkpoint saved at step 24928 to

exp_2_setting_1_glove//glove_model_step_24928.pt

Printing closest words

new: ['people', 'following', 'since', 'when', 'UNK', 'no', 'times',
'film']

were: ['most', 'while', 'some', 'first', 'he', 'four', 'last',
'where']

that: ['is', 'strenuous', 'appointed', 'your', 'borrowings', 'was',
'reduced', 'for']

called: ['maltered', 'UNK', 'UNK', 'wird', 'UNK', 'turque', 'available',
'friendship']

b: ['late', 'f', 'actor', 'algol', 'i', 'around', 'england', 'august']

which: ['he', 'due', 'UNK', 'UNK', 'UNK', 'cafes', 'UNK',
'podcasting']

some: ['first', 'all', 'he', 'last', 'would', 'were', 'most', 'where']

these: ['overruled', 'node', 'perpetuated', 'prooem', 'UNK', 'UNK',
'UNK', 'bolax']

from: ['meritocracy', 'UNK', 'translucency', 'castellanos', 'UNK',
'dnipropetrovsk', 'excluded', 'organisation']

they: ['not', 'but', 'after', 'been', 'had', 'being', 'there', 'used']

not: ['they', 'but', 'been', 'used', 'could', 'them', 'who', 'known']

had: ['being', 'after', 'than', 'other', 'what', 'they', 'could',
'man']

at: ['as', 'but', 'has', 'was', 's', 'history', 'than', 'it']

world: ['zero', 'the', 'six', 'of', 'is', 'enclave', 'earthly', 'two']

time: ['another', 'large', 'being', 'including', 'written', 'form',
'than', 'what']

only: ['however', 'UNK', 'UNK', 'UNK', 'hails', 'duillier', 'UNK',
'incitements']

Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 3116/3116 [00:16<00:00, 187.89it/s, Batch
Loss=1.05, Avg Loss=0.345]

Epoch 9 completed - Average Loss: 0.2820

Checkpoint saved at step 28044 to

exp_2_setting_1_glove//glove_model_step_28044.pt

Printing closest words

new: ['work', 'city', 'metaphorical', 'kinetic', 'under', 'hearts', 'UNK', 'chapin']

were: ['have', 'also', 'all', 'be', 'when', 'often', 'called', 'see']

that: ['for', 'six', 'nine', 'zero', 'has', 'in', 'around', 'not']

called: ['when', 'all', 'have', 'if', 'based', 'now', 'be', 'had']

b: ['both', 'own', 'great', 'its', 'world', 'so', 'old', 'can']

which: ['it', 'be', 'are', 'even', 'year', 'no', 'UNK', 'at']

some: ['screenwriting', 'sieberiana', 'yoshokai', 'small', 'UNK', 'UNK', 'mayor', 'UNK']

these: ['when', 'no', 'where', 'even', 'see', 'at', 'who', 'all']

from: ['were', 'have', 'the', 'eight', 'other', 'be', 'in', 'an']

they: ['b', 'include', 'both', 'old', 'been', 'own', 'england', 'world']

not: ['a', 'eight', 'other', 'had', 'his', 'who', 'the', 'at']

had: ['other', 'than', 'have', 'time', 'their', 'to', 'what', 'called']

at: ['are', 'his', 'who', 'form', 'it', 'year', 'during', 'isbn']

world: ['b', 'include', 'own', 'so', 'been', 'its', 'this', 'almost']

time: ['other', 'had', 'all', 'called', 'have', 'be', 'one', 'than']

only: ['however', 'god', 'systems', 'hawza', 'british', 'complex', 'so', 'there']

Starting Epoch 10/10

Epoch 10/10: 100%[██████████] 3116/3116 [00:16<00:00, 189.19it/s, Batch Loss=1.01, Avg Loss=0.331]

Epoch 10 completed - Average Loss: 0.2689

Checkpoint saved at step 31160 to

exp_2_setting_1_glove//glove_model_step_31160.pt

Printing closest words

new: ['state', 'time', 'being', 'general', 'people', 'east', 'west', 'while']

were: ['be', 'he', 'some', 'all', 'when', 'first', 'while', 'because']

that: ['domicile', 'seinfeld', 'UNK', 'nonprofit', 'khar', 'UNK', 'baird', 'nightfall']

called: ['known', 'some', 'while', 'because', 'were', 'been', 'himself', 'over']

b: ['actor', 'years', 'only', 'apollo', 'states', 'french', 'm', 'u']

which: ['or', 'he', 'be', 'first', 'this', 'but', 'people', 'with']

some: ['first', 'can', 'only', 'when', 'been', 'him', 'all', 'he']

these: ['ignatians', 'volcanos', 'UNK', 'UNK', 'dirac', 'tanha', 'diol', 'in']

from: ['menem', 'ostensible', 'equations', 'somnum', 'ravenna', 'somali', 'their', 'inflorescence']

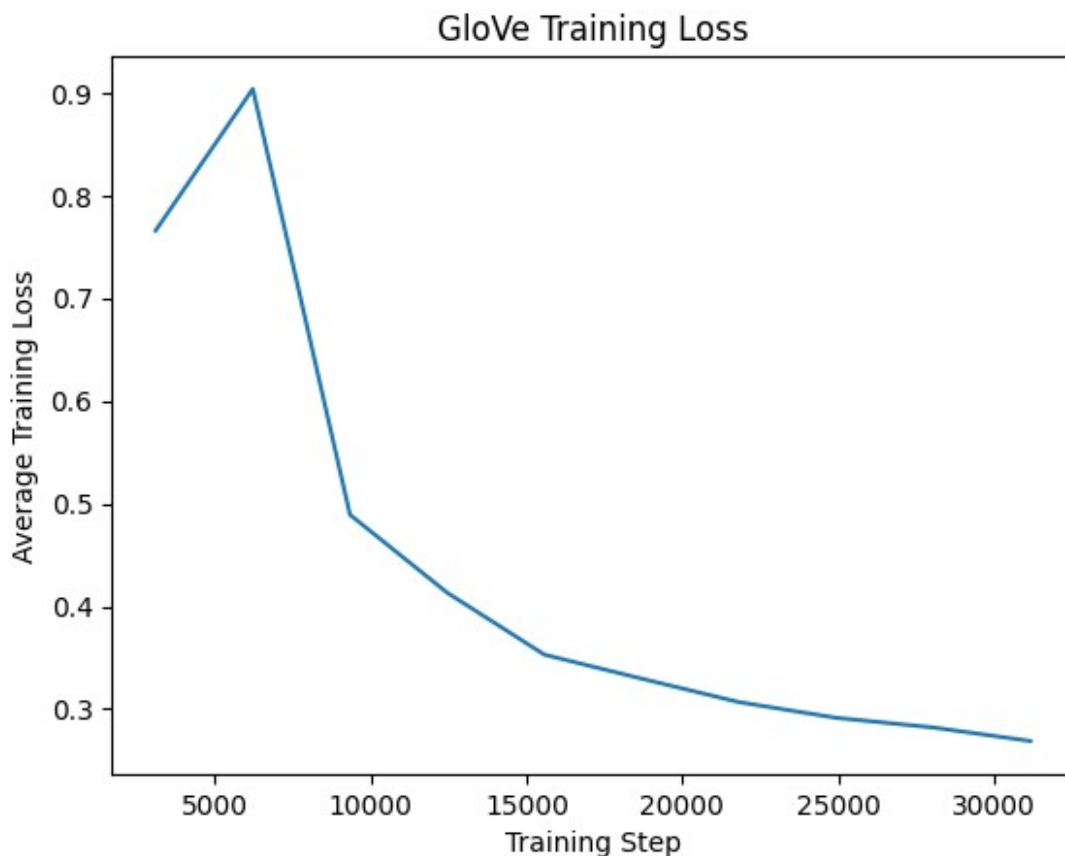
they: ['may', 'some', 'only', 'can', 'him', 'but', 'while', 'no']

not: ['are', 'UNK', 'UNK', 'it', 'suspending', 'nighthorse', 'tonio',

```

'pulitzer']
had: ['eight', 'than', 'after', 'has', 'who', 'perpetuity', 'idea',
'are']
at: ['his', 'theatreversions', 'techically', 'UNK', 'narrowed',
'ordain', 'stavanger', 'venn']
world: ['apollo', 'times', 'particular', 'b', 'philosophy',
'national', 'years', 'jewish']
time: ['may', 'all', 'or', 'which', 'information', 'though', 'while',
'people']
only: ['however', 'him', 'first', 'out', 'made', 'then', 'while',
'up']
Training completed after 10 epochs - Final Average Loss: 0.2689
Created path: exp_2_setting_1_final_model
Final model saved at exp_2_setting_1_final_model/glove_final_model.pt
Embeddings saved at exp_2_setting_1_final_model/glove_embeddings.pkl

```



Hyperparameter Setting 2 for Experiment 2

```

# TODO(student): start

lr = best_lr # Learning rate (fixed) - from experiment 1
checkpoint_model_path = "exp_2_setting_2" # Path to save checkpoints
final_model_path = "exp_2_setting_2_final_model" # Path to save the

```

final model and embeddings

Experiment 2: Vary the window size

You can run this with different window sizes (e.g., 2, 3, 5, 10)

window_size = 5

```
run_training(  
    epochs=10, # Number of epochs to train  
    lr=lr, # Learning rate  
    checkpoint_model_path=checkpoint_model_path, # Directory to save  
model checkpoints  
    final_model_path=final_model_path, # Directory to save the final  
model and embeddings  
    vocab_size=int(1e5), # Vocabulary size  
    batch_size=512, # Batch size for training  
    embedding_size=128, # Embedding size (vector dimension)  
    window_size=window_size, # Context window size for co-occurrence  
    checkpoint_step=5000, # Save checkpoints after every 2000 steps  
)  
  
# TODO(student): end
```

Created path: exp_2_setting_2_glove/

Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]

Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]

['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']

Generating co-occurrence matrix...

Building cooccurrence matrix: on line 0/1000000

Building cooccurrence matrix: on line 100000/1000000

Building cooccurrence matrix: on line 200000/1000000

Building cooccurrence matrix: on line 300000/1000000

Building cooccurrence matrix: on line 400000/1000000

Building cooccurrence matrix: on line 500000/1000000

Building cooccurrence matrix: on line 600000/1000000

Building cooccurrence matrix: on line 700000/1000000

Building cooccurrence matrix: on line 800000/1000000

Building cooccurrence matrix: on line 900000/1000000

Co-occurrence matrix generated.

Printing Validation Set Words

Word: see, ID: 71

Word: this, ID: 36

Word: states, ID: 76

Word: these, ID: 65

Word: he, ID: 32

Word: used, ID: 63

Word: the, ID: 1

Word: two, ID: 11

Word: people, ID: 79

Word: eight, ID: 13
Word: other, ID: 43
Word: on, ID: 25
Word: an, ID: 31
Word: which, ID: 34
Word: their, ID: 45
Word: than, ID: 70
Device: cuda:0
Total Batches per Epoch: 6900
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 6900/6900 [00:36<00:00, 191.18it/s, Batch Loss=0.132, Avg Loss=0.188]

Epoch 1 completed - Average Loss: 0.4807

Checkpoint saved at step 6900 to
exp_2_setting_2_glove//glove_model_step_6900.pt

Printing closest words

see: ['has', 'during', 'j', 'game', 'their', 'UNK', 'UNK',
'thrombocyte']

this: ['including', 'adding', 'are', 'who', 'issues', 'its', 'group',
'united']

states: ['duby', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK', 'gwyneth']

these: ['people', 'e', 'february', 'him', 'under', 'early', 'are',
'been']

he: ['an', 'then', 'it', 'when', 'as', 'UNK', 'that', 'and']

used: ['but', 'friedrich', 'UNK', 'internship', 'death', 'UNK',
'waf', 'while']

the: ['and', 'UNK', 'an', 'UNK', 'michelle', 'UNK', 'adjoins', 'died']

two: ['UNK', 'UNK', 'faoi', 'coherent', 'UNK', 'kemmler', 'UNK',
'UNK']

people: ['these', 'ildico', 'containing', 'UNK', 'UNK', 'anthracene',
'UNK', 'UNK']

eight: ['was', 'vipsania', 'musician', 'UNK', 'UNK', 'UNK', 'ak',
'nine']

other: ['although', 'what', 'against', 'vali', 'until', 'day', 'most',
'UNK']

on: ['an', 'his', 's', 'world', 'as', 'UNK', 'UNK', 'court']

an: ['at', 'some', 'he', 'on', 'including', 'who', 'their', 'man']

which: ['are', 'been', 'has', 'de', 'first', 'when', 'kyu', 'school']

their: ['UNK', 'an', 'no', 'over', 'world', 'UNK', 'been', 'public']

than: ['UNK', 'navy', 'g', 'order', 'much', 'UNK', 'when', 'UNK']

Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 6900/6900 [00:36<00:00, 189.19it/s, Batch Loss=1.41, Avg Loss=0.95]

Epoch 2 completed - Average Loss: 0.4779

Checkpoint saved at step 13800 to
exp_2_setting_2_glove//glove_model_step_13800.pt


```
Printing closest words
see: ['name', 'major', 'national', 'd', 'most', 'took', 'become',
'an']
this: ['between', 'out', 'united', 'including', 'same', 'book',
'last', 'economic']
states: ['many', 'under', 'use', 'time', 'more', 'about', 'country',
'day']
these: ['received', 'alkanes', 'another', 'completely', 'sender',
'activate', 'releasing', 'UNK']
he: ['that', 'UNK', 'no', 'compactification', 'has', 'UNK', 'stock',
'incarnation']
used: ['major', 'national', 'should', 'instead', 'never', 'held',
'name', 'using']
the: ['generally', 'only', 'when', 'UNK', 'many', 'union',
'australian', 'UNK']
two: ['UNK', 'UNK', 'seven', 'lito', 'UNK', 'UNK', 'UNK', 'UNK']
people: ['only', 'under', 'known', 'states', 'more', 'new', 'through',
'time']
eight: ['six', 'after', 'under', 'only', 'would', 'general', 'were',
'about']
other: ['but', 'are', 'into', 'well', 'until', 'most', 'english',
'may']
on: ['his', 'into', 'an', 'over', 'or', 's', 'classical', 'as']
an: ['major', 'national', 'another', 'most', 'although', 'name', 'd',
'have']
which: ['at', 'not', 'their', 'general', 'from', 'only', 'all', 'war']
their: ['not', 'which', 'they', 'at', 'general', 'only', 'after',
'end']
than: ['would', 'many', 'day', 'more', 'us', 'following', 'time',
'were']
Starting Epoch 3/10
```

```
Epoch 3/10: 100%|██████████| 6900/6900 [00:36<00:00, 188.60it/s, Batch
Loss=0.492, Avg Loss=0.399]
```

```
Epoch 3 completed - Average Loss: 0.2672
Checkpoint saved at step 20700 to
exp_2_setting_2_glove//glove_model_step_20700.pt
Printing closest words
see: ['been', 'first', 'during', 'using', 'european', 'run', 'line',
'list']
this: ['between', 'her', 'home', 'that', 'united', 'some', 'three',
'capital']
states: ['only', 'both', 'after', 'sometimes', 'word', 'great', 'new',
'north']
these: ['used', 'be', 'its', 'them', 'between', 'due', 'some', 'that']
he: ['for', 'its', 'writer', 'act', 'that', 'retributive', 'them',
'how']
used: ['these', 'term', 'although', 'later', 'them', 'british',
'united', 'due']
```

the: ['of', 'seem', 'neatly', 'maritain', 'and', 'UNK', 'householder', 'example']
two: ['was', 'nine', 'by', 'to', 'a', 'that', 'with', 'one']
people: ['time', 'government', 'more', 'would', 'm', 'when', 'not', 'since']
eight: ['lauriston', 'bistro', 'UNK', 'UNK', 'seas', 'homolytic', 'nadder', 'UNK']
other: ['not', 'but', 'over', 'then', 'when', 'were', 'science', 'back']
on: ['s', 'as', 'his', 'seven', 'five', 'for', 'may', 'an']
an: ['for', 'was', 'have', 'with', 'nine', 'in', 'two', 'by']
which: ['at', 'after', 'new', 'so', 'often', 'through', 'great', 'where']
their: ['were', 'they', 'more', 'has', 'can', 'not', 'may', 'this']
than: ['more', 'example', 'because', 'body', 'state', 'computer', 'year', 'her']
Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 6900/6900 [00:36<00:00, 187.53it/s, Batch Loss=1.05, Avg Loss=0.653]

Epoch 4 completed - Average Loss: 0.3084

Checkpoint saved at step 27600 to

exp_2_setting_2_glove//glove_model_step_27600.pt

Printing closest words

see: ['been', 'into', 'first', 'being', 'it', 'there', 'had', 'he']
this: ['were', 'such', 'some', 'used', 'also', 'may', 'he', 'many']
states: ['him', 'war', 'there', 'into', 'if', 'each', 'which', 'made']
these: ['its', 'between', 'british', 'known', 'many', 'g', 'some', 'used']
he: ['also', 'into', 'it', 'being', 'may', 'first', 'been', 'see']
used: ['later', 'some', 'this', 'same', 'different', 'may', 'language', 'between']
the: ['of', 'is', 'take', 'one', 'has', 'UNK', 'on', 'UNK']
two: ['zero', 'with', 'was', 'acid', 'a', 'prime', 'video', 'centuries']
people: ['new', 'became', 'found', 'about', 'those', 'under', 'what', 'because']
eight: ['s', 'on', 'for', 'as', 'four', 'his', 'six', 'five']
other: ['were', 'but', 'when', 'may', 'also', 'not', 'he', 'which']
on: ['his', 'as', 's', 'for', 'five', 'four', 'are', 'is']
an: ['has', 'itself', 'system', 'who', 'acupuncture', 'in', 'world', 'historical']
which: ['had', 'it', 'also', 'states', 'into', 'first', 'war', 'were']
their: ['who', 'co', 'historical', 'book', 'today', 'any', 'works', 'act']
than: ['life', 'year', 'government', 'state', 'century', 'right', 'full', 'time']
Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 6900/6900 [00:36<00:00, 187.18it/s, Batch Loss=0.557, Avg Loss=0.424]

Epoch 5 completed - Average Loss: 0.2388

Checkpoint saved at step 34500 to

exp_2_setting_2_glove//glove_model_step_34500.pt

Printing closest words

see: ['were', 'such', 'may', 'i', 'can', 'when', 'e', 'into']

this: ['when', 'such', 'between', 'during', 'which', 'i', 'first', 'some']

states: ['also', 'him', 'war', 'over', 'america', 'their', 'open', 'king']

these: ['including', 'a', 'south', 'what', 'production', 'he', 'less', 's']

he: ['s', 'a', 'in', 'was', 'to', 'as', 'are', 'and']

used: ['been', 'being', 'its', 'language', 'or', 'played', 'called', 'name']

the: ['of', 'and', 'a', 'in', 'to', 'by', 'nine', 'was']

two: ['by', 'that', 'nine', 'or', 'three', 'from', 'as', 'all']

people: ['europe', 'special', 'south', 'empire', 'less', 'what', 'nations', 'rather']

eight: ['or', 'by', 'two', 'nine', 'as', 'three', 'four', 'it']

other: ['including', 'had', 'as', 'were', 'first', 'incapacitated', 'western', 'efficient']

on: ['five', 'his', 'which', 'with', 'for', 'i', 'is', 'six']

an: ['model', 'gorillas', 'chemically', 'player', 'withdrawing', 'actress', 'UNK', 'UNK']

which: ['were', 'i', 'when', 'such', 'during', 'e', 'can', 'this']

their: ['has', 'then', 'states', 'america', 'before', 'world', 'much', 'also']

than: ['government', 'year', 'attempt', 'you', 'old', 'make', 'believe', 'important']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 6900/6900 [00:36<00:00, 187.60it/s, Batch Loss=0.788, Avg Loss=0.567]

Epoch 6 completed - Average Loss: 0.2697

Checkpoint saved at step 41400 to

exp_2_setting_2_glove//glove_model_step_41400.pt

Printing closest words

see: ['adelaide', 'using', 'into', 'military', 'sign', 'analysis', 'led', 'games']

this: ['in', 'for', 'an', 'with', 'was', 'as', 'is', 'other']

states: ['about', 'used', 'had', 'under', 'been', 'military', 'system', 'land']

these: ['people', 'there', 'him', 'work', 'then', 'called', 'he', 'american']

he: ['five', 'american', 'from', 'bc', 'c', 'seven', 'two', 'these']

used: ['been', 'being', 'about', 'popular', 'its', 'own', 'under',

'different']
the: ['one', 'of', 'is', 'zero', 's', 'in', 'other', 'to']
two: ['was', 'he', 's', 'from', 'all', 'as', 'their', 'often']
people: ['united', 'can', 'first', 'history', 'found', 'known', 'who', 'both']
eight: ['bear', 'believed', 'believe', 'why', 'term', 'series', 'family', 'kind']
other: ['in', 'an', 'with', 'for', 'united', 'from', 'was', 'is']
on: ['was', 's', 'some', 'for', 'filtration', 'five', 'be', 'he']
an: ['from', 'with', 'for', 'which', 'was', 'six', 'who', 'b']
which: ['with', 'an', 'first', 'six', 'not', 'who', 'can', 'united']
their: ['however', 'from', 'not', 'an', 'world', 'most', 'have', 'more']
than: ['example', 'various', 'large', 'side', 'novels', 'capital', 'find', 'fact']
Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 6900/6900 [00:36<00:00, 188.68it/s, Batch Loss=0.602, Avg Loss=0.435]

Epoch 7 completed - Average Loss: 0.2309

Checkpoint saved at step 48300 to

exp_2_setting_2_glove/glove_model_step_48300.pt

Printing closest words

see: ['into', 'd', 'this', 'would', 'about', 'were', 'when', 'after']

this: ['by', 'on', 'many', 'first', 'a', 'his', 'nine', 'most']

states: ['though', 'example', 'became', 'population', 'philosophy', 'office', 'energy', 'national']

these: ['do', 'form', 'no', 'place', 'without', 'until', 'just', 'sometimes']

he: ['it', 'this', 'their', 'on', 'like', 'by', 'but', 'while']

used: ['a', 'it', 'such', 'name', 'on', 'frequently', 'many', 'into']

the: ['of', 'and', 'nine', 'a', 'on', 'four', 'his', 'are']

two: ['three', 'eight', 'by', 'nine', 'on', 'four', 'his', 'a']

people: ['apollo', 'end', 'work', 'early', 'both', 'form', 'during', 'so']

eight: ['three', 'two', 'nine', 'four', 'by', 'on', 'his', 'of']

other: ['this', 'by', 'where', 'it', 'he', 'were', 'on', 'are']

on: ['by', 'his', 'four', 'this', 'nine', 'are', 'a', 'it']

an: ['also', 'like', 'their', 'people', 'who', 'most', 'so', 'found']

which: ['an', 'like', 'also', 'he', 'while', 'who', 'about', 'what']

their: ['but', 'first', 'many', 'most', 'them', 'while', 'this', 'until']

than: ['were', 'generally', 'into', 'information', 'since', 'what', 'use', 'because']

Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 6900/6900 [00:36<00:00, 187.60it/s, Batch Loss=0.846, Avg Loss=0.531]

Epoch 8 completed - Average Loss: 0.2565
Checkpoint saved at step 55200 to
exp_2_setting_2_glove//glove_model_step_55200.pt
Printing closest words
see: ['or', 'time', 'were', 'than', 'her', 'at', 'its', 'has']
this: ['on', 'some', 'dagny', 'lifts', 'pyrotechnic', 'superscript',
'mercy', 'UNK']
states: ['being', 'new', 'example', 'long', 'theory', 'united',
'usually', 'france']
these: ['against', 'most', 'still', 'them', 'austria', 'their',
'only', 'major']
he: ['or', 'have', 'his', 'it', 'are', 'see', 'at', 'to']
used: ['use', 'name', 'been', 'time', 'known', 'more', 'her', 'until']
the: ['one', 'to', 'is', 'as', 'zero', 'of', 'in', 'or']
two: ['was', 'lapis', 'by', 'eight', 'on', 'conscriptum', 'volume',
'eixample']
people: ['like', 'apollo', 'even', 'both', 'life', 'city', 'found',
'itself']
eight: ['opioids', 'mammalia', 'on', 'two', 'UNK', 'capita', 'UNK',
'flip']
other: ['but', 'were', 'her', 'when', 'man', 'more', 'thus', 'until']
on: ['was', 'this', 'four', 'some', 'eight', 'mired', 'ayurvedic',
'grinling']
an: ['over', 'any', 'into', 'united', 'he', 'number', 'various',
'numbers']
which: ['although', 'their', 'about', 'major', 'before', 'these',
'most', 'fourth']
their: ['first', 'like', 'most', 'life', 'even', 'about', 'them',
'war']
than: ['time', 'were', 'see', 'because', 'her', 'name', 'thus', 'use']
Starting Epoch 9/10

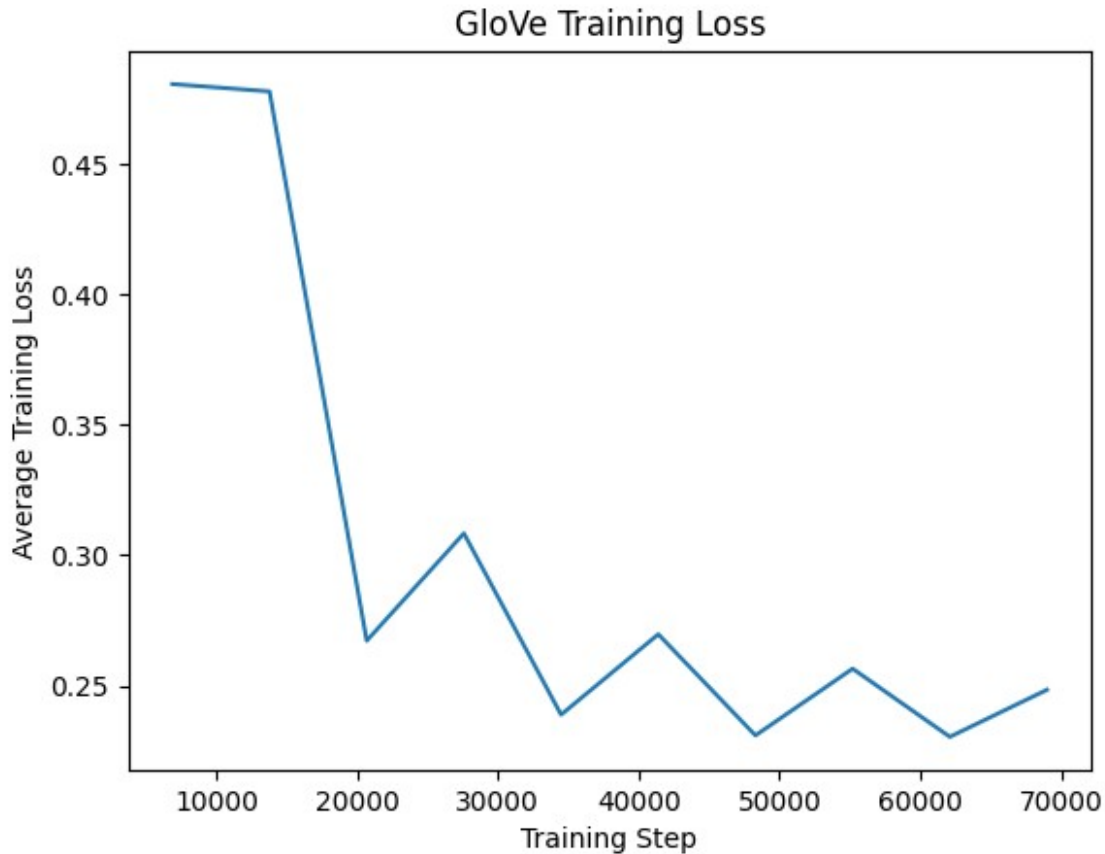
Epoch 9/10: 100%[██████████] 6900/6900 [00:36<00:00, 187.97it/s, Batch
Loss=0.572, Avg Loss=0.443]

Epoch 9 completed - Average Loss: 0.2303
Checkpoint saved at step 62100 to
exp_2_setting_2_glove//glove_model_step_62100.pt
Printing closest words
see: ['seven', 'five', 'ural', 'his', 'are', 'hh', 'mississippian',
'gregor']
this: ['that', 'an', 'by', 'be', 'nine', 'with', 'than', 'was']
states: ['new', 'being', 'more', 'between', 'can', 'may', 'various',
'theory']
these: ['no', 'only', 'all', 'would', 'well', 'often', 'people',
'first']
he: ['took', 'it', 'at', 'under', 'is', 'given', 'used', 'state']
used: ['more', 'who', 'known', 'can', 'use', 'name', 'may', 'all']
the: ['and', 'of', 'nine', 'two', 'a', 'that', 'was', 'be']
two: ['eight', 'four', 'nine', 'three', 'was', 'and', 'that', 'the']

```
people: ['even', 'city', 'war', 'them', 'no', 'first', 'only',
'state']
eight: ['two', 'four', 'three', 'was', 'nine', 'that', 'by', 'with']
other: ['UNK', 'allemani', 'UNK', 'UNK', 'were', 'farhud', 'UNK',
'UNK']
on: ['cactus', 'meran', 'anatomy', 'UNK', 'anacondas', 'gastrula',
'nubia', 'UNK']
an: ['that', 'three', 'nine', 'two', 'be', 'was', 'more', 'with']
which: ['be', 'that', 'their', 'who', 'all', 'then', 'them', 'often']
their: ['first', 'them', 'which', 'all', 'people', 'still', 'who',
'even']
than: ['that', 'two', 'was', 'four', 'can', 'name', 'who', 'more']
Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 6900/6900 [00:37<00:00, 186.48it/s,
Batch Loss=0.799, Avg Loss=0.513]

Epoch 10 completed - Average Loss: 0.2484
Checkpoint saved at step 69000 to
exp_2_setting_2_glove//glove_model_step_69000.pt
Printing closest words
see: ['over', 'between', 'i', 'people', 'god', 'history', 'used',
'natural']
this: ['for', 'as', 'is', 'are', 'five', 'some', 'but', 'it']
states: ['being', 'there', 'year', 'then', 'some', 'left', 'after',
'often']
these: ['after', 'then', 'some', 'new', 'been', 'b', 'law', 'both']
he: ['it', 'this', 'then', 'very', 'some', 's', 'states', 'same']
used: ['can', 'i', 'name', 'over', 'between', 'people', 'possible',
'known']
the: ['one', 'to', 'zero', 'is', 'for', 'this', 'not', 'number']
two: ['that', 'which', 'four', 'six', 'more', 'nine', 'all',
'missions']
people: ['even', 'between', 'name', 'until', 'over', 'battle', 'list',
'historical']
eight: ['s', 'UNK', 'so', 'very', 'in', 'four', 'be', 'lincoln']
other: ['part', 'lies', 'UNK', 'starhawk', 'battista', 'UNK', 'UNK',
'bibel']
on: ['is', 's', 'as', 'in', 'to', 'zero', 'five', 'number']
an: ['it', 'many', 'on', 'very', 'common', 'against', 'tragicomedy',
'UNK']
which: ['two', 'that', 'world', 'be', 'also', 'th', 'founded', 'four']
their: ['first', 'at', 'who', 'i', 'were', 'does', 'from', 'economic']
than: ['that', 'roadrunner', 'mirroring', 'jellinek', 'UNK', 'zeros',
'katie', 'ollie']
Training completed after 10 epochs - Final Average Loss: 0.2484
Created path: exp_2_setting_2_final_model
Final model saved at exp_2_setting_2_final_model/glove_final_model.pt
Embeddings saved at exp_2_setting_2_final_model/glove_embeddings.pkl
```



Hyperparameter Setting 3 for Experiment 2

TODO(student): start

```
lr = best_lr # Learning rate (fixed) - from experiment 1
checkpoint_model_path = "exp_2_setting_3" # Path to save checkpoints
final_model_path = "exp_2_setting_3_final_model" # Path to save the final model and embeddings
```

Experiment 2: Vary the window size

You can run this with different window sizes (e.g., 2, 3, 5, 10)

```
window_size = 10
```

```
run_training(
    epochs=10, # Number of epochs to train
    lr=lr, # Learning rate
    checkpoint_model_path=checkpoint_model_path, # Directory to save model checkpoints
    final_model_path=final_model_path, # Directory to save the final model and embeddings
    vocab_size=int(1e5), # Vocabulary size
    batch_size=512, # Batch size for training
    embedding_size=128, # Embedding size (vector dimension)
    window_size=window_size, # Context window size for co-occurrence
```

```

    checkpoint_step=5000, # Save checkpoints after every 2000 steps
)

# TODO(student): end

Created path: exp_2_setting_3_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
Co-occurrence matrix generated.
Printing Validation Set Words
Word: has, ID: 41
Word: this, ID: 36
Word: some, ID: 50
Word: so, ID: 93
Word: these, ID: 65
Word: war, ID: 75
Word: which, ID: 34
Word: they, ID: 49
Word: have, ID: 39
Word: all, ID: 51
Word: american, ID: 48
Word: no, ID: 80
Word: use, ID: 90
Word: in, ID: 5
Word: with, ID: 24
Word: he, ID: 32
Device: cuda:0
Total Batches per Epoch: 11804
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 11804/11804 [01:02<00:00, 189.50it/s,
Batch Loss=0.237, Avg Loss=0.153]

Epoch 1 completed - Average Loss: 0.3408
Checkpoint saved at step 11804 to
exp_2_setting_3_glove//glove_model_step_11804.pt

```


Printing closest words

has: ['UNK', 'UNK', 'anatolian', 'jews', 'omniglot', 'five', 'with', 'UNK']
this: ['just', 'an', 'all', 'itself', 'those', 'make', 'two', 'mesozoic']
some: ['will', 'such', 'system', 'UNK', 'its', 'UNK', 'including', 'hamedan']
so: ['terrail', 'quo', 'UNK', 'UNK', 'herrion', 'occurring', 'UNK', 'podio']
these: ['UNK', 'UNK', 'will', 'first', 'UNK', 'early', 'time', 'bayan']
war: ['ladar', 'solingen', 'israel', 'firebat', 'vol', 'fire', 'loc', 'UNK']
which: ['from', 'he', 'called', 'in', 'social', 'north', 'american', 'large']
they: ['but', 'defend', 'UNK', 'with', 'five', 'shrew', 'oldman', 'decompression']
have: ['at', 'called', 'he', 'that', 'because', 'be', 't', 'b']
all: ['an', 'this', 'but', 'more', 'i', 'her', 'those', 'power']
american: ['was', 'system', 'which', 'six', 'just', 'national', 'most', 'four']
no: ['same', 'her', 'UNK', 'UNK', 'there', 'peacetime', 'where', 'will']
use: ['UNK', 'standby', 'UNK', 'albrech', 'UNK', 'vaefar', 'unpleasant', 'swift']
in: ['a', 'which', 'from', 'large', 'widely', 'was', 'called', 'apollo']
with: ['four', 'is', 'it', 'd', 'but', 'five', 'above', 'telefonica']
he: ['which', 'have', 'nine', 'other', 'more', 'from', 'his', 'such']
Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 11804/11804 [01:02<00:00, 188.97it/s, Batch Loss=1.04, Avg Loss=0.634]

Epoch 2 completed - Average Loss: 0.3026

Checkpoint saved at step 23608 to

exp_2_setting_3_glove//glove_model_step_23608.pt

Printing closest words

has: ['its', 'also', 'had', 's', 'been', 'after', 'it', 'united']
this: ['but', 'some', 'it', 'on', 'also', 'any', 'been', 'other']
some: ['but', 'other', 'any', 'this', 'world', 'on', 'life', 'must']
so: ['while', 'use', 'well', 'under', 'about', 'however', 'being', 'out']
these: ['who', 'be', 'god', 'more', 'made', 'that', 'either', 'created']
war: ['use', 'known', 'under', 'during', 'about', 'while', 'may', 'between']
which: ['from', 'a', 'was', 'most', 'how', 'builder', 'warrant', 'dentine']
they: ['their', 'for', 'or', 'by', 's', 'would', 'government',

'based']
have: ['applause', 'porus', 'UNK', 'UNK', 'kampf', 'UNK', 'gustavus', 'swayed']
all: ['called', 'there', 'using', 'took', 'd', 'because', 'british', 'state']
american: ['there', 'called', 'british', 'practice', 'd', 'all', 'force', 'first']
no: ['both', 'about', 'between', 'time', 'being', 'its', 'may', 'very']
use: ['while', 'known', 'however', 'during', 'out', 'being', 'under', 'may']
in: ['eight', 'material', 'new', 'riding', 'natural', 'UNK', 'amarynthus', 'footwear']
with: ['five', 'on', 'but', 'for', 'life', 'other', 'would', 'h']
he: ['who', 'an', 'all', 'made', 'many', 'origin', 'UNK', 'hydaspes']
Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 11804/11804 [01:02<00:00, 187.77it/s, Batch Loss=0.482, Avg Loss=0.31]

Epoch 3 completed - Average Loss: 0.1954

Checkpoint saved at step 35412 to

exp_2_setting_3_glove//glove_model_step_35412.pt

Printing closest words

has: ['had', 'they', 'also', 'subject', 'ancient', 'now', 'last', 'alexander']

this: ['also', 'must', 'for', 'on', 'some', 'than', 'as', 'then']

some: ['for', 'not', 'his', 'on', 'at', 'also', 'such', 'had']

so: ['great', 'century', 'became', 'however', 'form', 'order', 'become', 'theory']

these: ['him', 'more', 'between', 'including', 'although', 'language', 'each', 'united']

war: ['th', 'however', 'great', 'so', 'german', 'century', 'became', 'could']

which: ['at', 'have', 'his', 'who', 'such', 'than', 'from', 'four']

they: ['many', 'now', 'has', 'alexander', 'did', 'been', 'had', 'still']

have: ['which', 'at', 'such', 'who', 'than', 'when', 'his', 'four']

all: ['not', 'there', 'have', 'many', 'which', 'when', 'he', 'at']

american: ['four', 'two', 'seven', 'is', 'was', 'six', 'at', 'new']

no: ['only', 'while', 'e', 'especially', 'film', 'national', 'both', 'roman']

use: ['may', 'however', 'name', 'known', 'modern', 'being', 'then', 'could']

in: ['and', 'a', 'to', 'one', 'is', 'nine', 'two', 'as']

with: ['for', 'on', 'four', 'two', 'is', 'such', 'as', 'was']

he: ['not', 'at', 'his', 'which', 'six', 'from', 'for', 'such']

Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.05it/s,
Batch Loss=0.657, Avg Loss=0.518]

Epoch 4 completed - Average Loss: 0.2158

Checkpoint saved at step 47216 to

exp_2_setting_3_glove//glove_model_step_47216.pt

Printing closest words

has: ['it', 'this', 's', 'or', 'who', 'up', 'all', 'first']

this: ['which', 'who', 'be', 'all', 'five', 'or', 'eight', 'at']

some: ['had', 'been', 'after', 'may', 'see', 'american', 'used',
'their']

so: ['later', 'century', 'than', 'being', 'which', 'this', 'who',
'book']

these: ['more', 'him', 'united', 'like', 'it', 'd', 'up', 'number']

war: ['may', 'after', 'new', 'french', 'play', 'some', 'earth',
'years']

which: ['this', 'at', 'all', 'who', 'that', 'not', 'five', 'be']

they: ['warner', 'comparison', 'deal', 'ensured', 'UNK', 'UNK',
'shaft', 'hopelessly']

have: ['such', 'when', 'for', 'which', 'his', 'not', 'also', 'at']

all: ['which', 'first', 'this', 'be', 'at', 'who', 'five', 'not']

american: ['see', 'some', 'world', 'after', 'new', 'been', 'had',
'may']

no: ['only', 'm', 'any', 'their', 'l', 'general', 'english', 'people']

use: ['great', 'last', 'n', 'death', 'around', 'th', 'out', 'since']

in: ['same', 'other', 'non', 'UNK', 'nomads', 'distillers', 'whether',
'sim']

with: ['for', 'on', 'and', 'a', 'three', 'second', 'have', 'follow']

he: ['be', 'at', 'called', 'which', 'all', 'also', 'such', 'on']

Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 11804/11804 [01:02<00:00, 187.55it/s,
Batch Loss=0.364, Avg Loss=0.378]

Epoch 5 completed - Average Loss: 0.1856

Checkpoint saved at step 59020 to

exp_2_setting_3_glove//glove_model_step_59020.pt

Printing closest words

has: ['this', 'which', 'two', 'b', 'i', 'being', 'century', 'number']

this: ['which', 'two', 'has', 'i', 'b', 'being', 'god', 'seen']

some: ['used', 'after', 'their', 'see', 'would', 'many', 'they',
'were']

so: ['during', 'history', 'under', 'all', 'called', 'them', 'through',
'will']

these: ['all', 'states', 'more', 'him', 'he', 'her', 'through',
'very']

war: ['english', 'been', 'public', 'new', 'good', 'actor', 'after',
'between']

which: ['two', 'this', 'b', 'has', 'i', 'zero', 'century', 'being']

they: ['their', 'but', 'were', 'most', 'to', 'many', 'he', 'some']

have: ['elmslie', 'although', 'columella', 'followed', 'person',
'vertebrate', 'UNK', 'UNK']
all: ['first', 'he', 'during', 'so', 'also', 'part', 'these', 'to']
american: ['six', 'was', 'states', 'is', 'to', 'history', 'a',
'during']
no: ['if', 'government', 'only', 'became', 'then', 'public', 'name',
'people']
use: ['death', 'since', 'where', 'power', 'greek', 'another', 'out',
'support']
in: ['is', 'to', 'are', 'and', 'were', 'most', 'he', 'was']
with: ['from', 'out', 's', 'his', 'being', 'great', 'four', 'b']
he: ['to', 'all', 'were', 'is', 'also', 'but', 'was', 'in']
Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.06it/s,
Batch Loss=0.439, Avg Loss=0.496]

Epoch 6 completed - Average Loss: 0.1973

Checkpoint saved at step 70824 to
exp_2_setting_3_glove//glove_model_step_70824.pt

Printing closest words

has: ['who', 'six', 'zero', 'for', 'it', 'his', 'than', 'united']
this: ['great', 'high', 'down', 'will', 'theory', 'could',
'aristotle', 'that']
some: ['but', 'they', 'see', 'after', 'many', 'have', 'time', 'be']
so: ['during', 'when', 'will', 'states', 'all', 'between', 'd',
'existence']
these: ['he', 'more', 'UNK', 'their', 'apostles', 'were', 'also',
'UNK']
war: ['between', 'years', 'english', 'when', 'd', 'new', 'those',
'form']
which: ['zero', 'every', 'seven', 'a', 'be', 'the', 'and', 'been']
they: ['many', 'but', 'some', 'used', 'time', 'see', 'after', 'most']
have: ['be', 'its', 'or', 'such', 'on', 'at', 'from', 'three']
all: ['first', 'when', 'd', 'during', 'will', 'so', 'states', 'fact']
american: ['eight', 'with', 'three', 'as', 'two', 'for', 'are', 'on']
no: ['only', 'known', 'if', 'name', 'after', 'state', 'such', 'him']
use: ['since', 'could', 'later', 'university', 'great', 'external',
'same', 'research']
in: ['UNK', 'UNK', 'their', 'interspaced', 'firearms', 'cryolite',
'that', 'guangdong']
with: ['his', 's', 'three', 'are', 'on', 'two', 'as', 'was']
he: ['these', 'their', 'were', 'also', 'sridhara', 'an',
'embarrassment', 'invaginates']
Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 11804/11804 [01:03<00:00, 186.76it/s,
Batch Loss=0.621, Avg Loss=0.415]

Epoch 7 completed - Average Loss: 0.1814
Checkpoint saved at step 82628 to
exp_2_setting_3_glove//glove_model_step_82628.pt
Printing closest words
has: ['had', 'this', 'that', 'an', 'also', 'there', 'which', 'seven']
this: ['also', 'that', 'an', 'has', 'which', 'first', 'had', 'he']
some: ['sometimes', 'support', 'they', 'dead', 'long', 'great', 'out',
'especially']
so: ['during', 'lincoln', 'times', 'fact', 'culture', 'production',
'region', 'articles']
these: ['their', 'religious', 'old', 'does', 'well', 'would', 'using',
'under']
war: ['between', 'years', 'english', 'i', 'new', 'those', 'including',
'century']
which: ['an', 'this', 'seven', 'also', 'he', 'has', 'have', 'that']
they: ['many', 'have', 'such', 'into', 'like', 'his', 'however',
'known']
have: ['which', 'six', 'may', 'such', 'they', 'other', 'he', 'many']
all: ['first', 'i', 'new', 'this', 'between', 'also', 'war', 'years']
american: ['r', 'UNK', 'gnat', 'UNK', 'from', 'UNK', 'judaic',
'stimulation']
no: ['leonardo', 'melody', 'UNK', 'UNK', 'duo', 'calumet', 'UNK',
'UNK']
use: ['he', 'different', 'that', 'seven', 'people', 'has', 'by',
'there']
in: ['seven', 'of', 'has', 'an', 'one', 'that', 'by', 's']
with: ['from', 'for', 'to', 's', 'at', 'large', 'UNK', 'autistic']
he: ['an', 'that', 'also', 'this', 'six', 'seven', 'which', 'had']
Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 11804/11804 [01:03<00:00, 186.80it/s,
Batch Loss=0.612, Avg Loss=0.492]

Epoch 8 completed - Average Loss: 0.1931
Checkpoint saved at step 94432 to
exp_2_setting_3_glove//glove_model_step_94432.pt
Printing closest words
has: ['also', 'only', 'united', 'that', 'he', 'had', 'where',
'people']
this: ['many', 'into', 'b', 'd', 'however', 'other', 'time', 'such']
some: ['victory', 'that', 'he', 'from', 'not', 'by', 'the', 'also']
so: ['later', 'these', 'number', 'part', 'their', 'death', 'produced',
'made']
these: ['their', 'would', 'been', 'three', 'its', 'but', 'will',
'not']
war: ['new', 'years', 'century', 'e', 'life', 'same', 'can', 'had']
which: ['six', 'between', 'known', 'other', 'UNK', 'they', 'such',
'with']
they: ['other', 'many', 'into', 'such', 'name', 'this', 'than',
'through']

```
have: ['that', 'had', 'he', 'even', 'it', 'by', 'not', 'be']
all: ['jewish', 'the', 'those', 'held', 'people', 'UNK', 'out',
'symbols']
american: ['b', 'd', 'this', 'like', 'particularly', 'al', 'means',
'alexander']
no: ['will', 'been', 'often', 'use', 'these', 'not', 'on', 'be']
use: ['will', 'been', 'when', 'often', 'not', 'can', 'no', 'were']
in: ['with', 'littoral', 'speciality', 'chords', 'for', 'six', 'UNK',
'UNK']
with: ['for', 'in', 'caen', 'dismal', 'which', 'UNK', 'drafting',
'multilateration']
he: ['by', 'it', 'that', 'have', 'was', 'also', 'not', 'had']
Starting Epoch 9/10
```

```
Epoch 9/10: 100%|██████████| 11804/11804 [01:03<00:00, 186.22it/s,
Batch Loss=0.566, Avg Loss=0.437]
```

```
Epoch 9 completed - Average Loss: 0.1829
```

```
Checkpoint saved at step 106236 to
exp_2_setting_3_glove//glove_model_step_106236.pt
```

```
Printing closest words
```

```
has: ['have', 'only', 'there', 'also', 'not', 'probably', 'levels',
'following']
```

```
this: ['many', 'an', 'at', 'for', 'most', 'are', 'other', 'used']
```

```
some: ['used', 'most', 'not', 'also', 'after', 'his', 'time',
'called']
```

```
so: ['indignation', 'UNK', 'lobelia', 'pioneer', 'doctrinal', 'UNK',
'betray', 'zimbabweans']
```

```
these: ['no', 'been', 'after', 'later', 'used', 'form', 'area',
'born']
```

```
war: ['new', 'century', 'had', 'been', 'e', 'no', 'born', 'it']
```

```
which: ['at', 'his', 'from', 'five', 'are', 'for', 'an', 'it']
```

```
they: ['an', 'other', 'for', 'are', 'his', 'which', 'five', 'at']
```

```
have: ['also', 'not', 'new', 'it', 'other', 'has', 'book', 'are']
```

```
all: ['third', 'subject', 'within', 'view', 'held', 'must', 'UNK',
'UNK']
```

```
american: ['his', 'seven', 'most', 'at', 'some', 'are', 'five',
'this']
```

```
no: ['been', 'these', 'those', 'later', 'form', 'same', 'born',
'called']
```

```
use: ['life', 'character', 'people', 'class', 'what', 'again', 'able',
'day']
```

```
in: ['one', 'five', 'seven', 'for', 'nine', 's', 'an', 'the']
```

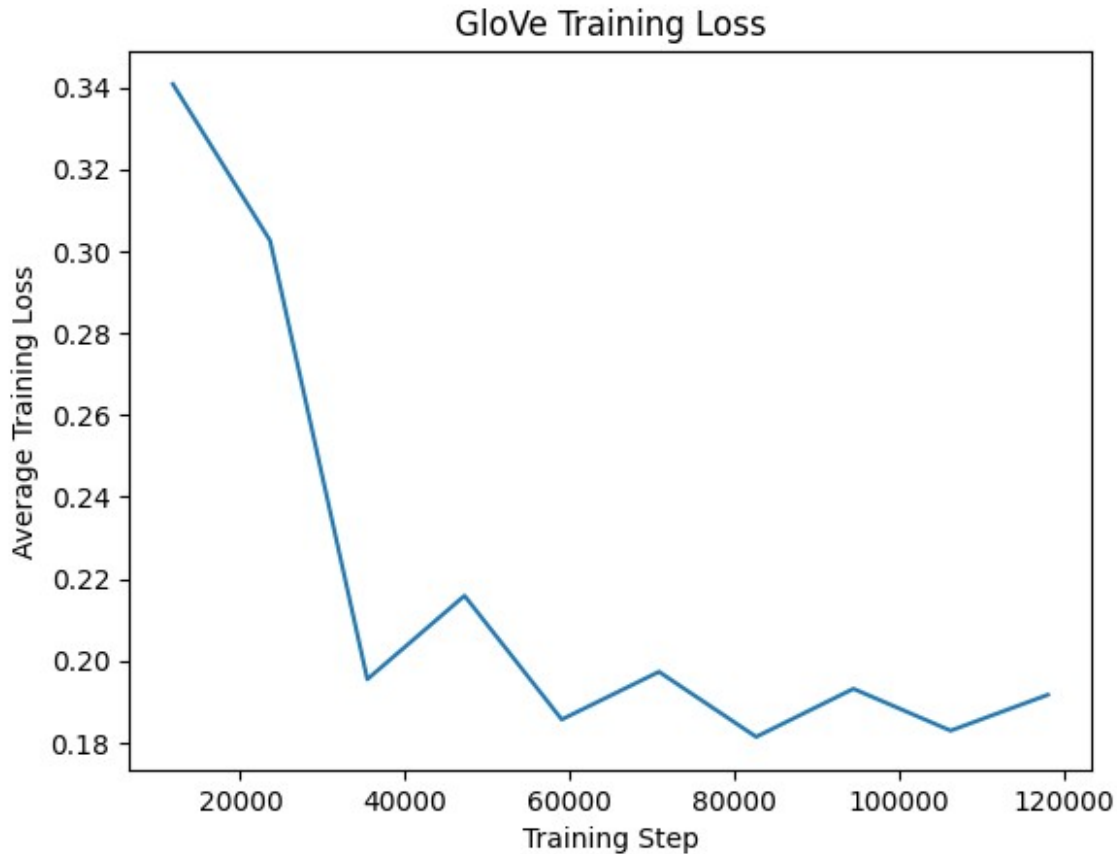
```
with: ['his', 'are', 'from', 'at', 'for', 'also', 'five', 's']
```

```
he: ['on', 'be', 'two', 'their', 'eight', 'zero', 'but', 'were']
```

```
Starting Epoch 10/10
```

```
Epoch 10/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.13it/s,
Batch Loss=0.616, Avg Loss=0.495]
```

```
Epoch 10 completed - Average Loss: 0.1916
Checkpoint saved at step 118040 to
exp_2_setting_3_glove//glove_model_step_118040.pt
Printing closest words
has: ['knots', 'UNK', 'brilliantly', 'scooping', 'UNK', 'montju',
'UNK', 'kook']
this: ['time', 'an', 'world', 'more', 'only', 'many', 'about', 'when']
some: ['UNK', 'choppiness', 'UNK', 'contractors', 'UNK', 'UNK', 'UNK',
'postpartum']
so: ['both', 'before', 'very', 'often', 'than', 'known', 'although',
'well']
these: ['also', 'well', 'may', 'much', 'seven', 'there', 'used', 'so']
war: ['all', 'into', 'no', 'which', 'it', 'they', 'there', 'such']
which: ['as', 'other', 'it', 's', 'all', 'they', 'is', 'into']
they: ['into', 'other', 'such', 'all', 'there', 'which', 'as', 'it']
have: ['also', 'other', 'which', 'there', 'that', 'it', 'but', 's']
all: ['such', 'first', 'into', 'they', 'other', 'eight', 'on', 'as']
american: ['used', 'these', 'four', 'most', 'who', 'trevor', 'past',
'UNK']
no: ['well', 'later', 'war', 'may', 'them', 'although', 'states',
'part']
use: ['st', 'demosthenes', 'based', 'there', 'h', 'way', 'ii', 'UNK']
in: ['of', 'one', 'another', 'states', 'and', 'now', 'it', 'the']
with: ['for', 'an', 'four', 'most', 'more', 'only', 'after', 'six']
he: ['on', 'was', 'it', 'eight', 'nine', 'two', 'is', 'its']
Training completed after 10 epochs - Final Average Loss: 0.1916
Created path: exp_2_setting_3_final_model
Final model saved at exp_2_setting_3_final_model/glove_final_model.pt
Embeddings saved at exp_2_setting_3_final_model/glove_embeddings.pkl
```



What is the best window_size for the GloVe model and why? What did you learn from the plots?

TODO(students): start

The best window_size is 10. because from the above experiments, as window_size increases, the avg total loss of glove model decreases. From plots, the avg loss of each epoch is decreasing. initially, since weight vectors are totally untrained, the avg loss is high in each experiment for the first epoch. And then, avg loss decreases slowly through each epoch. So, as epoch's increases, the avg loss decreases from the above plots.

TODO(students): end

```
## TODO(students): start  
  
best_window_size = 10  
  
## TODO(students): end
```

Experiment 3

For Experiment 3 in the GloVe model, we will set the best learning rate and window size from the previous experiments (Experiments 1 and 2). Now, we will vary the embedding size to explore how the dimensionality of the word vectors affects the quality of the embeddings.

Hyperparameter Setting 1 for Experiment 3

```
# TODO(student): start

#use best_lr & best_window_size (from previous experiments)
lr = best_lr
window_size = best_window_size

checkpoint_model_path = "exp_3_setting_1" # Path to save checkpoints
final_model_path = "exp_3_setting_1_final_model" # Path to save the
final model and embeddings

# Experiment 3: Vary the embedding size
# Try different embedding sizes like 50, 128, 300 to observe how it
affects the embeddings
embedding_size = 50

run_training(
    epochs=10, # Number of epochs to train
    lr=lr, # Best learning rate from Experiment 1
    checkpoint_model_path=checkpoint_model_path, # Directory to save
model checkpoints
    final_model_path=final_model_path, # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size (fixed)
    batch_size=512, # Batch size for training (fixed)
    embedding_size=embedding_size, # Vary this parameter for
different runs (e.g., 50, 128, 300)
    window_size=window_size, # Best window size from Experiment 2
    checkpoint_step=5000, # Save checkpoints after every 2000 steps
)

# TODO(student): end

Created path: exp_3_setting_1_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
```

Co-occurrence matrix generated.

Printing Validation Set Words

Word: name, ID: 97

Word: with, ID: 24

Word: years, ID: 86

Word: states, ID: 76

Word: their, ID: 45

Word: people, ID: 79

Word: when, ID: 67

Word: in, ID: 5

Word: three, ID: 14

Word: between, ID: 81

Word: some, ID: 50

Word: who, ID: 54

Word: or, ID: 29

Word: see, ID: 71

Word: an, ID: 31

Word: four, ID: 21

Device: cuda:0

Total Batches per Epoch: 11804

Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 11804/11804 [00:52<00:00, 222.88it/s,
Batch Loss=0.0647, Avg Loss=0.0606]

Epoch 1 completed - Average Loss: 0.1618

Checkpoint saved at step 11804 to

exp_3_setting_1_glove//glove_model_step_11804.pt

Printing closest words

name: ['themselves', 'precede', 'UNK', 'tuned', 'assented', 'UNK',
'mai', 'hemolitic']

with: ['have', 'for', 'six', 'UNK', 'is', 'UNK', 'from', 'national']

years: ['bonaduce', 'UNK', 'UNK', 'UNK', 'UNK', 'john', 'UNK', 'UNK']

states: ['miyazawa', 'then', 'convergence', 'dawkins',
'multidimensional', 'parma', 'strabo', 'lordly']

their: ['omnium', 'roxb', 'UNK', 'UNK', 'retaliated', 'UNK', 'bit',
'UNK']

people: ['UNK', 'algerie', 'valois', 'asexual', 'UNK', 'UNK', 'yury',
'centaurus']

when: ['undertakings', 'disjointed', 'saintcharlesalgiers', 'UNK',
'under', 'UNK', 'urs', 'UNK']

in: ['most', 'open', 's', 'UNK', 'UNK', 'semitism', 'national',
'auditory']

three: ['UNK', 'for', 'gunma', 'townspeople', 'is', 'two', 'UNK',
'frugal']

between: ['moved', 'UNK', 'rj', 'lines', 'UNK', 'deerfoot', 'lower',
'jure']

some: ['UNK', 'coachella', 'twinkies', 'symbiosis', 'UNK', 'basement',
'index', 'UNK']

who: ['meidani', 'squealer', 'musineon', 'religiously', 'eurasiatic',

'UNK', 'distinguishing', 'surges']
or: ['UNK', 'elitist', 'UNK', 'UNK', 'tecumseh', 'UNK', 'UNK', 'UNK']
see: ['retire', 'UNK', 'shakyamuni', 'hordes', 'these',
'codification', 'UNK', 'judoka']
an: ['war', 'this', 'travers', 'two', 'which', 'UNK', 'first', 'it']
four: ['UNK', 'UNK', 'theravada', 'verge', 'pointwise', 'UNK',
'aliat', 'UNK']
Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 11804/11804 [00:52<00:00, 223.29it/s,
Batch Loss=0.146, Avg Loss=0.0849]

Epoch 2 completed - Average Loss: 0.0798
Checkpoint saved at step 23608 to
exp_3_setting_1_glove//glove_model_step_23608.pt
Printing closest words
name: ['she', 'we', 'now', 'both', 'could', 'still', 'each', 'than']
with: ['UNK', 'for', 'galactics', 'scotty', 'UNK', 'UNK', 'praeger',
'reclusive']
years: ['more', 'then', 'only', 'people', 'her', 'general', 'had',
'the']
states: ['then', 'no', 'often', 'works', 'people', 'between',
'different', 'sometimes']
their: ['which', 'known', 'however', 'has', 'he', 'new', 'all', 'not']
people: ['between', 'about', 'years', 'often', 'no', 'city', 'her',
'only']
when: ['most', 'into', 'made', 'using', 'term', 'change', 'distinct',
'UNK']
in: ['to', 'many', 'was', 'and', 'of', 'world', 'zero', 'city']
three: ['victory', 'the', 'named', 'small', 'show', 'novel', 'long',
'any']
between: ['her', 'no', 'death', 'people', 'north', 'long', 'often',
'well']
some: ['such', 'been', 'had', 'more', 'only', 'can', 'also', 'large']
who: ['after', 'had', 'been', 'such', 'all', 'some', 'him', 'has']
or: ['to', 'called', 'was', 'he', 'been', 'nine', 'had', 'be']
see: ['these', 'been', 'called', 'all', 'also', 'more', 'only', 'has']
an: ['produce', 'UNK', 'UNK', 'are', 'control', 'UNK', 'UNK', 'woman']
four: ['all', 'which', 'at', 'eight', 'his', 'he', 'after', 'been']
Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 11804/11804 [00:52<00:00, 223.11it/s,
Batch Loss=0.0687, Avg Loss=0.0673]

Epoch 3 completed - Average Loss: 0.0697
Checkpoint saved at step 35412 to
exp_3_setting_1_glove//glove_model_step_35412.pt
Printing closest words
name: ['against', 'while', 'work', 'still', 'about', 'between', 'she',
'public']

with: ['for', 'is', 'as', 'from', 'are', 'six', 'an', 'two']
years: ['her', 'had', 'between', 'can', 'about', 'use', 'such', 'up']
states: ['then', 'called', 'often', 'because', 'since', 'this',
'only', 'case']
their: ['not', 'which', 'he', 'UNK', 'all', 'UNK', 'UNK', 'UNK']
people: ['between', 'while', 'about', 'these', 'such', 'way', 'up',
'food']
when: ['made', 'parryi', 'UNK', 'governor', 'ciconiiformes',
'deficit', 'UNK', 'must']
in: ['that', 'athens', 'medical', 'pygmy', 'afghanistan', 'stol',
'to', 'famous']
three: ['two', 'as', 'six', 'with', 'for', 'seven', 'five', 's']
between: ['people', 'about', 'up', 'use', 'while', 'her', 'no',
'these']
some: ['had', 'can', 'such', 'many', 'more', 'be', 'also', 'american']
who: ['after', 'its', 'world', 'this', 'him', 'for', 'from', 'with']
or: ['more', 'nine', 'four', 'had', 'now', 'UNK', 'which',
'tantamount']
see: ['also', 'more', 'some', 'rittson', 'been', 'UNK', 'UNK', 'UNK']
an: ['are', 'with', 'on', 'for', 'from', 'as', 'it', 'six']
four: ['nine', 'eight', 'which', 'all', 'also', 'zero', 'UNK', 'UNK']
Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 11804/11804 [00:53<00:00, 221.04it/s,
Batch Loss=0.0893, Avg Loss=0.0737]

Epoch 4 completed - Average Loss: 0.0672

Checkpoint saved at step 47216 to

exp_3_setting_1_glove//glove_model_step_47216.pt

Printing closest words

name: ['still', 'about', 'over', 'people', 'out', 'between', 'both',
'these']
with: ['for', 'are', 'from', 'is', 'UNK', 'two', 'snowcone', 'UNK']
years: ['returned', 'south', 'captured', 'estimated', 'italy',
'reported', 'UNK', 'persons']
states: ['published', 'found', 'life', 'may', 'd', 'called', 'new',
'works']
their: ['UNK', 'syrup', 'myles', 'regularly', 'perfectness', 'UNK',
'skeleton', 'mittelalters']
people: ['about', 'between', 'name', 'these', 'even', 'no', 'still',
'history']
when: ['most', 'has', 'hand', 'during', 'history', 'even', 'first',
'into']
in: ['zero', 'that', 'to', 'was', 'nine', 'on', 'his', 'were']
three: ['UNK', 'dhimmis', 'UNK', 'aldehydes', 'disprove', 'UNK',
'UNK', 'administers']
between: ['about', 'these', 'people', 'no', 'such', 'name', 'over',
'even']
some: ['been', 'such', 'had', 'has', 'these', 'which', 'also',
'there']

who: ['after', 'its', 'known', 'world', 'however', 'than', 'used', 'him']
or: ['his', 'not', 'which', 'was', 'has', 'been', 'also', 'eight']
see: ['there', 'which', 'all', 'these', 'four', 'also', 'history', 'both']
an: ['it', 'but', 'd', 'b', 'on', 'church', 'six', 's']
four: ['eight', 'which', 'his', 'also', 'nine', 'has', 'or', 'not']
Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 11804/11804 [00:53<00:00, 221.82it/s,
Batch Loss=0.0752, Avg Loss=0.0702]

Epoch 5 completed - Average Loss: 0.0641

Checkpoint saved at step 59020 to

exp_3_setting_1_glove//glove_model_step_59020.pt

Printing closest words

name: ['during', 'over', 'work', 'people', 'those', 'against', 'made', 'when']

with: ['an', 'for', 's', 'two', 'by', 'six', 'are', 'is']

years: ['while', 'them', 'named', 'later', 'south', 'much', 'more', 'only']

states: ['evidence', 'case', 'small', 'd', 'british', 'published', 'i', 'home']

their: ['this', 'at', 'he', 'three', 'for', 'seven', 'five', 'being']

people: ['name', 'over', 'history', 'work', 'there', 'up', 'during', 'against']

when: ['into', 'during', 'or', 'history', 'they', 'there', 'name', 'some']

in: ['is', 'an', 'other', 'a', 'or', 'the', 'one', 'for']

three: ['two', 'five', 'for', 'as', 'by', 'their', 'at', 'is']

between: ['process', 'names', 'evidence', 'any', 'case', 'standard', 'animals', 'theory']

some: ['into', 'had', 'there', 'when', 'he', 'see', 'so', 'or']

who: ['kska', 'whittier', 'it', 'turingismus', 'UNK', 'UNK', 'bao', 'UNK']

or: ['also', 'he', 'an', 'when', 'into', 'see', 'a', 's']

see: ['he', 'an', 'or', 'for', 'made', 'at', 'since', 'this']

an: ['it', 's', 'by', 'for', 'with', 'or', 'see', 'are']

four: ['b', 'six', 'eight', 'english', 'also', 'or', 'seven', 'UNK']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 11804/11804 [00:53<00:00, 221.53it/s,
Batch Loss=0.0519, Avg Loss=0.0716]

Epoch 6 completed - Average Loss: 0.0617

Checkpoint saved at step 70824 to

exp_3_setting_1_glove//glove_model_step_70824.pt

Printing closest words

name: ['into', 'when', 'during', 'operation', 'mainly', 'made', 'there', 'her']

with: ['have', 'was', 'he', 'they', 'as', 'other', 'eight', 'be']
years: ['later', 'them', 'while', 'than', 'if', 'much', 'because', 'great']
states: ['became', 'new', 'well', 'him', 'then', 'world', 'early', 'though']
their: ['m', 'when', 'five', 'at', 'is', 'there', 'so', 'made']
people: ['being', 'against', 'UNK', 'dunstan', 'example', 'indicative', 'house', 'despise']
when: ['into', 'during', 'there', 'so', 'their', 'time', 'important', 'made']
in: ['to', 'of', 'zero', 'was', 'that', 'be', 'on', 'nine']
three: ['five', 'seven', 'their', 'is', 'at', 'by', 'into', 'main']
between: ['about', 'general', 'west', 'current', 'original', 'main', 'life', 'type']
some: ['be', 'other', 'not', 'have', 'had', 'which', 'such', 'on']
who: ['had', 'other', 'before', 'war', 'no', 'some', 'which', 'even']
or: ['geometry', 'UNK', 'cowell', 'UNK', 'agincourt', 'shetterly', 'UNK', 'farley']
see: ['when', 'm', 'their', 'b', 'during', 'made', 'there', 'system']
an: ['its', 'by', 'from', 'their', 'seven', 'are', 'is', 'at']
four: ['film', 'e', 'before', 'all', 'he', 'one', 'had', 'word']
Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 11804/11804 [00:53<00:00, 220.57it/s, Batch Loss=0.0824, Avg Loss=0.0703]

Epoch 7 completed - Average Loss: 0.0600

Checkpoint saved at step 82628 to

exp_3_setting_1_glove//glove_model_step_82628.pt

Printing closest words

name: ['her', 'may', 'between', 'through', 'i', 'history', 'made', 'into']
with: ['as', 'that', 'for', 'are', 'by', 's', 'at', 'is']
years: ['early', 'much', 'c', 'states', 'no', 'year', 'seven', 'has']
states: ['state', 'early', 'war', 'c', 'world', 'after', 'him', 'known']
their: ['bauer', 'active', 'mcclellan', 'frequent', 'begin', 'sex', 'standing', 'featured']
people: ['has', 'it', 'not', 'or', 'that', 'first', 'no', 'being']
when: ['there', 'into', 'time', 'at', 'during', 'also', 'five', 'b']
in: ['to', 'was', 'rulership', 'of', 'americanism', 'on', 'other', 'through']
three: ['for', 'two', 'as', 'five', 'or', 'are', 's', 'it']
between: ['about', 'life', 'name', 'her', 'through', 'made', 'general', 'greek']
some: ['be', 'they', 'used', 'he', 'raison', 'had', 'waggoner', 'several']
who: ['that', 'not', 'all', 'could', 'no', 'states', 'also', 'years']
or: ['it', 'three', 'for', 'five', 'at', 'are', 'six', 's']
see: ['called', 'into', 's', 'often', 'for', 'many', 'five', 'six']

an: ['UNK', 'tocqueville', 'UNK', 'nemean', 'diepold', 'UNK', 'nw', 'archetypal']
four: ['seven', 'years', 'both', 'by', 'at', 'that', 'also', 'c']
Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 11804/11804 [00:53<00:00, 220.24it/s, Batch Loss=0.0516, Avg Loss=0.0707]

Epoch 8 completed - Average Loss: 0.0587

Checkpoint saved at step 94432 to

exp_3_setting_1_glove//glove_model_step_94432.pt

Printing closest words

name: ['may', 'i', 'both', 'ii', 'history', 'them', 'through', 'those']

with: ['and', 'as', 'two', 's', 'khair', 'flounders', 'belle', 'blackmail']

years: ['no', 'them', 'year', 'world', 'him', 'even', 'states', 'name']

states: ['people', 'him', 'history', 'world', 'them', 'then', 'may', 'war']

their: ['its', 'but', 'which', 'this', 'similar', 'have', 'use', 'under']

people: ['history', 'states', 'all', 'no', 'has', 'him', 'g', 'first']

when: ['being', 'up', 'there', 'time', 'at', 'however', 'also', 'have']

in: ['to', 'zero', 'nine', 'was', 'eight', 'of', 'seven', 'a']

three: ['five', 'is', 'or', 'six', 'such', 'dulfer', 'rrna', 's']

between: ['b', 'called', 'about', 'later', 'UNK', 'UNK', 'UNK', 'tranquilliser']

some: ['leeks', 'would', 'UNK', 'were', 'UNK', 'UNK', 'dengizik', 'sucinum']

who: ['would', 'over', 'example', 'just', 'though', 'early', 'christendom', 'while']

or: ['five', 'six', 'three', 's', 'strong', 'such', 'do', 'hungary']

see: ['there', 'time', 'at', 'many', 'these', 'so', 'seven', 'however']

an: ['which', 'this', 'he', 'was', 'from', 'be', 'have', 'his']

four: ['seven', 'at', 'eight', 'by', 'united', 'also', 'there', 'rorke']

Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 11804/11804 [00:53<00:00, 220.86it/s, Batch Loss=0.0979, Avg Loss=0.0706]

Epoch 9 completed - Average Loss: 0.0586

Checkpoint saved at step 106236 to

exp_3_setting_1_glove//glove_model_step_106236.pt

Printing closest words

name: ['her', 'still', 'both', 'those', 'then', 'reigned', 'does', 'written']

with: ['fetched', 'thrillers', 'samaritan', 'infidelity', 'listeria',
'UNK', 'vaishashik', 'electrolytic']
years: ['even', 'no', 'year', 'states', 'world', 'war', 'model',
'him']
states: ['even', 'no', 'years', 'set', 'fact', 'him', 'year', 'space']
their: ['he', 'its', 'at', 'UNK', 'UNK', 'UNK', 'UNK', 'apok']
people: ['can', 'all', 'may', 'i', 'history', 'not', 'about', 'some']
when: ['see', 'called', 'either', 'however', 'held', 'view',
'popular', 'cases']
in: ['to', 'a', 'of', 'would', 'nine', 'large', 'aiwass', 'god']
three: ['five', 'is', 'as', 'six', 'that', 'one', 'by', 'two']
between: ['them', 'made', 'called', 'b', 'see', 'shape', 'when',
'make']
some: ['who', 'used', 'it', 'all', 'not', 'i', 'same', 'they']
who: ['some', 'all', 'used', 'about', 'not', 'american', 'i', 'it']
or: ['jamaican', 'organised', 'courthouse', 'UNK', 'UNK', 'city',
'UNK', 'fitfully']
see: ['however', 'when', 'these', 'either', 'made', 'there', 'south',
'york']
an: ['on', 'his', 'was', 'two', 'are', 'by', 'is', 'that']
four: ['zero', 'his', 'two', 'one', 'three', 'by', 'five', 'external']
Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 11804/11804 [00:53<00:00, 221.37it/s,
Batch Loss=0.0758, Avg Loss=0.0706]

Epoch 10 completed - Average Loss: 0.0578

Checkpoint saved at step 118040 to

exp_3_setting_1_glove//glove_model_step_118040.pt

Printing closest words

name: ['still', 'then', 'those', 'often', 'them', 'both', 'having',
'same']

with: ['he', 'or', 'which', 'eight', 'also', 'such', 'but', 'seven']
years: ['him', 'them', 'later', 'states', 'called', 'often', 'world',
'when']

states: ['him', 'when', 'world', 'called', 'between', 'them', 'also',
'but']

their: ['its', 'kalvik', 'reforms', 'UNK', 'UNK', 'return', 'UNK',
'declarations']

people: ['all', 'well', 'i', 'sometimes', 'so', 'found', 'd',
'glaucescens']

when: ['called', 'between', 'states', 'them', 'often', 'later', 'see',
'but']

in: ['a', 'to', 's', 'they', 'for', 'with', 'zero', 'four']

three: ['city', 'six', 'during', 'first', 'article', 'nine', 'UNK',
'september']

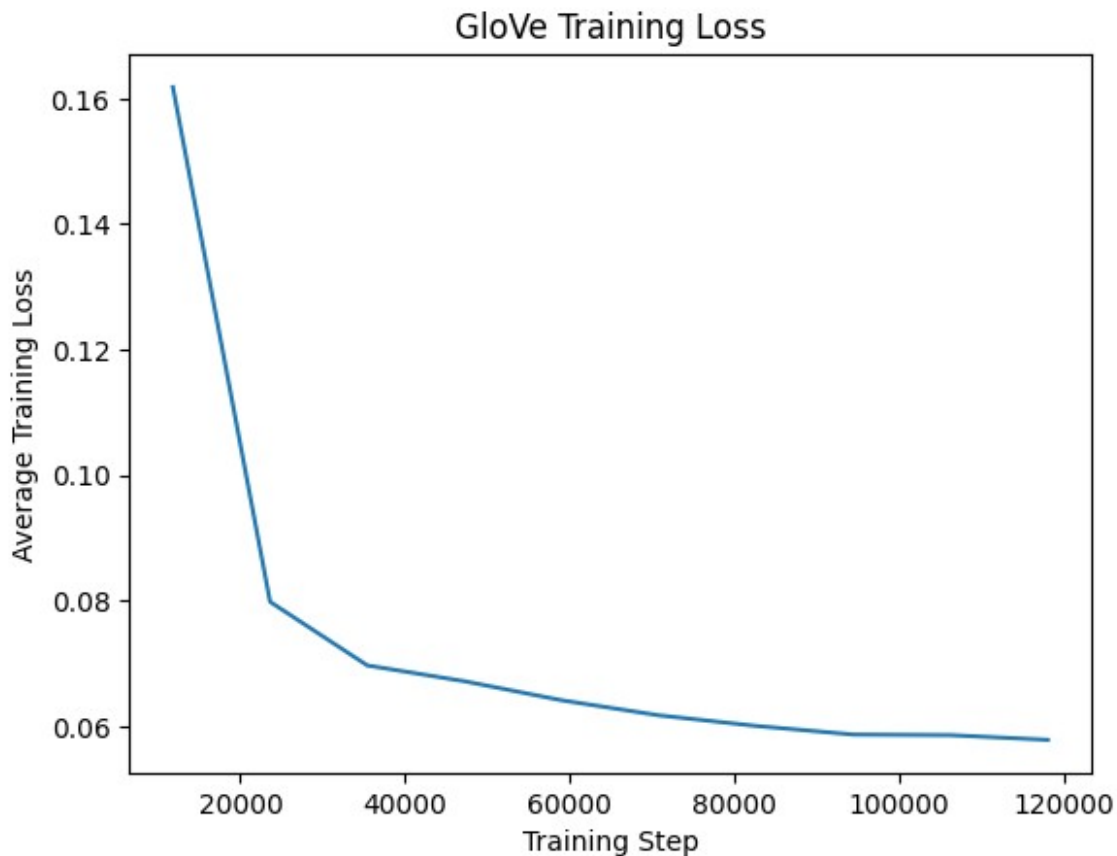
between: ['when', 'them', 'called', 'states', 'later', 'b', 'see',
'him']

some: ['UNK', 'pneuma', 'exhibiting', 'hurting', 'leading', 'besides',
'revolte', 'engage']


```

who: ['time', 'not', 'american', 'on', 'used', 'had', 'UNK', 'after']
or: ['which', 'such', 'be', 'with', 'other', 'also', 'but', 'seven']
see: ['when', 'later', 'b', 'between', 'which', 'seven', 'with',
      'called']
an: ['his', 'this', 'which', 'for', 'may', 's', 'four', 'from']
four: ['seven', 'zero', 'two', 'eight', 's', 'which', 'such', 'with']
Training completed after 10 epochs - Final Average Loss: 0.0578
Created path: exp_3_setting_1_final_model
Final model saved at exp_3_setting_1_final_model/glove_final_model.pt
Embeddings saved at exp_3_setting_1_final_model/glove_embeddings.pkl

```



Hyperparameter Setting 2 for Experiment 2

```

# TODO(student): start

#use best_lr & best_window_size (from previous experiments)
lr = best_lr
window_size = best_window_size

checkpoint_model_path = "exp_3_setting_2" # Path to save checkpoints
final_model_path = "exp_3_setting_2_final_model" # Path to save the
final model and embeddings

```

```

# Experiment 3: Vary the embedding size
# Try different embedding sizes like 50, 128, 300 to observe how it
affects the embeddings
embedding_size = 128

run_training(
    epochs=10, # Number of epochs to train
    lr=lr, # Best learning rate from Experiment 1
    checkpoint_model_path=checkpoint_model_path, # Directory to save
model checkpoints
    final_model_path=final_model_path, # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size (fixed)
    batch_size=512, # Batch size for training (fixed)
    embedding_size=embedding_size, # Vary this parameter for
different runs (e.g., 50, 128, 300)
    window_size=window_size, # Best window size from Experiment 2
    checkpoint_step=5000, # Save checkpoints after every 2000 steps
)

# TODO(student): end

```

```

Created path: exp_3_setting_2_glove/
Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789),
('and', 25238), ('one', 24679)]
Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]
['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first',
'used', 'against']
Generating co-occurrence matrix...
Building cooccurrence matrix: on line 0/1000000
Building cooccurrence matrix: on line 100000/1000000
Building cooccurrence matrix: on line 200000/1000000
Building cooccurrence matrix: on line 300000/1000000
Building cooccurrence matrix: on line 400000/1000000
Building cooccurrence matrix: on line 500000/1000000
Building cooccurrence matrix: on line 600000/1000000
Building cooccurrence matrix: on line 700000/1000000
Building cooccurrence matrix: on line 800000/1000000
Building cooccurrence matrix: on line 900000/1000000
Co-occurrence matrix generated.
Printing Validation Set Words
Word: was, ID: 15
Word: being, ID: 99
Word: into, ID: 62
Word: world, ID: 66
Word: has, ID: 41
Word: were, ID: 42
Word: united, ID: 85
Word: this, ID: 36
Word: so, ID: 93

```

Word: b, ID: 59
Word: it, ID: 30
Word: as, ID: 12
Word: no, ID: 80
Word: also, ID: 38
Word: during, ID: 94
Word: the, ID: 1
Device: cuda:0
Total Batches per Epoch: 11804
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 11804/11804 [01:02<00:00, 189.06it/s,
Batch Loss=0.134, Avg Loss=0.153]

Epoch 1 completed - Average Loss: 0.3330

Checkpoint saved at step 11804 to
exp_3_setting_2_glove/glove_model_step_11804.pt

Printing closest words

was: ['that', 'music', 'as', 's', 'of', 'their', 'braunau', 'often']
being: ['he', 'called', 'rather', 'commonly', 'UNK', 'music', 'blyth',
'UNK']
into: ['with', 'any', 'be', 'empire', 'ecclesial', 'is', 'english',
'for']
world: ['UNK', 'related', 'manga', 'iona', 'UNK', 'UNK', 'openstep',
'UNK']
has: ['different', 'though', 'zero', 'orients', 'war', 'UNK', 'UNK',
'alternate']
were: ['are', 'which', 'but', 'an', 'einstein', 'region', 'such',
'side']
united: ['cranmer', 'nassenes', 'UNK', 'cgpm', 'UNK', 'UNK', 'an',
'UNK']
this: ['some', 'may', 'number', 'result', 'messina', 'alternate',
'general', 'about']
so: ['auberon', 'UNK', 'called', 'states', 'rasis', 'west', 'number',
'fooled']
b: ['for', 'd', 'their', 'french', 'american', 'new', 'who',
'between']
it: ['UNK', 'between', 'this', 'like', 'he', 'war', 'not', 'body']
as: ['s', 'was', 'that', 'on', 'UNK', 'frost', 'UNK', 'UNK']
no: ['st', 'such', 'd', 'english', 'UNK', 'caregivers', 'drove',
'imitate']
also: ['new', 'film', 'both', 'UNK', 'their', 'war', 'tdd', 'is']
during: ['cohesion', 'system', 'UNK', 'often', 'above',
'justification', 'film', 'states']
the: ['based', 'UNK', 'uncomplimentary', 'UNK', 'unacceptably',
'messenger', 'vespers', 'sensor']

Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 11804/11804 [01:02<00:00, 188.32it/s,
Batch Loss=1.18, Avg Loss=0.633]

Epoch 2 completed - Average Loss: 0.2985
Checkpoint saved at step 23608 to
exp_3_setting_2_glove//glove_model_step_23608.pt
Printing closest words
was: ['on', 'their', 'in', 'influence', 'are', 'catholic', 'terms',
'anime']
being: ['any', 'this', 'century', 'take', 'made', 'about', 'greek',
'based']
into: ['which', 'with', 'first', 'the', 'all', 'five', 'over',
'united']
world: ['such', 'than', 'their', 'where', 'which', 'great', 'life',
'death']
has: ['be', 'most', 'been', 'war', 'not', 'other', 'however',
'against']
were: ['be', 's', 'not', 'government', 'on', 'however', 'are', 'was']
united: ['over', 'all', 'during', 'than', 'such', 'first', 'state',
'life']
this: ['some', 'any', 'that', 'being', 'have', 'may', 'word', 'had']
so: ['given', 'total', 'william', 'british', 'making', 'down',
'could', 'based']
b: ['some', 'general', 'american', 'that', 'i', 'this', 'five',
'first']
it: ['support', 'who', 'british', 'based', 'd', 'but', 'similar',
'current']
as: ['s', 'be', 'most', 'are', 'for', 'would', 'terms', 'been']
no: ['support', 'data', 'would', 'almost', 'states', 'actually', 'it',
'because']
also: ['including', 'years', 'in', 'their', 'since', 'such', 'north',
'how']
during: ['name', 'under', 'state', 'over', 'only', 'will', 'all',
'united']
the: ['five', 'such', 'which', 'two', 'first', 'than', 'all', 'world']
Starting Epoch 3/10

Epoch 3/10: 100%[██████████] 11804/11804 [01:02<00:00, 188.28it/s,
Batch Loss=0.352, Avg Loss=0.309]

Epoch 3 completed - Average Loss: 0.1830
Checkpoint saved at step 35412 to
exp_3_setting_2_glove//glove_model_step_35412.pt
Printing closest words
was: ['s', 'three', 'on', 'for', 'as', 'his', 'seven', 'or']
being: ['generally', 'current', 'this', 'physical', 'features',
'europe', 'followed', 'greek']
into: ['with', 'had', 'bishop', 'five', 'eight', 'john', 'way',
'meaning']
world: ['existence', 'could', 'referred', 'animals', 'almost',
'vinson', 'describing', 'move']
has: ['be', 'though', 'particular', 'successful', 'head', 'church',
'having', 'military']

were: ['but', 'its', 'states', 'american', 'after', 'other',
'through', 'only']
united: ['than', 'him', 'such', 'found', 'we', 'area', 'early',
'then']
this: ['have', 'some', 'history', 'use', 'all', 'john', 'art',
'particular']
so: ['both', 'could', 'given', 'making', 'many', 'example', 'almost',
'th']
b: ['d', 'father', 'actor', 'UNK', 'ad', 'synelcosciadium', 'howard',
'joseph']
it: ['any', 'at', 'or', 'example', 'his', 'between', 'far', 'england']
as: ['was', 's', 'for', 'three', 'on', 'his', 'to', 'or']
no: ['refer', 'kingdom', 'though', 'sound', 'character',
'programming', 'fire', 'related']
also: ['which', 'were', 'up', 'their', 'since', 'after', 'said',
'other']
during: ['had', 'although', 'system', 'used', 'including', 'usually',
'may', 'countries']
the: ['of', 'in', 'life', 'or', 'as', 'his', 's', 'it']
Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.12it/s,
Batch Loss=0.829, Avg Loss=0.52]

Epoch 4 completed - Average Loss: 0.2119

Checkpoint saved at step 47216 to

exp_3_setting_2_glove//glove_model_step_47216.pt

Printing closest words

was: ['by', 'for', 'three', 'with', 'from', 'will', 'which', 'six']

being: ['include', 'greek', 'order', 'having', 'sense', 'another',
'still', 'elements']

into: ['such', 'against', 'number', 'books', 'abortion', 'abraham',
'arabic', 'today']

world: ['states', 'most', 'would', 'english', 'first', 'while',
'also', 'base']

has: ['be', 'not', 'all', 'from', 'first', 'however', 'there', 'most']

were: ['their', 'its', 'are', 'nabla', 'on', 'breaking', 'was',
'second']

united: ['time', 'great', 'such', 'believe', 'long', 'abraham',
'rand', 'against']

this: ['all', 'has', 'have', 'be', 'six', 'an', 'had', 'there']

so: ['both', 'ancient', 'could', 'these', 'take', 'part', 'at',
'while']

b: ['UNK', 'UNK', 'stir', 'so', 'UNK', 'denoted', 'years', 'rowland']

it: ['itself', 'king', 'century', 'bc', 'claim', 'produced', 'belief',
'land']

as: ['three', 'he', 'for', 'even', 's', 'other', 'well', 'language']

no: ['christian', 'british', 'however', 'english', 'about', 'economy',
'people', 'technology']

also: ['which', 'most', 'first', 'other', 'from', 'but', 'been',

'states']
during: ['way', 'made', 'may', 'history', 'ii', 'later', 'then',
'year']
the: ['and', 'of', 'in', 'zero', 'by', 'is', 'several', 'place']
Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 11804/11804 [01:02<00:00, 187.40it/s,
Batch Loss=0.336, Avg Loss=0.378]

Epoch 5 completed - Average Loss: 0.1858

Checkpoint saved at step 59020 to

exp_3_setting_2_glove//glove_model_step_59020.pt

Printing closest words

was: ['by', 'three', 'as', 'with', 'from', 'in', 'eight', 'but']

being: ['often', 'work', 'where', 'him', 'use', 'under', 'number',
'given']

into: ['time', 'only', 'some', 'more', 'such', 'have', 'they', 'who']

world: ['english', 'space', 'england', 'external', 'year', 'UNK',
'UNK', 'state']

has: ['be', 'an', 'this', 'have', 'can', 'no', 'because', 'however']

were: ['been', 'its', 'them', 'has', 'over', 'all', 'even', 'may']

united: ['time', 'being', 'number', 'into', 'play', 'above', 'act',
'government']

this: ['have', 'an', 'be', 'on', 'who', 'has', 'can', 'are']

so: ['before', 'full', 'both', 'himself', 'came', 'thought', 'with',
'evidence']

b: ['actor', 'u', 'won', 'both', 'eight', 'states', 'followed', 'so']

it: ['they', 'western', 'more', 'where', 'on', 'individual', 'term',
'being']

as: ['by', 'was', 'with', 'three', 'in', 'from', 'for', 'according']

no: ['be', 'have', 'this', 'has', 'an', 'because', 'however', 'art']

also: ['other', 'which', 'but', 'first', 'east', 'she', 'several',
'using']

during: ['may', 'all', 'part', 'over', 'UNK', 'war', 'later',
'called']

the: ['of', 'and', 'a', 'four', 'on', 'seven', 's', 'are']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 11804/11804 [01:02<00:00, 187.62it/s,
Batch Loss=0.86, Avg Loss=0.495]

Epoch 6 completed - Average Loss: 0.1949

Checkpoint saved at step 70824 to

exp_3_setting_2_glove//glove_model_step_70824.pt

Printing closest words

was: ['by', 'he', 'three', 'as', 'were', 'from', 'their', 'without']

being: ['them', 'however', 'more', 'north', 'became', 'new', 'called',
'between']

into: ['time', 'its', 'they', 'i', 'that', 'when', 'number', 'often']

world: ['he', 'not', 'after', 'were', 'which', 'i', 'its', 'people']

has: ['no', 'been', 'more', 'however', 'can', 'an', 'them', 'her']
were: ['he', 'their', 'at', 'which', 'from', 'not', 'its', 'i']
united: ['often', 'time', 'made', 'under', 'century', 'him', 'into', 'early']
this: ['that', 'be', 'its', 'they', 'who', 'into', 'were', 'by']
so: ['any', 'found', 'which', 'he', 'from', 'now', 'later', 'during']
b: ['both', 'ii', 'received', 'off', 'even', 'using', 'politician', 'over']
it: ['seven', 'distilling', 'an', 'tengmalm', 'unallocated', 'UNK', 'zampino', 'steerable']
as: ['that', 'was', 'by', 'he', 'three', 'from', 'at', 'were']
no: ['has', 'been', 'however', 'her', 'north', 'them', 'more', 'church']
also: ['which', 'their', 'history', 'he', 'not', 'were', 'from', 'five']
during: ['later', 'made', 'against', 'after', 'god', 'like', 'century', 'general']
the: ['and', 'with', 'its', 'not', 'culture', 'their', 'that', 'he']
Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.32it/s, Batch Loss=0.392, Avg Loss=0.416]

Epoch 7 completed - Average Loss: 0.1798

Checkpoint saved at step 82628 to

exp_3_setting_2_glove//glove_model_step_82628.pt

Printing closest words

was: ['eight', 'in', 'found', 'philosophical', 'until', 'assyria', 'corea', 'ference']

being: ['over', 'where', 'even', 'just', 'north', 'example', 'what', 'only']

into: ['i', 'often', 'its', 'number', 'like', 'made', 'about', 'university']

world: ['after', 'first', 'many', 'known', 'states', 'there', 'people', 'century']

has: ['it', 'an', 'other', 'been', 'have', 'this', 'there', 'but']

were: ['he', 'be', 'i', 'into', 'five', 'about', 'its', 'made']

united: ['states', 'second', 'old', 'her', 'president', 'area', 'civil', 'whose']

this: ['s', 'at', 'it', 'have', 'on', 'with', 'an', 'to']

so: ['many', 'out', 'then', 'world', 'still', 'history', 'century', 'known']

b: ['m', 'seven', 'v', 'actor', 'year', 'english', 'actress', 'th']

it: ['an', 'on', 'have', 'four', 'this', 'at', 's', 'but']

as: ['that', 'wikis', 'of', 'repentance', 't', 'UNK', 'ninth', 'dudinka']

no: ['just', 'being', 'over', 'only', 'even', 'father', 'example', 'different']

also: ['glory', 'sabb', 'UNK', 'already', 'UNK', 'metallurgical', 'UNK', 'UNK']

during: ['up', 'made', 'often', 'early', 'against', 'now', 'may', 'british']
the: ['of', 'and', 'to', 'zero', 'nine', 'four', 'a', 'two']
Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 11804/11804 [01:02<00:00, 187.71it/s, Batch Loss=0.794, Avg Loss=0.491]

Epoch 8 completed - Average Loss: 0.1895

Checkpoint saved at step 94432 to

exp_3_setting_2_glove//glove_model_step_94432.pt

Printing closest words

was: ['by', 'three', 'as', 'eight', 'that', 'also', 'be', 'five']

being: ['only', 'over', 'where', 'even', 'them', 'just', 'no', 'may']

into: ['about', 'england', 'make', 'school', 'production', 'took', 'under', 'high']

world: ['known', 'some', 'had', 'all', 'been', 'more', 'however', 'name']

has: ['there', 'many', 'their', 'most', 'first', 'other', 'way', 'these']

were: ['he', 'are', 'it', 'from', 'his', 'by', 'more', 's']

united: ['states', 'all', 'world', 'm', 'american', 'some', 'used', 'd']

this: ['s', 'six', 'an', 'it', 'for', 'seven', 'which', 'have']

so: ['known', 'then', 'each', 'do', 'those', 'own', 'world', 'century']

b: ['there', 'him', 'most', 'has', 'way', 'first', 'such', 'x']

it: ['s', 'his', 'more', 'have', 'are', 'had', 'some', 'world']

as: ['was', 'by', 'be', 'that', 'eight', 'in', 'three', 'and']

no: ['only', 'even', 'part', 'where', 'being', 'over', 'various', 'another']

also: ['been', 'that', 'had', 'more', 'three', 'be', 'was', 'can']

during: ['up', 'first', 'way', 'support', 'others', 'local', 'major', 'novel']

the: ['and', 'in', 'from', 'to', 'by', 'of', 'be', 'eight']

Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.28it/s, Batch Loss=0.423, Avg Loss=0.441]

Epoch 9 completed - Average Loss: 0.1831

Checkpoint saved at step 106236 to

exp_3_setting_2_glove//glove_model_step_106236.pt

Printing closest words

was: ['two', 'to', 'had', 'years', 'of', 'that', 'for', 'four']

being: ['UNK', 'animations', 'over', 'UNK', 'parents', 'africa', 'UNK', 'easter']

into: ['i', 'their', 'most', 'about', 'see', 'if', 'were', 'during']

world: ['states', 'some', 'more', 'early', 'much', 'same', 'century', 'when']

has: ['not', 'their', 'most', 'other', 'first', 'on', 'he', 'at']
were: ['not', 'other', 'than', 'i', 'which', 'on', 'their', 'had']
united: ['word', 'more', 'so', 'french', 'large', 'government',
'often', 'some']
this: ['UNK', 'UNK', 'guggenheim', 'bloodier', 'kingdom', 'UNK', 'in',
'wok']
so: ['more', 'modern', 'some', 'do', 'much', 'century', 'another',
'government']
b: ['nobel', 'actor', 'nine', 'seven', 'immensely', 'allende',
'physicist', 'two']
it: ['from', 'are', 'december', 'UNK', 'indiscriminately', 'russia',
'lens', 'amingdola']
as: ['a', 'nine', 'to', 'two', 'four', 'zero', 'or', 'an']
no: ['some', 'varlam', 'daoud', 'after', 'people', 'more', 'browns',
'UNK']
also: ['five', 'be', 'while', 'between', 'children', 'after', 'based',
'century']
during: ['up', 'later', 'major', 'country', 'man', 'if', 'i', 'than']
the: ['one', 'of', 'he', 'six', 'two', 'four', 'to', 'is']
Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 11804/11804 [01:03<00:00, 187.03it/s,
Batch Loss=0.812, Avg Loss=0.491]

Epoch 10 completed - Average Loss: 0.1910

Checkpoint saved at step 118040 to

exp_3_setting_2_glove//glove_model_step_118040.pt

Printing closest words

was: ['three', 'eight', 'as', 'by', 'in', 'it', 's', 'and']

being: ['out', 'built', 'l', 'them', 'through', 'earth', 'region',
'using']

into: ['about', 'th', 'most', 'during', 'however', 'over', 'has',
'times']

world: ['states', 'known', 'had', 'music', 'another', 'been', 'all',
'before']

has: ['most', 'first', 'which', 'during', 's', 'into', 'if', 'th']

were: ['had', 'all', 'its', 'their', 'been', 'not', 'see', 'world']

united: ['french', 'while', 'early', 'time', 'when', 'film', 'will',
'more']

this: ['from', 'it', 's', 'and', 'has', 'but', 'b', 'his']

so: ['century', 'people', 'jewish', 'end', 'based', 'modern',
'together', 'believe']

b: ['such', 'many', 'there', 'also', 'it', 'however', 'back', 'could']

it: ['be', 'also', 'as', 'by', 's', 'was', 'but', 'eight']

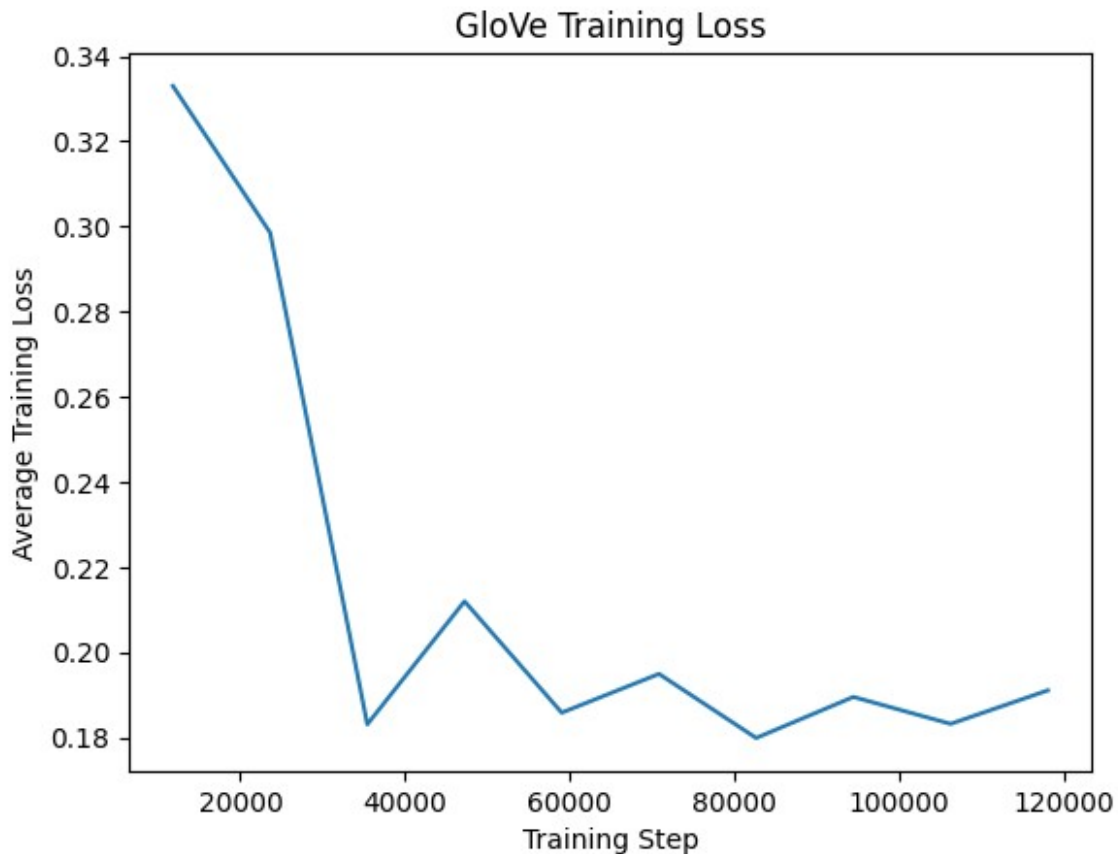
as: ['by', 'was', 'eight', 'three', 'it', 's', 'in', 'and']

no: ['these', 'when', 'such', 'also', 'often', 'there', 'very',
'while']

also: ['but', 'be', 'it', 'such', 'there', 'they', 'after', 'used']

during: ['later', 'up', 'if', 'great', 'th', 'major', 'however',
'most']

```
the: ['and', 'in', 's', 'a', 'eight', 'it', 'three', 'by']
Training completed after 10 epochs - Final Average Loss: 0.1910
Created path: exp_3_setting_2_final_model
Final model saved at exp_3_setting_2_final_model/glove_final_model.pt
Embeddings saved at exp_3_setting_2_final_model/glove_embeddings.pkl
```



Hyperparameter Setting 3 for Experiment 3

```
# TODO(student): start

#use best_lr & best_window_size (from previous experiments)
lr = best_lr
window_size = best_window_size

checkpoint_model_path = "exp_3_setting_3" # Path to save checkpoints
final_model_path = "exp_3_setting_3_final_model" # Path to save the
final model and embeddings

# Experiment 3: Vary the embedding size
# Try different embedding sizes like 50, 128, 300 to observe how it
affects the embeddings
embedding_size = 300
```

```

run_training(
    epochs=10, # Number of epochs to train
    lr=lr, # Best learning rate from Experiment 1
    checkpoint_model_path=checkpoint_model_path, # Directory to save
model checkpoints
    final_model_path=final_model_path, # Directory to save the final
model and embeddings
    vocab_size=int(1e5), # Vocabulary size (fixed)
    batch_size=512, # Batch size for training (fixed)
    embedding_size=embedding_size, # Vary this parameter for
different runs (e.g., 50, 128, 300)
    window_size=window_size, # Best window size from Experiment 2
    checkpoint_step=5000, # Save checkpoints after every 2000 steps
)

# TODO(student): end

```

Created path: exp_3_setting_3_glove/

Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789), ('and', 25238), ('one', 24679)]

Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]

['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first', 'used', 'against']

Generating co-occurrence matrix...

Building cooccurrence matrix: on line 0/1000000

Building cooccurrence matrix: on line 100000/1000000

Building cooccurrence matrix: on line 200000/1000000

Building cooccurrence matrix: on line 300000/1000000

Building cooccurrence matrix: on line 400000/1000000

Building cooccurrence matrix: on line 500000/1000000

Building cooccurrence matrix: on line 600000/1000000

Building cooccurrence matrix: on line 700000/1000000

Building cooccurrence matrix: on line 800000/1000000

Building cooccurrence matrix: on line 900000/1000000

Co-occurrence matrix generated.

Printing Validation Set Words

Word: four, ID: 21

Word: and, ID: 3

Word: when, ID: 67

Word: or, ID: 29

Word: two, ID: 11

Word: eight, ID: 13

Word: th, ID: 91

Word: its, ID: 46

Word: in, ID: 5

Word: all, ID: 51

Word: only, ID: 74

Word: seven, ID: 23

Word: also, ID: 38

Word: can, ID: 64

Word: of, ID: 2
Word: there, ID: 61
Device: cuda:0
Total Batches per Epoch: 11804
Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 11804/11804 [01:19<00:00, 147.73it/s,
Batch Loss=0.335, Avg Loss=0.444]

Epoch 1 completed - Average Loss: 0.8164

Checkpoint saved at step 11804 to
exp_3_setting_3_glove//glove_model_step_11804.pt

Printing closest words

four: ['was', 'had', 'six', 'war', 'legal', 'two', 'abledata', 'work']
and: ['zero', 'a', 'poles', 'three', 'under', 'were', 'or',
'prentice']

when: ['people', 'UNK', 'free', 'as', 'zero', 'such', 'movement',
'conocophillips']

or: ['zero', 'as', 'cannes', 'to', 'committees', 'and', 'UNK', 'UNK']

two: ['three', 'this', 'national', 'her', 'part', 'much', 'eight',
'australia']

eight: ['since', 'less', 'two', 'like', 'ii', 'UNK', 'this',
'australia']

th: ['may', 'father', 'made', 'will', 'this', 'its', 'also', 'which']

its: ['people', 'single', 'used', 'other', 'will', 'since', 'been',
'never']

in: ['to', 'zero', 'two', 'three', 's', 'as', 'an', 'for']

all: ['great', 'he', 'were', 'it', 'live', 'down', 'later', 'two']

only: ['could', 'do', 'under', 'time', 'both', 'being', 'although',
'present']

seven: ['an', 'their', 'has', 'two', 'he', 'national', 'six', 'rull']

also: ['such', 'early', 'main', 'his', 'recently', 'by', 'gentian',
'men']

can: ['UNK', 'UNK', 'raousset', 'chastity', 'encephalitis', 'UNK',
'had', 'noma']

of: ['folks', 'art', 'scribners', 'UNK', 'UNK', 'all', 'UNK',
'tempered']

there: ['his', 'it', 'england', 'not', 'you', 'also', 'where', 'most']

Starting Epoch 2/10

Epoch 2/10: 100%|██████████| 11804/11804 [01:20<00:00, 147.04it/s,
Batch Loss=8.15, Avg Loss=4.28]

Epoch 2 completed - Average Loss: 1.8515

Checkpoint saved at step 23608 to
exp_3_setting_3_glove//glove_model_step_23608.pt

Printing closest words

four: ['five', 'as', 'state', 'used', 'for', 'at', 'with', 'who']

and: ['to', 'in', 'of', 'at', 'while', 'five', 'found', 'time']

when: ['i', 'states', 'known', 'first', 'there', 'for', 'have', 'be']

or: ['from', 'for', 'at', 'with', 'be', 'have', 'are', 'five']
two: ['by', 'was', 'three', 'stressful', 'UNK', 'priests', 'UNK',
'UNK']
eight: ['knights', 'climate', 'vendsyssel', 'UNK', 'psychoanal',
'ein', 'military', 'duncan']
th: ['national', 'use', 'important', 'union', 'very', 'australia',
'city', 'generally']
its: ['have', 'first', 'be', 'who', 'more', 'or', 'can', 'from']
in: ['to', 'and', 'as', 'while', 'time', 'at', 'number', 'best']
all: ['it', 'time', 'some', 'while', 'most', 'had', 'using', 'early']
only: ['i', 'into', 'general', 'great', 'much', 'at', 'known',
'history']
seven: ['is', 'three', 'maric', 'UNK', 'majority', 'UNK', 'zero',
'UNK']
also: ['were', 'out', 'other', 'this', 'that', 'these', 'then', 'her']
can: ['however', 'many', 'world', 'even', 'up', 'have', 'there',
'them']
of: ['and', 'to', 'in', 'this', 'one', 'sentiment', 'while',
'phyllotaxis']
there: ['first', 'after', 'they', 'who', 'can', 'known', 'into', 'i']
Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 11804/11804 [01:19<00:00, 147.56it/s,
Batch Loss=1.48, Avg Loss=1.48]

Epoch 3 completed - Average Loss: 0.7474

Checkpoint saved at step 35412 to

exp_3_setting_3_glove//glove_model_step_35412.pt

Printing closest words

four: ['six', 's', 'seven', 'eight', 'an', 'also', 'that', 'not']

and: ['in', 'one', 'to', 'a', 'of', 'is', 'as', 'three']

when: ['there', 'after', 'used', 'being', 'will', 'such', 'these',
'between']

or: ['zero', 'to', 'in', 'was', 'a', 'are', 'by', 'his']

two: ['was', 'by', 'three', 'eight', 's', 'nine', 'zero', 'that']

eight: ['s', 'by', 'was', 'two', 'three', 'six', 'that', 'seven']

th: ['then', 'east', 'national', 'later', 'way', 'within', 'theory',
'alexander']

its: ['first', 'be', 'other', 'not', 'also', 'six', 'their', 'who']

in: ['a', 'to', 'one', 'and', 'is', 'zero', 'as', 'was']

all: ['most', 'than', 'where', 'up', 'then', 'not', 'into', 'however']

only: ['history', 'there', 'will', 'much', 'against', 'being', 'less',
'when']

seven: ['s', 'eight', 'by', 'six', 'three', 'two', 'was', 'that']

also: ['other', 'not', 'six', 's', 'be', 'four', 'an', 'its']

can: ['many', 'states', 'first', 'world', 'not', 'should', 'great',
's']

of: ['and', 'in', 'as', 'one', 'to', 'is', 'a', 'for']

there: ['after', 'when', 'history', 'only', 'will', 'used', 'they',

'against']

Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 11804/11804 [01:20<00:00, 146.64it/s,
Batch Loss=4.01, Avg Loss=3.43]

Epoch 4 completed - Average Loss: 1.1103

Checkpoint saved at step 47216 to

exp_3_setting_3_glove//glove_model_step_47216.pt

Printing closest words

four: ['with', 'for', 'on', 'he', 'from', 'five', 'years', 'or']

and: ['the', 'of', 'is', 'one', 'that', 'he', 'UNK', 'nyse']

when: ['such', 'follower', 'these', 'been', 'united', 'year',
'genius', 'esnault']

or: ['on', 'from', 'five', 'with', 'for', 'they', 'all', 'he']

two: ['nine', 'three', 'was', 'to', 'outside', 'largely', 'kirsch',
'six']

eight: ['a', 'nine', 'it', 'four', 'for', 'he', 'th', 'last']

th: ['years', 'very', 'roman', 'year', 'united', 'west', 'he',
'party']

its: ['not', 'an', 'many', 'first', 'hinge', 'UNK', 'exchanging',
'pedestal']

in: ['to', 'of', 'three', 'as', 'an', 'a', 'obolus', 'solf']

all: ['most', 'from', 'than', 'no', 'other', 'also', 'with', 'about']

only: ['history', 'would', 'being', 'use', 'there', 'while',
'against', 'often']

seven: ['kbrj', 'UNK', 'disappointment', 'UNK', 'UNK', 'UNK', 'nine',
'UNK']

also: ['be', 'from', 'other', 'all', 'most', 'time', 'five', 'with']

can: ['all', 'no', 'other', 'most', 'than', 'also', 'term', 'time']

of: ['the', 'and', 'that', 'five', 'for', 'a', 'with', 'years']

there: ['only', 'after', 'they', 'used', 'known', 'being', 'would',
'history']

Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 11804/11804 [01:20<00:00, 146.59it/s,
Batch Loss=2.15, Avg Loss=1.98]

Epoch 5 completed - Average Loss: 0.8227

Checkpoint saved at step 59020 to

exp_3_setting_3_glove//glove_model_step_59020.pt

Printing closest words

four: ['with', 'their', 'be', 'are', 'before', 'all', 'other', 'said']

and: ['one', 'in', 'to', 'of', 'the', 'a', 'zero', 'as']

when: ['british', 'these', 's', 'will', 'world', 'early', 'seven',
'although']

or: ['it', 'but', 'be', 'cultural', 'were', 'other', 'where', 'there']

two: ['was', 'seven', 's', 'by', 'as', 'to', 'eight', 'for']

eight: ['two', 'by', 'as', 'was', 's', 'to', 'seven', 'for']

th: ['then', 'states', 'later', 'about', 'before', 'all', 'still',

```
'south']
its: ['not', 'was', 's', 'seven', 'two', 'as', 'by', 'more']
in: ['to', 'and', 'zero', 'one', 'as', 'the', 'two', 'eight']
all: ['there', 'about', 'states', 'their', 'before', 'since', 'then',
'very']
only: ['no', 'may', 'considered', 'each', 'her', 'out', 'they',
'than']
seven: ['was', 's', 'two', 'by', 'as', 'not', 'at', 'for']
also: ['have', 'from', 'than', 'no', 'they', 'at', 'seven', 's']
can: ['most', 'also', 'no', 'than', 'state', 'over', 'because',
'both']
of: ['the', 'and', 'to', 'one', 'in', 'a', 'zero', 'is']
there: ['all', 'their', 'before', 'states', 'it', 'used', 'but',
'about']
```

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 11804/11804 [01:20<00:00, 145.74it/s,
Batch Loss=4.13, Avg Loss=3.22]

Epoch 6 completed - Average Loss: 1.0126

Checkpoint saved at step 70824 to

exp_3_setting_3_glove//glove_model_step_70824.pt

Printing closest words

```
four: ['that', 'five', 'or', 'are', 'it', 'his', 'this', 'three']
and: ['of', 'the', 'nine', 'four', 'that', 'on', 'three', 'this']
when: ['at', 'period', 'since', 'adelaide', 'alaska', 'even', 'given',
'moon']
or: ['that', 'are', 'it', 'four', 'five', 'an', 'this', 'his']
two: ['s', 'UNK', 'UNK', 'UNK', 'plausible', 'UNK', 'ane', 'UNK']
eight: ['on', 'four', 'to', 'five', 'six', 'by', 'an', 'three']
th: ['then', 'australia', 'called', 'him', 'since', 'up', 'south',
'books']
its: ['not', 'this', 'who', 'has', 'many', 'been', 'first', 'an']
in: ['was', 'for', 'zero', 'as', 'at', 'release', 'from', 'also']
all: ['were', 'states', 'it', 'that', 'are', 'which', 'or', 'did']
only: ['name', 'often', 'can', 'alexander', 'some', 'new', 'they',
'she']
seven: ['six', 'an', 'were', 'UNK', 'a', 'UNK', 'game', 'extremists']
also: ['can', 'some', 'they', 'new', 'time', 'considered', 'was',
'force']
can: ['also', 'new', 'they', 'some', 'name', 'often', 'own', 'only']
of: ['the', 'and', 'nine', 'a', 'is', 'on', 'to', 'that']
there: ['see', 'other', 'first', 'used', 'states', 'later', 'has',
'were']
```

Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 11804/11804 [01:20<00:00, 146.43it/s,
Batch Loss=2.86, Avg Loss=2.3]

Epoch 7 completed - Average Loss: 0.8464

Checkpoint saved at step 82628 to

exp_3_setting_3_glove//glove_model_step_82628.pt

Printing closest words

four: ['an', 'not', 'many', 'other', 'more', 'may', 'both', 'during']

and: ['the', 'one', 'of', 'a', 'in', 'to', 'that', 'eight']

when: ['divisive', 'esher', 'uniting', 'converging', 'UNK', 'duckworth', 'suppletive', 'alsace']

or: ['intermolecular', 'ivanisevic', 'burdensome', 'ron', 'unrecognised', 'UNK', 'UNK', 'esa']

two: ['s', 'eight', 'was', 'as', 'seven', 'for', 'with', 'by']

eight: ['seven', 'by', 's', 'as', 'three', 'two', 'with', 'are']

th: ['where', 'century', 'years', 'until', 'united', 'states', 'well', 'book']

its: ['been', 'catholic', 'most', 'pinnacle', 'were', 'herb', 'often', 'published']

in: ['the', 'zero', 'as', 'and', 'for', 'of', 'eight', 'two']

all: ['seven', 'eight', 's', 'has', 'had', 'two', 'but', 'by']

only: ['war', 'known', 'than', 'often', 'about', 'new', 'can', 'name']

seven: ['s', 'by', 'be', 'eight', 'with', 'are', 'for', 'as']

also: ['they', 'any', 'between', 'alexander', 'more', 'may', 'church', 'an']

can: ['known', 'these', 'name', 'several', 'but', 'often', 'new', 'about']

of: ['the', 'to', 'and', 'a', 'in', 'eight', 'with', 'by']

there: ['has', 'states', 'but', 'see', 'would', 'world', 'even', 'had']

Starting Epoch 8/10

Epoch 8/10: 100%|██████████| 11804/11804 [01:20<00:00, 146.51it/s, Batch Loss=5.27, Avg Loss=3.18]

Epoch 8 completed - Average Loss: 0.9979

Checkpoint saved at step 94432 to

exp_3_setting_3_glove//glove_model_step_94432.pt

Printing closest words

four: ['on', 'five', 'his', 'or', 'an', 'which', 'zero', 'not']

and: ['the', 'of', 'a', 'to', 'with', 'six', 'this', 'have']

when: ['copies', 'tretow', 'lobes', 'UNK', 'paleolithic', 'UNK', 'aias', 'wetumpka']

or: ['on', 'his', 'which', 'an', 'five', 'four', 'zero', 'this']

two: ['one', 'and', 'veer', 'nine', 'for', 'shyness', 'UNK', 'paucifolia']

eight: ['by', 'are', 'in', 'people', 'general', 'who', 'mostly', 'waves']

th: ['through', 'name', 'day', 'states', 'well', 'year', 'century', 'much']

its: ['been', 'also', 'which', 'have', 'they', 'were', 'or', 'but']

in: ['by', 'are', 'eight', 'has', 'law', 'member', 'all', 'result']

all: ['are', 'has', 'in', 'arthurccclarke', 'those', 'called', 'there',


```
'balto']
only: ['than', 'new', 'however', 'after', 'but', 'have', 'war',
'been']
seven: ['six', 'on', 'his', 'five', 'which', 'have', 'with', 'this']
also: ['they', 'which', 'or', 'have', 'an', 'on', 'not', 'into']
can: ['have', 'some', 'this', 'than', 'which', 'been', 'new', 'its']
of: ['the', 'and', 'a', 'to', 'four', 'five', 'with', 'on']
there: ['protozoa', 'real', 'these', 'see', 'hayley', 'united',
'those', 'denoting']
Starting Epoch 9/10
```

```
Epoch 9/10: 100%|██████████| 11804/11804 [01:20<00:00, 146.02it/s,
Batch Loss=2.72, Avg Loss=2.5]
```

Epoch 9 completed - Average Loss: 0.8796

Checkpoint saved at step 106236 to

exp_3_setting_3_glove//glove_model_step_106236.pt

Printing closest words

```
four: ['also', 'on', 'the', 'other', 'it', 'by', 'that', 'were']
and: ['the', 'of', 'a', 'to', 'in', 'was', 'from', 'on']
when: ['more', 'first', 'there', 'most', 'known', 'that', 'by', 'all']
or: ['are', 'when', 'norfolk', 'empire', 'arrhenius', 'by',
'hypothesized', 'rope']
two: ['as', 'by', 'was', 'eight', 'that', 'are', 'it', 'from']
eight: ['by', 'that', 'was', 'in', 'from', 'are', 'it', 'as']
th: ['day', 'what', 'century', 'art', 'often', 'series', 'apollo',
'atheism']
its: ['an', 'swallowing', 'have', 'this', 'degradation',
'desalination', 'UNK', 'alone']
in: ['eight', 'the', 'by', 'was', 'that', 'of', 'are', 'and']
all: ['first', 'most', 'were', 'such', 'it', 'other', 'time', 'who']
only: ['war', 'than', 'were', 'about', 'if', 'many', 'considered',
'found']
seven: ['with', 'have', 'six', 'an', 'their', 'what', 'stimulations',
'this']
also: ['other', 'that', 'it', 'not', 'were', 'they', 'by', 'from']
can: ['only', 'were', 'some', 'many', 'they', 'under', 'which',
'than']
of: ['the', 'and', 'a', 'to', 'in', 'from', 'was', 'eight']
there: ['these', 'more', 'first', 'other', 'when', 'though', 'many',
'that']
```

Starting Epoch 10/10

```
Epoch 10/10: 100%|██████████| 11804/11804 [01:20<00:00, 146.36it/s,
Batch Loss=4.36, Avg Loss=3.19]
```

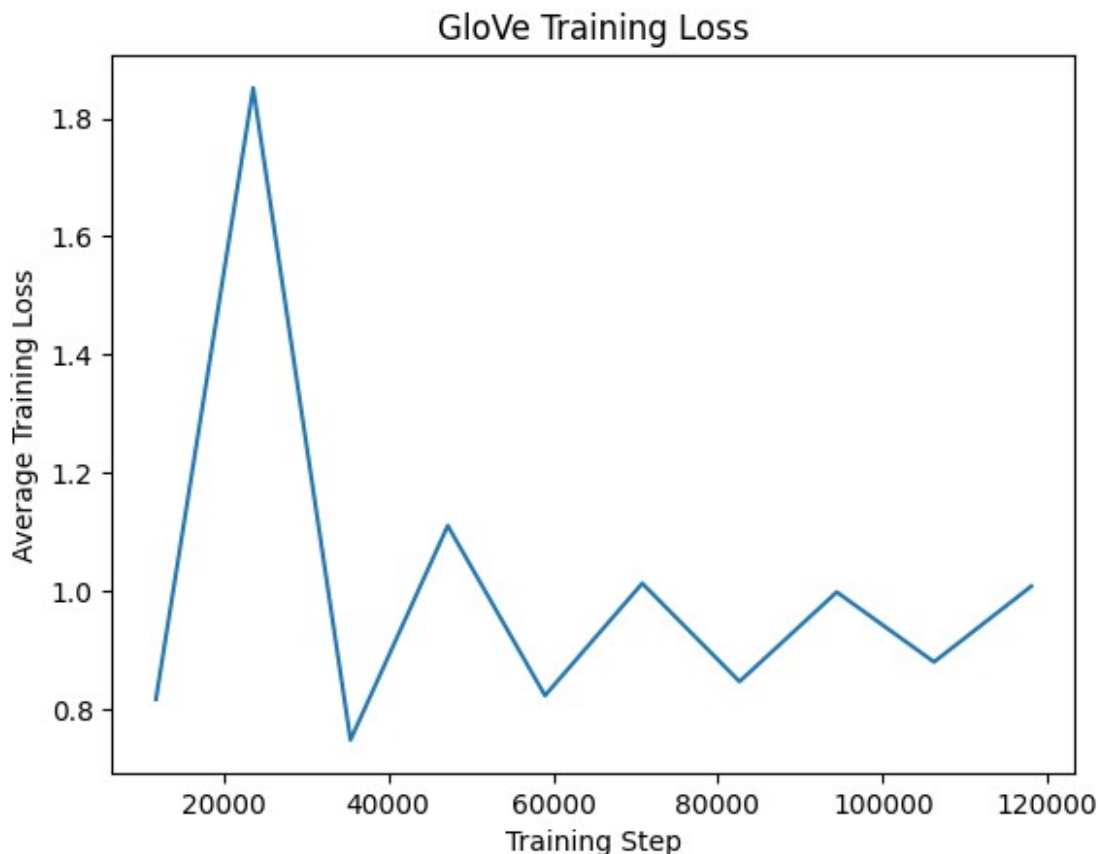
Epoch 10 completed - Average Loss: 1.0077

Checkpoint saved at step 118040 to

exp_3_setting_3_glove//glove_model_step_118040.pt

Printing closest words

```
four: ['was', 'in', 'from', 'book', 'on', 'not', 'would', 'thys']
and: ['the', 'of', 'a', 'to', 'two', 'be', 'are', 'were']
when: ['there', 'first', 'united', 'most', 'made', 'so', 'use',
'under']
or: ['an', 'seven', 's', 'are', 'two', 'he', 'have', 'to']
two: ['to', 's', 'is', 'seven', 'or', 'a', 'an', 'be']
eight: ['is', 'two', 'as', 'zero', 'a', 'to', 's', 'an']
th: ['up', 'out', 'both', 'm', 'film', 'south', 'early', 'including']
its: ['have', 'has', 'been', 'an', 'or', 'be', 'into', 'were']
in: ['was', 'four', 'from', 'not', 'over', 'UNK', 'could', 'that']
all: ['were', 'they', 'six', 'first', 'a', 'to', 'two', 'most']
only: ['when', 'most', 'her', 'work', 'were', 'first', 'm', 'there']
seven: ['an', 'he', 'or', 'be', 'two', 'six', 's', 'have']
also: ['are', 'were', 'he', 'at', 'six', 'or', 'to', 'an']
can: ['they', 'were', 'area', 'first', 'been', 'has', 'however',
'either']
of: ['and', 'the', 'two', 'to', 'is', 'be', 'a', 'seven']
there: ['when', 'united', 'first', 'under', 'any', 'made', 'most',
'name']
Training completed after 10 epochs - Final Average Loss: 1.0077
Created path: exp_3_setting_3_final_model
Final model saved at exp_3_setting_3_final_model/glove_final_model.pt
Embeddings saved at exp_3_setting_3_final_model/glove_embeddings.pkl
```



What is the best embedding size to use for GloVe model and why? What did you learn from the plots?

TODO(students): start

Best embedding size is 50 because the avg total loss of all experiments is less for embedding size when it is 50 than 128 and 300. And from plots, for emb_size = 50, the avgtotal loss is decreasing gradually through the experiment. For other embedding experiments, the overall loss is decreasing but for each epoch, the avg loss is first increasing and then decreasing. It shows that the weights are already in stable state so for next epoch, the avg loss is first increasing and then decreasing more than prev epoch. So, we cannot much change the avg loss for emb_size = 128 and 300.

TODO(students): end

```
## TODO(students): start

best_embedding_size = 50

## TODO(students): end
```

Training the best models

Now, you should use the best hyperparameter settings for the above three experiments to train your best GloVe model.

```
# TODO(student): start

# Best hyperparameters identified from previous experiments
lr = best_lr # Replace with the best learning rate identified
window_size = best_window_size # Replace with the best window size identified
embedding_size = best_embedding_size # Replace with the best embedding size identified

checkpoint_model_path = "best_params_model_settings"
final_model_path = "best_params_model"

# Training the best GloVe model with the identified hyperparameters
run_training(
    epochs=10, # Train for more epochs to achieve better performance
    lr=lr, # Best learning rate from previous experiments
    checkpoint_model_path=checkpoint_model_path, # Directory to save model checkpoints
    final_model_path=final_model_path, # Directory to save the final model and embeddings
    vocab_size=int(1e5), # Vocabulary size (fixed)
    batch_size=512, # Batch size for training (fixed)
    embedding_size=embedding_size, # Best embedding size from previous experiments
```

```
    window_size=window_size, # Best window size from previous
    experiments
    checkpoint_step=5000, # Save checkpoints after every 5000 steps
)
```

```
# TODO(student): end
```

Created path: best_params_model_settings_glove/

Most common words (+UNK) [['UNK', 0], ('the', 62827), ('of', 36789), ('and', 25238), ('one', 24679)]

Sample data: [632, 3668, 12, 6, 173, 2, 4288, 47, 63, 150]

['anarchism', 'originated', 'as', 'a', 'term', 'of', 'abuse', 'first', 'used', 'against']

Generating co-occurrence matrix...

Building cooccurrence matrix: on line 0/1000000

Building cooccurrence matrix: on line 100000/1000000

Building cooccurrence matrix: on line 200000/1000000

Building cooccurrence matrix: on line 300000/1000000

Building cooccurrence matrix: on line 400000/1000000

Building cooccurrence matrix: on line 500000/1000000

Building cooccurrence matrix: on line 600000/1000000

Building cooccurrence matrix: on line 700000/1000000

Building cooccurrence matrix: on line 800000/1000000

Building cooccurrence matrix: on line 900000/1000000

Co-occurrence matrix generated.

Printing Validation Set Words

Word: world, ID: 66

Word: its, ID: 46

Word: has, ID: 41

Word: he, ID: 32

Word: war, ID: 75

Word: they, ID: 49

Word: the, ID: 1

Word: all, ID: 51

Word: most, ID: 53

Word: would, ID: 77

Word: two, ID: 11

Word: was, ID: 15

Word: over, ID: 88

Word: american, ID: 48

Word: one, ID: 4

Word: b, ID: 59

Device: cuda:0

Total Batches per Epoch: 11804

Starting Epoch 1/10

Epoch 1/10: 100%|██████████| 11804/11804 [00:52<00:00, 223.62it/s, Batch Loss=0.0467, Avg Loss=0.0604]

Epoch 1 completed - Average Loss: 0.1712
Checkpoint saved at step 11804 to
best_params_model_settings_glove//glove_model_step_11804.pt
Printing closest words
world: ['m', 'for', 'ethereal', 'control', 'his', 'population',
'because', 'an']
its: ['four', 'an', 'lttl', 'absolute', 'because', 'taxonomists',
'fiction', 'pyloric']
has: ['bonaparte', 'their', 'bureaucrat', 'UNK', 'for', 'UNK', 'not',
'UNK']
he: ['were', 'such', 'this', 'water', 'other', 'pareja', 'nivat',
'both']
war: ['murch', 'system', 'if', 'messianic', 'this', 'apex',
'contians', 'UNK']
they: ['no', 'this', 'UNK', 'chlorus', 'last', 'UNK', 'cyrus', 'UNK']
the: ['diablerie', 'opossum', 'UNK', 'adjudged', 'anabaptist', 'UNK',
'ounces', 'UNK']
all: ['before', 'are', 'this', 'germany', 'based', 'b', 'UNK', 'it']
most: ['real', 'system', 'saltpetre', 'scofield', 'UNK', 'UNK', 'UNK',
'marshalling']
would: ['UNK', 'UNK', 'aloud', 'gaps', 'before', 'bc', 'scion', 'UNK']
two: ['on', 'm', 'four', 'that', 'of', 'an', 'have', 'as']
was: ['radiotelephones', 'UNK', 'which', 'feminists', 'niggers', 'on',
'in', 'control']
over: ['completed', 'sinister', 'fraudulent', 'nsync', 'UNK', 'UNK',
'UNK', 'UNK']
american: ['UNK', 'm', 'UNK', 'UNK', 'land', 'it', 'as', 'were']
one: ['s', 'four', 'and', 'nine', 'UNK', 'among', 'public', 'at']
b: ['brahe', 'common', 'UNK', 'soviet', 'which', 'all', 'this',
'diamonds']
Starting Epoch 2/10

Epoch 2/10: 100%[██████████] 11804/11804 [00:53<00:00, 222.61it/s,
Batch Loss=0.138, Avg Loss=0.0835]

Epoch 2 completed - Average Loss: 0.0832
Checkpoint saved at step 23608 to
best_params_model_settings_glove//glove_model_step_23608.pt
Printing closest words
world: ['m', 'personal', 'existence', 'because', 'phenomenon',
'translated', 'like', 'classic']
its: ['UNK', 'land', 'masoleum', 'UNK', 'UNK', 'theory', 'hush',
'UNK']
has: ['their', 'which', 'first', 'been', 'five', 'name', 'when',
'can']
he: ['from', 'usually', 'law', 'a', 'roman', 'it', 'was', 'america']
war: ['john', 'famous', 'part', 'true', 'UNK', 'sometimes', 'other',
'UNK']
they: ['this', 'any', 'god', 'while', 'after', 'how', 'usually',
'first']

the: ['also', 'under', 'usually', 'human', 'roman', 'began',
'government', 'non']
all: ['are', 'data', 'each', 'law', 'unseeable', 'water', 'due',
'pixodarus']
most: ['well', 'sometimes', 'include', 'only', 'for', 'great',
'banastre', 'england']
would: ['only', 'far', 'will', 'than', 'should', 'do', 'lingerie',
'believe']
two: ['four', 'zero', 'is', 'one', 'at', 'through', 'austrian', 'd']
was: ['he', 'in', 'nature', 'roman', 'or', 'a', 'also', 'as']
over: ['their', 'under', 'name', 'city', 'history', 'while', 'any',
'up']
american: ['century', 'uschi', 'added', 'allgood', 'b', 'anowski',
'symbology', 'UNK']
one: ['four', 'nobel', 'two', 'zero', 'actress', 'million', 'nine',
'he']
b: ['model', 'few', 'fire', 'foreign', 'l', 'nation', 'run',
'italian']

Starting Epoch 3/10

Epoch 3/10: 100%|██████████| 11804/11804 [00:53<00:00, 222.00it/s,
Batch Loss=0.0529, Avg Loss=0.0676]

Epoch 3 completed - Average Loss: 0.0724

Checkpoint saved at step 35412 to

best_params_model_settings_glove//glove_model_step_35412.pt

Printing closest words

world: ['if', 'see', 'into', 'b', 'him', 'new', 'war', 'use']
its: ['an', 'other', 'with', 'at', 'see', 'american', 'most', 'new']
has: ['this', 'during', 'it', 'out', 'had', 'agony', 'people',
'langworthy']
he: ['it', 'or', 'UNK', 'also', 'UNK', 'brickmaking', 'untill', 'UNK']
war: ['world', 'new', 'see', 'b', 'american', 'if', 'into', 'inner']
they: ['first', 'many', 'known', 'plains', 'UNK', 'about', 'dooagh',
'had']
the: ['fallopian', 'chamfort', 'epideixis', 'UNK', 'pentateuch',
'stoltz', 'UNK', 'UNK']
all: ['without', 'claims', 'many', 'addition', 'belief', 'occurs',
'highly', 'formed']
most: ['than', 'at', 'being', 'for', 'when', 'his', 'were',
'sometimes']
would: ['only', 'than', 'now', 'book', 'states', 'both', 'him', 'new']
two: ['nine', 'is', 'four', 'zero', 'one', 'six', 'of', 'for']
was: ['nothing', 'mushroom', 'plaintiffs', 'UNK', 'kitchen',
'zosimus', 'caret', 'UNK']
over: ['when', 'after', 'where', 'long', 'including', 'them', 'for',
'four']
american: ['b', 'see', 'new', 'i', 'were', 'born', 'there', 'with']
one: ['is', 'nine', 'zero', 'two', 'for', 'to', 'four', 'six']
b: ['american', 'british', 'world', 'i', 'actor', 'player', 'see',

'french']

Starting Epoch 4/10

Epoch 4/10: 100%|██████████| 11804/11804 [00:53<00:00, 220.09it/s,
Batch Loss=0.0844, Avg Loss=0.0735]

Epoch 4 completed - Average Loss: 0.0702

Checkpoint saved at step 47216 to

best_params_model_settings_glove//glove_model_step_47216.pt

Printing closest words

world: ['there', 'b', 'british', 'american', 'time', 'years',
'country', 'several']

its: ['i', 'were', 'new', 'each', 'there', 'modern', 'american',
'too']

has: ['UNK', 'feuding', 'UNK', 'tlc', 'crucifixion', 'developer',
'UNK', 'overwhelmingly']

he: ['also', 'it', 'be', 'or', 'from', 'all', 'may', 'have']

war: ['b', 'south', 'world', 'who', 'british', 'american', 'were',
'became']

they: ['or', 'it', 'between', 'more', 'be', 'upon', 'he', 'also']

the: ['from', 'was', 'as', 'of', 'five', 'seven', 'he', 'three']

all: ['also', 'system', 'he', 'international', 'many', 'history',
'various', 'it']

most: ['than', 'different', 'book', 'those', 'only', 'at', 'when',
'if']

would: ['only', 'than', 'book', 'states', 'now', 'players', 'them',
'most']

two: ['zero', 'nine', 'six', 'four', 'is', 'UNK', 'UNK', 'UNK']

was: ['from', 'three', 'by', 'as', 'five', 'seven', 'eight', 'he']

over: ['could', 'modern', 'its', 'each', 'i', 'on', 'this', 'were']

american: ['b', 'there', 'world', 'modern', 'are', 'years', 'time',
'south']

one: ['four', 'zero', 'on', 'for', 'american', 'a', 'are',
'detriment']

b: ['american', 'world', 'years', 'south', 'there', 'time', 'british',
'modern']

Starting Epoch 5/10

Epoch 5/10: 100%|██████████| 11804/11804 [00:53<00:00, 221.47it/s,
Batch Loss=0.12, Avg Loss=0.0692]

Epoch 5 completed - Average Loss: 0.0644

Checkpoint saved at step 59020 to

best_params_model_settings_glove//glove_model_step_59020.pt

Printing closest words

world: ['between', 'there', 'seven', 'e', 'by', 'british', 'became',
'death']

its: ['other', 'this', 'his', 'have', 'that', 'since', 'is', 'an']

has: ['this', 'such', 'his', 'which', 'other', 'no', 'had', 'that']

he: ['their', 'massaliote', 'be', 'five', 'postpartum', 'may', 'it',

'herzer']
war: ['UNK', 'inosinic', 'standardising', 'UNK', 'UNK', 'stricter',
'speciosa', 'consecrated']
they: ['upon', 'more', 'family', 'though', 'own', 'mother', 'times',
'body']
the: ['and', 'zero', 'to', 'army', 'is', 'nine', 'two', 'asia']
all: ['also', 'history', 'this', 'because', 'have', 'people', 'part',
'her']
most: ['been', 'only', 'would', 'both', 'if', 'at', 'which',
'sometimes']
would: ['only', 'most', 'if', 'will', 'been', 'both', 'after',
'states']
two: ['nine', 'is', 'zero', 'six', 'an', 's', 'one', 'other']
was: ['as', 'eight', 'three', 'red', 'UNK', 'b', 'still', 'disney']
over: ['famous', 'however', 'form', 'p', 'between', 'area', 'since',
'there']
american: ['b', 'south', 'i', 'more', 'modern', 'year', 'd', 'each']
one: ['to', 'zero', 'is', 'nine', 'two', 'by', 'other', 'an']
b: ['american', 'south', 'modern', 'i', 'more', 'year', 'each',
'name']

Starting Epoch 6/10

Epoch 6/10: 100%|██████████| 11804/11804 [00:53<00:00, 220.17it/s,
Batch Loss=0.0635, Avg Loss=0.0711]

Epoch 6 completed - Average Loss: 0.0647

Checkpoint saved at step 70824 to

best_params_model_settings_glove//glove_model_step_70824.pt

Printing closest words

world: ['between', 'part', 'several', 'based', 'natural', 'time',
'australia', 'since']
its: ['other', 'this', 'have', 'all', 'which', 'since', 'several',
'century']
has: ['been', 'had', 'most', 'after', 'history', 'but', 'known',
'about']
he: ['with', 'about', 'by', 'from', 'been', 'six', 'on', 'four']
war: ['first', 'called', 'used', 'who', 'these', 'english', 'while',
'not']
they: ['not', 'who', 'used', 'be', 'being', 'first', 'god', 'up']
the: ['to', 'in', 'of', 'a', 'and', 'british', 'are', 'with']
all: ['have', 'its', 'other', 'his', 'this', 'century', 'so', 'also']
most: ['been', 'after', 'has', 'had', 'would', 'during', 'about', 's']
would: ['only', 'known', 'been', 'common', 'most', 'about', 'after',
'considered']
two: ['zero', 'UNK', 'UNK', 'rightarrow', 'UNK', 'dstrup', 'UNK',
'oreca']
was: ['three', 'are', 'five', 'eight', 'on', 'first', 'with', 'not']
over: ['of', 'wyman', 'axelrod', 'webcam', 'UNK', 'carota', 'UNK',
'imitate']
american: ['UNK', 'i', 'crockett', 'days', 'b', 'UNK', 'floatation',


```
'UNK']
one: ['is', 'and', 'nine', 'converses', 's', 'giralddi', 'rickey',
'six']
b: ['though', 'each', 'right', 'south', 'times', 'including',
'committee', 'algae']
Starting Epoch 7/10

Epoch 7/10: 100%|██████████| 11804/11804 [00:54<00:00, 217.83it/s,
Batch Loss=0.0933, Avg Loss=0.0704]

Epoch 7 completed - Average Loss: 0.0622
Checkpoint saved at step 82628 to
best_params_model_settings_glove//glove_model_step_82628.pt
Printing closest words
world: ['part', 'between', 'since', 'british', 'similar', 'h',
'popular', 'philosophy']
its: ['UNK', 'pseudocoelomate', 'UNK', 'UNK', 'UNK', 'UNK', 'UNK',
'augusto']
has: ['most', 'after', 'but', 'history', 'her', 'number', 'these',
'about']
he: ['that', 'it', 'is', 'his', 'also', 'nine', 'two', 'zero']
war: ['than', 'he', 'on', 'when', 'zero', 'one', 'th', 'is']
they: ['used', 'be', 'some', 'than', 'when', 'not', 'were', 'that']
the: ['to', 'that', 'is', 'and', 'zero', 'his', 'one', 'in']
all: ['also', 'have', 'an', 'no', 'his', 'he', 'on', 'can']
most: ['has', 'after', 'but', 'see', 'these', 'history', 'about',
'number']
would: ['only', 'been', 'do', 'can', 'were', 'later', 'when',
'another']
two: ['nine', 'zero', 'or', 'is', 'it', 'at', 'one', 'which']
was: ['as', 'seven', 'on', 'forty', 'used', 'published', 'not', 'any']
over: ['de', 'transformations', 'six', 'three', 'between', 'long',
'fulton', 'sim']
american: ['b', 'or', 'i', 'nine', 'time', 'he', 'it', 'an']
one: ['nine', 'zero', 'is', 'two', 'and', 'that', 'it', 'to']
b: ['american', 'i', 'or', 'two', 'their', 'nine', 'at', 'time']
Starting Epoch 8/10
```

```
Epoch 8/10: 100%|██████████| 11804/11804 [00:53<00:00, 220.87it/s,
Batch Loss=0.083, Avg Loss=0.0699]
```

```
Epoch 8 completed - Average Loss: 0.0597
Checkpoint saved at step 94432 to
best_params_model_settings_glove//glove_model_step_94432.pt
Printing closest words
world: ['over', 'since', 'will', 'see', 'new', 'near', 'around',
'through']
its: ['s', 'other', 'by', 'such', 'most', 'for', 'has', 'see']
has: ['but', 'most', 'by', 'after', 's', 'see', 'are', 'its']
he: ['that', 'were', 'be', 'diffuse', 'former', 'french', 'UNK',
```

'still']
war: ['th', 'links', 'first', 'still', 'during', 'another', 'any',
'among']
they: ['UNK', 'ti', 'UNK', 'used', 'managed', 'according', 'cowhorn',
'koskalos']
the: ['a', 'and', 'of', 'into', 'to', 'for', 'in', 's']
all: ['also', 'their', 'it', 'who', 'found', 'which', 'would',
'other']
most: ['by', 'has', 'but', 'for', 'are', 's', 'see', 'with']
would: ['where', 'their', 'which', 'than', 'have', 'found', 'all',
'them']
two: ['zero', 'nine', 'paulo', 'th', 'tomography', 'sheat', 'any',
'wellington']
was: ['by', 'as', 's', 'eight', 'five', 'three', 'for', 'are']
over: ['world', 'will', 'new', 'see', 'through', 'since', 'became',
'six']
american: ['zero', 'result', 'nine', 'published', 'UNK', 'b', 'UNK',
'others']
one: ['caste', 'and', 'on', 'to', 'UNK', 'sidelined', 'UNK', 'UNK']
b: ['UNK', 'american', 'ballantine', 'dec', 'jacaranda', 'dominance',
'UNK', 'UNK']
Starting Epoch 9/10

Epoch 9/10: 100%|██████████| 11804/11804 [00:53<00:00, 219.97it/s,
Batch Loss=0.0967, Avg Loss=0.0706]

Epoch 9 completed - Average Loss: 0.0586

Checkpoint saved at step 106236 to

best_params_model_settings_glove//glove_model_step_106236.pt

Printing closest words

world: ['since', 'over', 'there', 'state', 'him', 'had', 'known',
'nine']
its: ['UNK', 'spiders', 'UNK', 'platensis', 'this', 'hers', 'cec',
'seamlessly']
has: ['at', 'other', 'all', 'like', 'new', 'when', 'this', 'than']
he: ['that', 'his', 'two', 'nine', 'it', 'or', 'be', 'zero']
war: ['links', 'american', 'th', 'first', 'century', 'civil',
'german', 'film']
they: ['used', 'name', 'some', 'been', 'can', 'she', 'had', 'more']
the: ['in', 'a', 'and', 'that', 'one', 'it', 'or', 'he']
all: ['other', 'than', 'same', 'no', 'at', 'under', 'this', 'has']
most: ['use', 'would', 'some', 'who', 'also', 'been', 'could',
'there']
would: ['use', 'most', 'there', 'known', 'been', 'often', 'who',
'some']
two: ['nine', 'zero', 'seven', 'he', 'that', 'his', 'or', 'eight']
was: ['by', 'to', 'UNK', 'usually', 'configurations', 'UNK', 's',
'gameplay']
over: ['since', 'there', 'him', 'god', 'so', 'world', 'known',
'state']

american: ['war', 'c', 'first', 'seven', 'century', 'nine', 'born', 'two']

one: ['and', 'in', 'nine', 'zero', 'two', 'or', 'it', 'he']

b: ['UNK', 'r', 'luxembourg', 'UNK', 'UNK', 'humorous', 'united', 'UNK']

Starting Epoch 10/10

Epoch 10/10: 100%|██████████| 11804/11804 [00:54<00:00, 218.16it/s, Batch Loss=0.0712, Avg Loss=0.0702]

Epoch 10 completed - Average Loss: 0.0567

Checkpoint saved at step 118040 to

best_params_model_settings_glove//glove_model_step_118040.pt

Printing closest words

world: ['after', 'were', 'on', 'he', 'called', 'which', 'during', 'alexander']

its: ['all', 'other', 'their', 'has', 'have', 'this', 'such', 'although']

has: ['at', 'have', 'all', 'this', 'which', 'but', 'other', 'its']

he: ['his', 'are', 'on', 'were', 'five', 's', 'three', 'which']

war: ['during', 'name', 'about', 'years', 'history', 'used', 'british', 'number']

they: ['name', 'used', 'being', 'during', 'about', 'she', 'history', 'years']

the: ['of', 'to', 'a', 'in', 'and', 'on', 'for', 's']

all: ['have', 'has', 'other', 'at', 'this', 'their', 'its', 'but']

most: ['would', 'however', 'after', 'of', 'his', 'are', 'there', 'main']

would: ['however', 'most', 'government', 'there', 'for', 'are', 'so', 'she']

two: ['tiananmen', 'granvelle', 'revitalized', 'echolalia', 'phrixus', 'armonico', 'endogenous', 'filmmaker']

was: ['as', 'by', 's', 'is', 'with', 'five', 'on', 'are']

over: ['bachman', 'anec', 'had', 'aedile', 'parallelized', 'UNK', 'UNK', 'UNK']

american: ['UNK', 'UNK', 'brahmagupta', 'arnaldo', 'three', 'four', 'UNK', 'schwantes']

one: ['in', 'and', 'a', 'he', 'of', 'war', 'were', 'six']

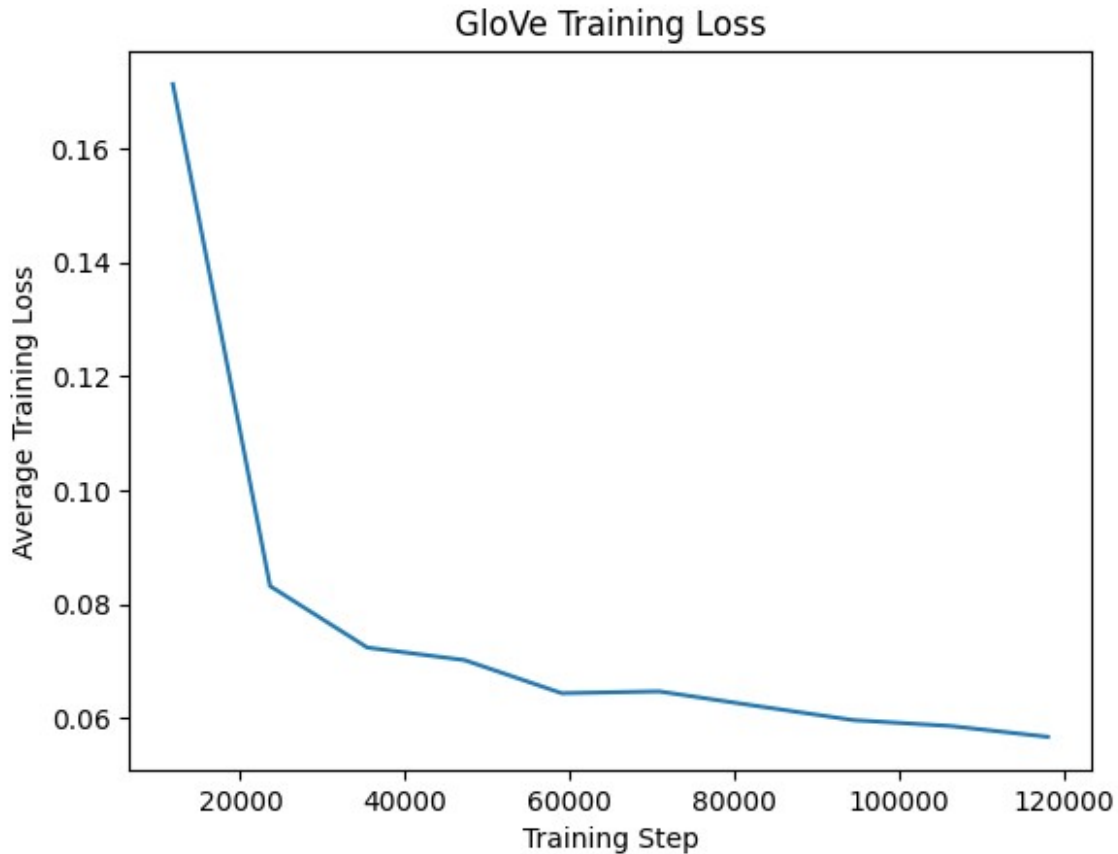
b: ['see', 'only', 'united', 'may', 'while', 'new', 'writer', 'its']

Training completed after 10 epochs - Final Average Loss: 0.0567

Created path: best_params_model

Final model saved at best_params_model/glove_final_model.pt

Embeddings saved at best_params_model/glove_embeddings.pkl



Testing Framework

Analogies using word vectors

You will use the word vectors you learned from both approaches in the following word analogy task.

Each question/task is in the following form.

Consider the following word pairs that share the same relation, R:

pilgrim:shrine, hunter:quarry, assassin:victim, climber:peak

Among these word pairs,

- (1) pig:mud
- (2) politician:votes
- (3) dog:bone
- (4) bird:worm

Q1. Which word pairs has the MOST illustrative(similar) example of the relation R?

Q2. Which word pairs has the LEAST illustrative(similar) example of the relation R?

For each question, there are examples pairs of a certain relation. Your task is to find the most/least illustrative word pair of the relation. One simple method to answer those questions will be measuring the similarities of difference vectors.

Recall that vectors are representing some direction in space. If (a, b) and (c, d) pairs are analogous pairs then the transformation from a to b (i.e., some x vector when added to a gives b: $a + x = b$) should be highly similar to the transformation from c to d (i.e., some y vector when added to c gives d: $c + y = d$). In other words, the difference vector (b-a) should be similar to difference vector (d-c).

This difference vector can be thought to represent the relation between the two words.

Please fill-in the TODO section below to implement the above mentioned task.

Due to the noisy annotation data, the expected accuracy is not high. You can use 33-36% as a general benchmark for GloVe's performance on the analogy task, aiming for improvements in that range. Improving this score 1~3% would be your goal.

Further implementation instructions:

- In the next 2 cells: You will write a code in the TODO section for evaluating relation between pairs of words -- called the [MaxDiff question](#). You will generate a file with your predictions following the format of `word_analogy_sample_predictions.txt`.
- `evaluate_word_analogy.pl`: This is a perl script to evaluate YOUR PREDICTIONS on development data. Use it as shown in the next cell. You do not need to submit prediction or score files related to the dev set.
- `word_analogy_dev.txt`: This is some data for development. Each line of this file is divided into "examples" and "choices" by "||". [examples]||[choices] "Examples" and "choices" are delimited by a comma. For example:
"tailor:suit","oracle:prophecy","baker:flour"
- `word_analogy_dev_sample_predictions.txt`: A sample prediction file. Pay attention to the format of this file. Your prediction file should follow this to use "score_maxdiff.pl" script. Each row is in this format:

The order of word pairs should match their original order found in `word_analogy_dev.txt`.

- `word_analogy_dev_mturk_answers.txt`: This is the answers collected using Amazon mechanical turk for `word_analogy_dev.txt`. The answers in this file is used as the correct answer and used to evaluate your analogy predictions. (using

"evaluate_word_analogy.pl") For your information, the answers here are a little bit noisy.

- `word_analogy_test.txt`: Test data file. When you are done experimenting with your model, you will generate predictions for this test data using your best model. You will not be able to run the evaluation script on the test set.

Make sure your submission files are named: `test_preds_glove.txt`

```
def read_data(file_path):
    with open(file_path, 'r') as f:
        data = f.readlines()

    candidate, test = [], []
    for line in data:
        a, b = line.strip().split("||")
        a = [i[1:-1].split(":") for i in a.split(",")]
        b = [i[1:-1].split(":") for i in b.split(",")]
        candidate.append(a)
        test.append(b)

    return candidate, test

def get_embeddings(examples, embeddings, dictionary):
    """
    For the word pairs in the 'examples' array, fetch embeddings and
    return.
    """
    norm = np.sqrt(np.sum(np.square(embeddings), axis=1,
    keepdims=True))
    normalized_embeddings = embeddings / norm

    embs = []
    for line in examples:
        temp = []
        for pairs in line:
            word1, word2 = pairs[0], pairs[1]

            # Get embeddings for word1 and word2 or fall back to 'UNK'
            if word1 in dictionary:
                emb1 = normalized_embeddings[dictionary[word1]]
            else:
                emb1 = normalized_embeddings[dictionary['UNK']]

            if word2 in dictionary:
                emb2 = normalized_embeddings[dictionary[word2]]
            else:
                emb2 = normalized_embeddings[dictionary['UNK']]

            # Append both embeddings as a pair
```

```

        temp.append([emb1, emb2])
        embs.append(temp)

    #result = np.array(embs)
    result = np.array(embs, dtype=object)
    return result

def evaluate_pairs(candidate_embs, test_embs):
    """
    Write code to evaluate a relation between pairs of words.
    Find the best and worst pairs and return that.
    """

    best_pairs = []
    worst_pairs = []

    ### TODO(students): start

    for i in range(len(candidate_embs)):

        cand_diffs_list = []
        for pair in candidate_embs[i]:
            v1 = np.array(pair[0], dtype=float)
            v2 = np.array(pair[1], dtype=float)
            cand_diffs_list.append(v1 - v2)

        # difference between the word embeddings for each pair in the
        # candidate & test sets
        candidate_diffs = np.mean(cand_diffs_list, axis=0)

        test_diffs_list = []
        for pair in test_embs[i]:
            v1 = np.array(pair[0], dtype=float)
            v2 = np.array(pair[1], dtype=float)
            test_diffs_list.append(v1 - v2)
        test_diffs = np.array(test_diffs_list, dtype=float)

        # compute the similarity
        numerator = np.dot(test_diffs, candidate_diffs)
        denominator = np.linalg.norm(test_diffs, axis=1) *
np.linalg.norm(candidate_diffs)
        similarity = numerator / (denominator + 1e-9)

        # rank similarities with highest ranked first
        rank = np.argsort(similarity)[::-1]

        # identify the best and worst pairs
        best, worst = rank[0], rank[-1]

```

```

        # append to the respective lists
        best_pairs.append(best)
        worst_pairs.append(worst)

    ### TODO(students): end

    return best_pairs, worst_pairs

def write_solution(best_pairs, worst_pairs, test, path):
    """
    Write best and worst pairs to a file, that can be evaluated by
    evaluate_word_analogy.pl
    """

    ans = []
    for i, line in enumerate(test):
        temp = [f'"{pairs[0]}:{pairs[1]}"' for pairs in line]
        temp.append(f'"{line[worst_pairs[i]][0]}:{line[worst_pairs[i]]
[1]}"')
        temp.append(f'"{line[best_pairs[i]][0]}:{line[best_pairs[i]]
[1]}"')
        ans.append(" ".join(temp))

    with open(path, 'w') as f:
        f.write("\n".join(ans))

def run_word_analogy_eval_glove(
    model_path='./final_model', # Path to the location where the
    GloVe model is stored
    input_filepath='word_analogy_dev.txt', # Word analogy file to
    evaluate on
    output_filepath='word_analogy_glove_results.txt', # Predicted
    results
):
    """
    Main function to evaluate GloVe embeddings on word analogy tasks.
    """
    print(f'Model file: {model_path}/glove_embeddings.pkl')
    model_filepath = os.path.join(model_path, 'glove_embeddings.pkl')

    with open(model_filepath, 'rb') as f:
        data = pickle.load(f)

    # Unpack the saved data
    dictionary = data['vocab_token_to_id']
    embeddings = data['center_embeddings']

    # Read the analogy data

```



```

candidate, test = read_data(input_filepath)

# Get embeddings for candidate and test examples
candidate_embs = get_embeddings(candidate, embeddings, dictionary)
test_embs = get_embeddings(test, embeddings, dictionary)

# Evaluate the relationships between word pairs
best_pairs, worst_pairs = evaluate_pairs(candidate_embs,
test_embs)

# Write the solution to the output file
print(f'Output file: {output_filepath}')
write_solution(best_pairs, worst_pairs, test, output_filepath)

```

Once the word_analogy code is complete with the TODO section, you can use the following function call to generate the results of the word analogy task.

```

!unzip "538-hw1-117411478-models.zip"
Archive: 538-hw1-117411478-models.zip
  inflating: glove_embeddings.pkl
replace __MACOSX/._glove_embeddings.pkl? [y]es, [n]o, [A]ll, [N]one,
[r]ename: y
  inflating: __MACOSX/._glove_embeddings.pkl
  inflating: glove.model
replace __MACOSX/._glove.model? [y]es, [n]o, [A]ll, [N]one, [r]ename:
y
  inflating: __MACOSX/._glove.model

run_word_analogy_eval_glove(
    model_path = './', # path to the location where the model being
evaluated is stored
    input_filepath = 'word_analogy_dev.txt', # Word analogy file to
evaluate on
    output_filepath = 'word_analogy_best_glove_dev_results.txt', #
predicted results
)

Model file: ../glove_embeddings.pkl
Output file: word_analogy_best_glove_dev_results.txt

# test for word_analogy_test.txt

run_word_analogy_eval_glove(
    model_path = '', # path to the location where the model being
evaluated is stored
    input_filepath = 'word_analogy_test.txt', # Word analogy file to
evaluate on
    output_filepath = 'test_preds_glove.txt', # predicted results
)

```

```
)
```

```
Model file: /glove_embeddings.pkl  
Output file: test_preds_glove.txt
```

The results can finally be converted into numeric metrics using the Perl script below. A demo score result is also provided for reference.

```
!chmod 777 evaluate_word_analogy.pl  
!./evaluate_word_analogy.pl word_analogy_dev_mturk_answers.txt  
word_analogy_best_glove_dev_results.txt demo_score_glove.txt  
  
reading the Mechanical Turk answers to MaxDiff questions  
word_analogy_dev_mturk_answers.txt ...  
... read 914 unique MaxDiff questions ...  
... done.  
reading the MaxDiff answers to be evaluated  
word_analogy_best_glove_dev_results.txt ...  
... read 914 answers ...  
... done.  
scoring the answers and writing a summary to demo_score_glove.txt ...  
... done.
```

Conclude the results of your experiments. Include a table in the notebook showing all your results.

TODO(students): start

Exp No	learning rate	window_size	embed_size	Avg_total_loss	comment
1.0(default)	0.01	3	128	0.2741	default settings
1.1	0.05	3	128	91.7088	loss exploding compared to default
1.2	0.1	3	128	1344.1214	loss exploding compared to 1.1
1.3	0.5	3	128	801748.5250	loss exploding more compared to 1.2
2.1	0.01	2	128	0.2689	best loss

Exp No	learning rate	window_size	embed_size	Avg_total_loss	comment
2.2	0.01	5	128	0.2484	compared to default window size loss decreased compared to 2.1
2.3	0.01	10	128	0.1916	loss decreased compared to 2.2
3.1	0.01	10	50	0.0578	best emb_size compared to 2.3(best of prev exp)
3.2	0.01	10	128	0.1910	loss increasing compared to 3.1
3.3	0.01	10	300	0.8164	loss increasing more compared to 3.3

TODO(students): end

Please make sure that you also test your best model on word_analogy_test.txt

Sentiment Analysis using word vectors

Remember, we had built and evaluated a logistic regression model in Assignment 1 for the task of sentiment classification. Similar to the word vectors we have just trained, we will first download word vectors trained on much larger corpora, and then use them to perform sentiment classification, instead of using the pre-built CountVectorizer function we had used last time.

With a much smaller input vector size, we can still achieve similar performance.

```
!wget http://nlp.stanford.edu/data/glove.6B.zip
--2026-02-28 20:44:43-- http://nlp.stanford.edu/data/glove.6B.zip
Resolving nlp.stanford.edu (nlp.stanford.edu)... 171.64.67.140
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:80...
```

```
connected.
HTTP request sent, awaiting response... 302 Found
Location: https://nlp.stanford.edu/data/glove.6B.zip [following]
--2026-02-28 20:44:43-- https://nlp.stanford.edu/data/glove.6B.zip
Connecting to nlp.stanford.edu (nlp.stanford.edu)|
171.64.67.140|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip
[following]
--2026-02-28 20:44:44--
https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip
Resolving downloads.cs.stanford.edu (downloads.cs.stanford.edu)...
171.64.64.22
Connecting to downloads.cs.stanford.edu (downloads.cs.stanford.edu)|
171.64.64.22|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 862182613 (822M) [application/zip]
Saving to: 'glove.6B.zip'

glove.6B.zip      100%[=====>] 822.24M  5.02MB/s   in
2m 39s
```

```
2026-02-28 20:47:23 (5.18 MB/s) - 'glove.6B.zip' saved
[862182613/862182613]
```

```
!unzip glove*.zip
```

```
Archive: glove.6B.zip
  inflating: glove.6B.50d.txt
  inflating: glove.6B.100d.txt
  inflating: glove.6B.200d.txt
  inflating: glove.6B.300d.txt
```

```
# load and index GloVe vectors for logistic regression experiments
```

```
def load_glove_vectors():
    embeddings = {}
    f = open('glove.6B.300d.txt', encoding='utf-8')
    for line in f:
        values = line.split()
        word = values[0]
        coefs = np.asarray(values[1:], dtype='float32')
        embeddings[word] = coefs
    f.close()

    print('Found %s word vectors.' % len(embeddings))

    return embeddings

def generate_input_output_for_logistic_regression_model(dataset):
    """
```

```

input: dataset -> List[Dict]
Each dict contains two keys - text and label.
text contains the text for which sentiment has to be labeled.
label is an integer denoting the gold sentiment.
output: inputs -> List, outputs -> List
inputs is a list of texts from dataset
outputs is a list of labels from dataset
"""

inputs, outputs = [], []
for element in dataset:
    inputs.append(element["text"])
    outputs.append(element["label"])

assert len(inputs) == len(outputs)

return inputs, outputs

def generate_data_slice(data, percentage):
    """
    input: data -> Any, percentage: float
    output: train_inputs -> List, train_labels -> List, dev_inputs ->
    List, dev_labels -> List
    """

    # generate data slice from train set
    # this will be split into train and dev set for experiments
    size = int(percentage*len(data["train"]))
    data_slice = data["train"].shuffle(seed=42).select([i for i in
list(range(size))])

    X, Y =
generate_input_output_for_logistic_regression_model(data_slice)

    # create dev split
    dev_split_idx = int(0.8*len(X))
    # first 80% of the data slice will become the train set
    train_inputs = X[:dev_split_idx]
    train_labels = Y[:dev_split_idx]
    # rest of the 20% will become the dev set
    dev_inputs = X[dev_split_idx:]
    dev_labels = Y[dev_split_idx:]

    return train_inputs, train_labels, dev_inputs, dev_labels

def calculate_accuracy(Y_test, predictions):
    """
    input: Y_test -> List, predictions -> List
    Y_test is a list of gold labels of the test set
    predictions is a list of model predictions for the test set
    """

```

output: accuracy -> float

The length of Y_test and predictions should be equal.

To calculate accuracy, count the number of times when the gold label matches the model prediction.

Multiply this count by 100 and divide it by the length of the list
"""

```
correct, total = 0, 0
for y, yp in zip(Y_test, predictions):
    total += 1
    if y == yp:
        correct += 1
accuracy = round(correct*100/total, 2)

return accuracy
```

```
def calculate_f1(Y_test, predictions):
    """
```

input: Y_test -> List, predictions -> List

Y_test is a list of gold labels of the test set

predictions is a list of model predictions for the test set

output: precision, recall, f1_score -> float

Make sure to only return each value upto 2 decimal places

The length of Y_test and predictions should be equal.

To calculate F1 score, first calculate precision and recall.

*F1 Score = 2 * precision * recall / (precision + recall)*

Precision = True Positive / (False Positive + True Positive)

Recall = True Positive / (False Negative + True Positive)

True positives for a class C are the instances of the class C for which the model also correctly predicts the same class C as the output label.

False positives for a class C are the instances that do not belong to class C, but the model incorrectly predicts to them to be belong to class C.

True negatives for a class C are the instances that do not belong to class C, for which the model also does not predict the class C (i.e. predicts some other class 'D').

False negatives for a class C are the instances of the class C for which the model incorrectly predicts a different class (say some D) as the output label.

```

    """
    true_positive, false_positive, true_negative, false_negative = 0,
0, 0, 0
    for y, yp in zip(Y_test, predictions):
        if (y == 1) and (yp == 1):
            true_positive += 1
        if (y == 0) and (yp == 0):
            true_negative += 1
        if (y == 1) and (yp == 0):
            false_negative += 1
        if (y == 0) and (yp == 1):
            false_positive += 1
    precision = true_positive / (false_positive+true_positive)
    recall = true_positive / (false_negative+true_positive)
    f1_score = 2 * precision * recall / (precision + recall)
    precision = round(precision, 2)
    recall = round(recall, 2)
    f1_score = round(f1_score, 2)

    return precision, recall, f1_score

class LogisticRegressionModel(nn.Module):

    # This method initializes the Logisititc Regression model.
    def __init__(self, input_dim, output_dim):
        super(LogisticRegressionModel, self).__init__()

        # Weight matrix
        self.weight = nn.Parameter(torch.randn(input_dim, output_dim,
requires_grad=True))

        #bias vector
        bias_value = torch.sqrt(torch.tensor([input_dim],
dtype=torch.float))
        self.bias = nn.Parameter(bias_value.repeat(output_dim),
requires_grad=True)

        # This is a special initialization that samples values
        # uniformly in a narrow range of values depending on the
number of
        # weights. You need this type of initialization in order to
ensure
        # the weights are set up in a way that enables good
convergence.
        init.kaiming_uniform_(self.weight, a=0, mode='fan_out',
nonlinearity='sigmoid')

        # Note: we can also use nn.Linear(input_dim, output_dim)
        # Don't uncomment this line. This is just for you to see how
        # this can be done.

```

```

        #self.linear = nn.Linear(input_dim, output_dim)

    # This method is invoked for producing the outputs of the
    classifier.
    # NOTE: This method should really return the probability of the
    classes.
    # However, we simply return the  $W^T.X + b$  instead of  $\text{softmax}(W^T.X + b)$  here.
    # This is because of stability issues.
    def forward(self, x):
        # Use the torch matrix multiply function to obtain x
        multiplied by W
        # NOTE: Use torch.mm to do the matrix multiplication.
        output = torch.mm(x, self.weight) + self.bias

        # The output can also be computed using the forward method of
        the self.linear
        # Don't uncomment this line.
        #output = self.linear(x)
        return output

# This is the method that trains the logistic regression model.
# train and dev loaders allow us to go over training and dev instances
in batches.
# model is the logisitic regression model class.
# criterion is the loss function (i.e., negative log-likelihood).
# optimizer is the optimization algorithm (e.g. a specific variant of
gradient descent called Adam)
# num_epochs is the maximum number of epochs you want to allow for
training.
def train_nn_model(train_loader, dev_loader, model, criterion,
optimizer, num_epochs):
    best_dev_loss = -9999
    patience = 0
    dev_losses = []
    train_losses = []

    torch.manual_seed(42)

    # train the model by iterating over the dataset for num_epochs
    epochs
    for epoch in range(num_epochs):

        # keep track of the training loss across batches in each epoch
        epoch_loss = 0

        # iterate over train_loader and each slice is a batch in the
        data
        # i is the batch counter, X is the inputs for a batch and Y is
        the outputs

```



```

        # the length of the loader represents the number of batches in
the data
        for i, (X, Y) in enumerate(train_loader):

            outputs = model(X)

            loss = criterion(outputs, Y)

            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

            # storing the loss of each batch
            epoch_loss += loss

        # calculating the average training loss across the batches
        avg_training_loss = epoch_loss/len(train_loader)
        train_losses.append(avg_training_loss.item())

        # Calculate the loss on the dev dataset
        # to calculate the loss over the dev set, iterate over
dev_loader
        # for each set of inputs in the batch, first generate model
predictions
        # calculate the loss by calling the criterion
        # add that to the dev_loss variable
        # after the loop ends, remember to divide the summed dev_loss
by the length of the dev_loader
        dev_loss = 0
        for xDev, yDev in dev_loader:
            outputs = model(xDev)

            dev_loss += criterion(outputs, yDev)

        dev_loss = dev_loss/len(dev_loader)
        dev_losses.append(dev_loss.item())

        print(f"Epoch: {epoch+1}; Train loss:
{avg_training_loss.item()}; Dev loss: {dev_loss.item()}")

    return train_losses, dev_losses

def visualize_losses(train_losses, dev_losses):
    """
    input: losses -> List
    take an array of loss on the dev set and visualize it using
matplotlib
    """

    plt.clf()

```

```

epochs = list(range(1, len(train_losses)+1))

plt.plot(epochs, train_losses, marker='x', label='Training Loss')
plt.plot(epochs, dev_losses, marker='+', label='Dev Loss')

# Add legend to the plot
plt.legend()
plt.xlabel('Epoch Number')
plt.ylabel('Loss')
plt.title('Change in Losses over Epochs in Training')

# Show the chart
plt.show()

def convert_to_embeddings(examples, dictionary, embeddings):
    """
    input: examples -> List, dictionary -> Dict, embeddings -> Dict
    examples is the list of sentences we want to get corresponding
    embeddings for
    dictionary is the word -> embedding dictionary we created in the
    glove model
    embeddings are the trained glove embeddings
    output: embedded_examples -> List
    embedded_examples is the list of sentence (word-averaged)
    embeddings for the input examples
    """

    embedded_examples = []
    for review in examples:
        words = review.lower().split(' ')
        word_vectors = []
        for word in words:
            try:
                word_vector = embeddings[dictionary[word]]
            except:
                # if word is not in vocab, treat it as UNK token
                word_vector = embeddings[dictionary['UNK']]
            word_vectors.append(word_vector)
        word_vectors = np.array(word_vectors)
        word_vectors = np.mean(word_vectors, axis=0)
        embedded_examples.append(word_vectors)

    return embedded_examples

def convert_to_glove_embeddings(examples, embeddings):
    """
    input: examples -> List, embeddings -> Dict
    examples is the list of sentences we want to get corresponding
    embeddings for
    embeddings are the trained GloVe embeddings
    """

```

```
output: embedded_examples -> List
embedded_examples is the list of sentence (word-averaged)
embeddings for the input examples
"""
```

```
# this is embedding from 6B.300d GloVe file for UNK token, see
reference here: https://stackoverflow.com/questions/49239941/what-is-unk-in-the-pretrained-glove-vector-files-e-g-glove-6b-50d-txt
```

```
unk_vector_string = '0.22418134 -0.28881392 0.13854356 0.00365387
-0.12870757 0.10243822 0.061626635 0.07318011 -0.061350107 -1.3477012
0.42037755 -0.063593924 -0.09683349 0.18086134 0.23704372 0.014126852
0.170096 -1.1491593 0.31497982 0.06622181 0.024687296 0.076693475
0.13851812 0.021302193 -0.06640582 -0.010336159 0.13523154 -
0.042144544 -0.11938788 0.006948221 0.13333307 -0.18276379 0.052385733
0.008943111 -0.23957317 0.08500333 -0.006894406 0.0015864656
0.063391194 0.19177166 -0.13113557 -0.11295479 -0.14276934 0.03413971
-0.034278486 -0.051366422 0.18891625 -0.16673574 -0.057783455
0.036823478 0.08078679 0.022949161 0.033298038 0.011784158 0.05643189
-0.042776518 0.011959623 0.011552498 -0.0007971594 0.11300405 -
0.031369694 -0.0061559738 -0.009043574 -0.415336 -0.18870236
0.13708843 0.005911723 -0.113035575 -0.030096142 -0.23908928 -
0.05354085 -0.044904727 -0.20228513 0.0065645403 -0.09578946 -
0.07391877 -0.06487607 0.111740574 -0.048649278 -0.16565254 -
0.052037314 -0.078968436 0.13684988 0.0757494 -0.006275573 0.28693774
0.52017444 -0.0877165 -0.33010918 -0.1359622 0.114895485 -0.09744406
0.06269521 0.12118575 -0.08026362 0.35256687 -0.060017522 -0.04889904
-0.06828978 0.088740796 0.003964443 -0.0766291 0.1263925 0.07809314 -
0.023164088 -0.5680669 -0.037892066 -0.1350967 -0.11351585 -
0.111434504 -0.0905027 0.25174105 -0.14841858 0.034635577 -0.07334565
0.06320108 -0.038343467 -0.05413284 0.042197507 -0.090380974 -
0.070528865 -0.009174437 0.009069661 0.1405178 0.02958134 -0.036431845
-0.08625681 0.042951006 0.08230793 0.0903314 -0.12279937 -0.013899368
0.048119213 0.08678239 -0.14450377 -0.04424887 0.018319942 0.015026873
-0.100526 0.06021201 0.74059093 -0.0016333034 -0.24960588 -0.023739101
0.016396184 0.11928964 0.13950661 -0.031624354 -0.01645025 0.14079992
-0.0002824564 -0.08052984 -0.0021310581 -0.025350995 0.086938225
0.14308536 0.17146006 -0.13943303 0.048792403 0.09274929 -0.053167373
0.031103406 0.012354865 0.21057427 0.32618305 0.18015954 -0.15881181
0.15322933 -0.22558987 -0.04200665 0.0084689725 0.038156632 0.15188617
0.13274793 0.113756925 -0.095273495 -0.049490947 -0.10265804 -
0.27064866 -0.034567792 -0.018810693 -0.0010360252 0.10340131
0.13883452 0.21131058 -0.01981019 0.1833468 -0.10751636 -0.03128868
0.02518242 0.23232952 0.042052146 0.11731903 -0.15506615 0.0063580726
-0.15429358 0.1511722 0.12745973 0.2576985 -0.25486213 -0.0709463
0.17983761 0.054027 -0.09884228 -0.24595179 -0.093028545 -0.028203879
0.094398156 0.09233813 0.029291354 0.13110267 0.15682974 -0.016919162
0.23927948 -0.1343307 -0.22422817 0.14634751 -0.064993896 0.4703685 -
0.027190214 0.06224946 -0.091360025 0.21490277 -0.19562101 -0.10032754
-0.09056772 -0.06203493 -0.18876675 -0.10963594 -0.27734384 0.12616494
```

```

-0.02217992 -0.16058226 -0.080475815 0.026953284 0.110732645
0.014894041 0.09416802 0.14299914 -0.1594008 -0.066080004 -0.007995227
-0.11668856 -0.13081996 -0.09237365 0.14741232 0.09180138 0.081735
0.3211204 -0.0036552632 -0.047030564 -0.02311798 0.048961394
0.08669574 -0.06766279 -0.50028914 -0.048515294 0.14144728 -
0.032994404 -0.11954345 -0.14929578 -0.2388355 -0.019883996 -
0.15917352 -0.052084364 0.2801028 -0.0029121689 -0.054581646 -
0.47385484 0.17112483 -0.12066923 -0.042173345 0.1395337 0.26115036
0.012869649 0.009291686 -0.0026459037 -0.075331464 0.017840583 -
0.26869613 -0.21820338 -0.17084768 -0.1022808 -0.055290595 0.13513643
0.12362477 -0.10980586 0.13980341 -0.20233242 0.08813751 0.3849736 -
0.10653763 -0.06199595 0.028849555 0.03230154 0.023856193 0.069950655
0.19310954 -0.077677034 -0.144811'
    unk_vector = np.array(unk_vector_string.split(" "),
dtype='float32')
    unk_vector = []
    for val in unk_vector_string.split(" "):
        unk_vector.append(float(val))

    embedded_examples = []
    for review in examples:
        words = review.lower().split(' ')
        word_vectors = []
        for word in words:
            try:
                word_vector = list(embeddings[word])
            except:
                # if word is not in vocab, treat it as UNK token
                word_vector = unk_vector
            word_vectors.append(word_vector)
        word_vectors = np.array(word_vectors)
        word_vectors = np.mean(word_vectors, axis=0)
        embedded_examples.append(word_vectors)

    return embedded_examples

def test_nn_model(test_loader, model):
    predictions, gold_labels = [], []
    for X, Y in test_loader:
        outputs = model(X)
        predicted = torch.argmax(outputs, dim = 1)
        predictions.extend(predicted.numpy().tolist())
        gold_labels.extend(Y.numpy().tolist())

    accuracy = calculate_accuracy(gold_labels, predictions)
    precision, recall, f1_score = calculate_f1(gold_labels,
predictions)
    print(f"\nAccuracy: {accuracy}; Precision: {precision}; Recall:
{recall}; F1 score: {f1_score}")

```

```

def run_neural_model_pipeline(data, hparams):
    """
    input: data -> Any, hparams -> Dict
    data is the downloaded dataset
    hparams is a dictionary of hyperparameters used to train the model
    """

    # extract hyperparameters from specified dictionary
    print(f"Hyperparameters: {hparams}")
    num_epochs = hparams["num_epochs"]
    batch_size = hparams["batch_size"]
    learning_rate = hparams["learning_rate"]
    input_size = hparams["input_size"]
    percentage = hparams["data_percentage"] # the fraction of training
    data to use for training.
    embedding_path = hparams["embedding_path"]
    output_dim = 2

    X_train, Y_train, X_dev, Y_dev = generate_data_slice(data,
percentage)
    X_test, Y_test =
generate_input_output_for_logistic_regression_model(data["test"])

    print(f"Training data size: {len(X_train)}")
    print(f"Dev data size: {len(X_dev)}")
    print(f"Test data size: {len(X_test)}")

    embeddings = load_glove_vectors()

    X_train = convert_to_glove_embeddings(X_train, embeddings)
    X_dev = convert_to_glove_embeddings(X_dev, embeddings)
    X_test = convert_to_glove_embeddings(X_test, embeddings)

    Y_train = np.array(Y_train)
    Y_dev = np.array(Y_dev)
    Y_test = np.array(Y_test)

    # convert the numpy arrays to tensors, create a tensor dataset
    objects and then torch dataloaders for easy batching
    X_train = torch.tensor(X_train).float()
    Y_train = torch.tensor(Y_train).long()

    X_dev = torch.tensor(X_dev).float()
    Y_dev = torch.tensor(Y_dev).long()

    X_test = torch.tensor(X_test).float()
    Y_test = torch.tensor(Y_test).long()

    train_tensorDataset = torch.utils.data.TensorDataset(X_train,
Y_train)
    dev_tensorDataset = torch.utils.data.TensorDataset(X_dev, Y_dev)

```

```

    test_tensorDataset = torch.utils.data.TensorDataset(X_test,
Y_test)

    train_loader =
torch.utils.data.DataLoader(dataset=train_tensorDataset,
batch_size=batch_size, shuffle=True)
    dev_loader =
torch.utils.data.DataLoader(dataset=dev_tensorDataset,
batch_size=batch_size, shuffle=False)
    test_loader =
torch.utils.data.DataLoader(dataset=test_tensorDataset,
batch_size=batch_size, shuffle=False)

    # setting up the model, loss function and optimizer
    model = LogisticRegressionModel(input_size, output_dim)
    criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=learning_rate)

    train_losses, dev_losses = train_nn_model(train_loader,
dev_loader, model, criterion, optimizer, num_epochs)
    visualize_losses(train_losses, dev_losses)
    test_nn_model(test_loader, model)

def create_hyperparameter_options(num_epochs, batch_size,
learning_rate, data_percentage, input_size, embedding_path):
    """
    input:
    num_epochs -> int
    batch_size -> int
    num_epochs -> int
    data_percentage -> float
    input_size -> int

    output: hparams -> Dict
    """

    hparams = {
        "num_epochs": num_epochs,
        "batch_size": batch_size,
        "learning_rate": learning_rate,
        "input_size": input_size,
        "data_percentage": data_percentage,
        "embedding_path": embedding_path
    }

    return hparams

# hyperparameters

# batch size represent the number of data points on which loss is

```

computed and backpropagated. Larger batch sizes (to a certain degree) often lead to more stable model training
learning rate represents a factor by which model parameters are updated after each batch is processed. Usually, lower learning rates (to a certain degree) lead to more stable model training
num epochs represents the maximum number of iterations over the training data a model performs. Usually, models improve over more epochs but there is a limit
input size represents the size of the vectors to be used. Higher the number, richer the representation. But that also brings along computational complexity
data percentage represents how much of the train split will be used for train+dev splits of the experiments. More data usually leads to better models

we store hyperparameters in options dictionary and pass it to the run_neural_model_pipeline function
the hparams dict should necessarily contain all the keys demonstrated below

```
num_epochs = 11
batch_size = 128
learning_rate = 0.001
input_size = 300
embedding_path = 'glove.6B.300d.txt'
```

```
data_percentage = 1
hparams = create_hyperparameter_options(num_epochs, batch_size,
learning_rate, data_percentage, input_size, embedding_path)
```

```
data = load_dataset("imdb")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
```

The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```
{"model_id": "d9694dd54fe646cfb6d9d736d8cfa66e", "version_major": 2, "version_minor": 0}
```

Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending

unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

```
{"model_id": "7d3921305b1245f69f8498a3ac24d066", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "1e6b0b40feff44318dd5cd0f6cc6c50f", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "4acd8fd28d554b4db36346f6e1a4b7f6", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "ae22dc5448af404bb09a6b15552ddd7f", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "88452cf147e94006b4d6c96cb03f01f9", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "73d8f1f4025748c5a2e46743d038b90c", "version_major": 2, "version_minor": 0}
```

```
run_neural_model_pipeline(data, hparams)
```

```
Hyperparameters: {'num_epochs': 11, 'batch_size': 128, 'learning_rate': 0.001, 'input_size': 300, 'data_percentage': 1, 'embedding_path': 'glove.6B.300d.txt'}
```

```
Training data size: 20000
```

```
Dev data size: 5000
```

```
Test data size: 25000
```

```
Found 400000 word vectors.
```

```
/tmp/ipython-input-2276/1453223946.py:36: UserWarning: Creating a tensor from a list of numpy.ndarrays is extremely slow. Please consider converting the list to a single numpy.ndarray with numpy.array() before converting to a tensor. (Triggered internally at /pytorch/torch/csrc/utils/tensor_new.cpp:253.)
```

```
X_train = torch.tensor(X_train).float()
```

```
Epoch: 1; Train loss: 0.6633991599082947; Dev loss: 0.6417169570922852
```

```
Epoch: 2; Train loss: 0.6217861175537109; Dev loss: 0.6104958057403564
```

```
Epoch: 3; Train loss: 0.5931899547576904; Dev loss: 0.589229941368103
```

```
Epoch: 4; Train loss: 0.5725799202919006; Dev loss: 0.5691683888435364
```

```
Epoch: 5; Train loss: 0.5559154152870178; Dev loss: 0.5546500086784363
```

```
Epoch: 6; Train loss: 0.5424410700798035; Dev loss: 0.5427017211914062
```

```
Epoch: 7; Train loss: 0.5306751132011414; Dev loss: 0.5320853590965271
```

```
Epoch: 8; Train loss: 0.5217226147651672; Dev loss: 0.5254524350166321
```

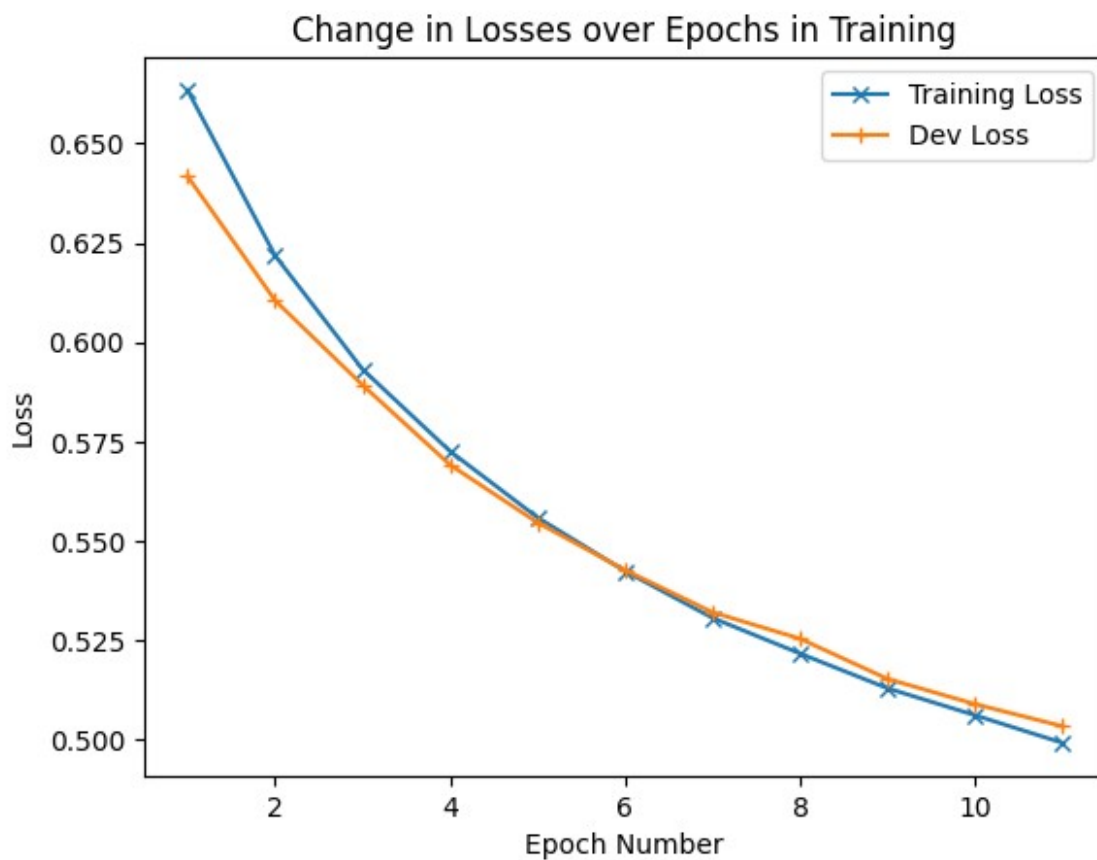
```
Epoch: 9; Train loss: 0.5129861235618591; Dev loss: 0.5153419375419617
```

```
Epoch: 10; Train loss: 0.5062018632888794; Dev loss:
```

```
0.5089990496635437
```

```
Epoch: 11; Train loss: 0.4992505609989166; Dev loss:
```

```
0.5034319162368774
```

Accuracy: 77.46; Precision: 0.81; Recall: 0.72; F1 score: 0.76

Submission Guidelines

Create a folder having your solution and should contain the following:

- This notebook with your solution
- All linked files provided in the 'Files to upload in notebook' folder
- A 'solution/' folder containing the files mentioned below

Files to be generated and submitted: Create a new folder called `submission/` and place the following files in it:

- `test_preds_glove.txt` - Your best Glove model predictions for `word_analogy_test.txt`
- `gdrive_link.txt` - Should contain a `wget` table to a folder that contains your best models. The model files should be named `glove.model`, and the folder should be named `538-hw1-<SBUID>-models`. Please make sure you provide the necessary permissions.
- `<SBUID>_Report.pdf` - A PDF report as detailed below.

Collaboration Guidelines

- You can collaborate to discuss ideas and to help each other for better understanding of concepts and math.
- You should NOT collaborate on the code level. This includes all implementation activities: design, coding, and debugging.
- You should NOT not use any code that you did not write to complete the assignment.
- The homework will be **cross-checked**. Do not cheat at all! It's worth doing the homework partially instead of cheating and copying your code and get 0 for the whole homework. In previous years, students have faced harsh disciplinary action as a result of the same.

Extra Notes

- If you add any code apart from the TODOs in the codebase (note that you don't need to), please mark it by commenting in the code itself. An example of the same could be: `# Adding some_global_var for XXX      some_global_var`
- General tips when you work on tensor computations:
 - Break the whole list of operations into smaller ones.
 - Write down the shapes of the tensors

Credits and Disclaimer

Credits: This code is part of the starter package of the assignment/s used in NLP course at Stony Brook University. This assignment has been designed, implemented and revamped as required by many NLP TAs to varying degrees. In chronological order of TAship they include Heeyoung Kwon, Jun Kang, Mohaddeseh Bastan, Harsh Trivedi, Matthew Matero, Nikita Soni, Sharvil Katariya, Yash Kumar Lal, Adithya V. Ganesan, Sounak Mondal, and Jasdeep Singh Grover. Thanks to all of them!

Disclaimer/License: This code is only for school assignment purpose, and **any version of this should NOT be shared publicly on github or otherwise even after semester ends**. Public availability of answers devalues usability of the assignment and work of several TAs who have contributed to this. We hope you'll respect this restriction.

```
!wget -O "538-hw1-117411478-models.zip"
"https://drive.google.com/file/d/1rcae0Iqjmw2cg4vxwZcvWhFoB1ccCnzg/
view?usp=drive_link"

--2026-03-03 17:14:41--
https://drive.google.com/file/d/1rcae0Iqjmw2cg4vxwZcvWhFoB1ccCnzg/
view?usp=drive_link
Resolving drive.google.com (drive.google.com)... 172.217.194.113,
172.217.194.138, 172.217.194.100, ...
Connecting to drive.google.com (drive.google.com)|
172.217.194.113|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: '538-hw1-117411478-models.zip'
```

538-hw1-117411478-m [<=>] 113.53K ---KB/s in
0.04s

2026-03-03 17:14:43 (2.56 MB/s) - '538-hw1-117411478-models.zip' saved
[116258]